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Introduction to Economic

Economics is the social science that analyzes the production, distribution, and consumption of goods and services. The term *economics* comes from the Ancient Greekοἰκονομία (*oikonomia*, "management of a household, administration") from οἶκος (*oikos*, "house") + νόμος (*nomos*, "custom" or "law"), hence "rules of the household.

"**Economics**" has been called "the science of how people make a living." We all have to make a living! Shouldn't we all understand something about economics? Often, though, when people talk about *Economics*, the words seem to get all twisted around! How do we even know what's being talked about?

Economics is a science which studies human behavior as a relationship between ends and scarce means which have alternative uses

Generally, economics is defined as a social science which covers the actions of individuals and groups of individuals in the processes of producing, exchanging and consuming of goods and services. In other words it is the "study of the allocation of scarce resources among alternative uses."

Illustrative Example

1. The term scarcity in economics refers to the fact that:
   A. economic wants are limited and resources are abused.
   B. even in the richest country some people goes hungry.
   C. no country can produce enough products to satisfy everybody's economic wants.
   D. it is impossible to produce too much of any particular good or service in a market economy

Explanation: Scarcity is a fundamental concept of economics. It indicates that less of a good in freely available than consumers would like to have. Because of scarcity societies cannot produce enough products to satisfy all societies' wants.

Answer: C

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2. The basic economic problem is essentially one of deciding how to make the best use of:
   A. limited resources to satisfy limited economic wants.
   B. limited resources to satisfy unlimited economic wants.
   C. unlimited resources to satisfy unlimited economic wants.
   D. unlimited resources to satisfy limited economic wants.

***Explanation*:** The starting point of any economic analysis is the existence of unlimited human wants and very limited means

Answer: B

3. The key economic concept that serves as the basis for the study of economics is:
   A. inflation.
   B. unemployment.
   C. money.
   D. scarcity

Explanation: Economics as a subject matter arises in this world is because of the existence of scarcity.

Answer: D

4. The main concern of economics is
   A. how to organize and run a business
   B. how individuals and societies organize their scarce resources
   C. how to protect the consumer
   D. government spending and taxing

***Explanation*:**

- The main concern of economics is about people and the choice they make.
- Scarce resources are to be allocated among different uses in the best possible manner.

Answer: B

5. What is Economics?

***Explanation*:** Economics is the study of how people choose to use their limited resources to produce, exchange consumer goods and services.

6. The crucial problem of economics is
   A. establishing a fair tax system.
   B. providing social goods and services.
   C. developing a price mechanism that reflects the relative scarcities of products and resources.
   D. allocating scarce productive resources to satisfy wants.

***Explanation*:** The economic problem is unlimited human wants and very limited resources to satisfy these wants.

Answer: D

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7. The primary focus of the study of economics is with:
   A. expanding the production of goods and services.
   B. equalizing the distribution of consumer income and wealth.
   C. making the most efficient use of scarce productive resources.
   D. reallocating resources from consumption to production in the economy.

***Explanation*:** The main concern of economics is the study of how peoples choose to use scarce resources in order to satisfy unlimited human wants.

Answer: C

8. What is the purpose of studying economics?

Explanation:

- Studying economics is an abstract way of thinking that allows the development of principles or tools that can be used to study complex social problems.
- It helps to understand the nature of any major economic problems such as inflation or mass poverty and thinking about any major national issues.

9. When economic growth occurs?

***Explanation*:** Economic growth occurs when society produces more, either by acquiring more resources or by learning to produce more with the existing resources.

10. What is the main subject matter of economics?

***Explanation*:**

- Economics is the social science. It is concerned with the solution of society's economic problems and human behavior as a relationship between ends and scarce means which have alternative uses.

11. The economizing problem involves the allocation of resources among competing wants. What are the economizing problems?

Explanation:

The central problems to allocate resources are:

- Unlimited wants
- Limited resources

12. What is economics about?

***Explanation*:** Economics is about people and the choice they make. The basic units of analysis in economics are individuals, firms and government are guided and directed by the choice they make.

1.2 Branch of economics:

The two major branches of economics are:

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1.2.1Microeconomics:

Microeconomics is the study of particular markets, and segments of the economy. It looks at issues such as consumer behavior, individual labor markets, and the theory of firms. It is the study of the economy in the small.

Micro economics is concerned with:

- Supply and demand in individual markets
- Individual consumer behavior. Example -Consumer choice theory
- Individual labor markets. Example - demand for labor, wage determination
- Externalities arising from production and consumption.

1.2.2 Macroeconomics:

Macroeconomics is the study of the economy in the large. Rather than dealing with individual markets and individual consumers and producers, macroeconomics deals with the economy as a whole.

Macroeconomics is concerned with

- Monetary / fiscal policy.

Example: what effect does interest rates have on whole economy?

- Reasons for inflation, and unemployment
- Economic Growth
- International trade and globalization
- Reasons for differences in living standards and economic growth between countries.
- Government borrowing

Illustrative Example

13. Microeconomics studies such topics as
    A. The factors that determine inflation.
    B. The forces that influence the price of shoes.
    C. The determinants of total output.
    D. The overall trend in stock market prices.

***Explanation*:** Microeconomics studies the decision making of individual economic units, such as the pricing decisions of individual shoe companies

Answer: B

14. Microeconomics is not concerned with:
    A. the quantities of each good produced.
    B. the size of aggregate money flows in the economy.
    C. the distribution of goods among households.
    D. the allocate function of changes in relative prices.

***Explanation*:** The size of aggregate money flows in the economy is the concern of macroeconomics.

Answer: B

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15. All of the following basic economic questions deal with microeconomics except:
    A. How is the supply of goods allocated among the members of the society?
    B. What goods and services are being produced and in what quantities?
    C. By what methods are goods and services being produced?
    D. Is the economy's capacity to produce goods growing over time?

Explanation:

- The central problems of microeconomics are: what to produce, how to produce and for whom to produce.
- The economic capacity to produce goods growing over time is the concern of macroeconomics.

Answer: D

16. The basic difference between macroeconomics and microeconomics is that:
    A. Macroeconomics looks at the forest (aggregate markets), while microeconomic is concerned with the individual trees (subcomponents).
    B. Macroeconomics is concerned with policy decisions, while microeconomics applies only to theory.
    C. Microeconomics is concerned with the forest (aggregate markets), while macroeconomics is concerned with the trees (components).
    D. opportunity cost is applicable to macroeconomics, and the fallacy of composition relates to microeconomics

Explanation:

- Microeconomics focus on microscopic units such as individual consumer, individual firms, individual price level of goods and services.
- While macroeconomics is concerned with highly aggregate units such as national output, total employment level, general price level etc.

Answer: A

17. Microeconomics is the branch of economics that focuses on
    A. national economic activity
    B. individual decision makers and markets
    C. deficit spending
    D. inflation and unemployment

Explanation:

- Microeconomics is concerned with microscopic study of the various elements of the economic system.

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- It is the study of the economic behavior of individual consumer and producer.

Answer: B

18. Macroeconomics is the study of economics from the standpoint of:
    A. individual economic units.
    B. a typical firm.
    C. a typical household.
    D. the overall economy

***Explanation*:** Macroeconomics takes an **overall view** of the economy.

Answer: D

19. Which of the following subjects fall under microeconomics and which ones under macroeconomics?
    A. What determine the overall level of employment in Ethiopian economy
    B. How many workers are employed in Kombolcha textile industry
    C. The price of computers increases from time to time
    D. Drought will affect the output of coffee
    E. Price of medical care
    F. Number of employees in Black Lion Hospitals
    G. Unemployment rate
    H. How many jobs exists in the economy
    I. How general price level is determined?
    J. What determines rate of economic growth of a country.
    K. Cement production in a year
    L. Output of cotton fabrics
    M. Level of national income
    N. How to produce

Answer:

Microeconomics concerns are: B, C, D, E, F, K, L, N

Macroeconomics concern: A, G, H, I, J, M

1.3 Positive and Normative economics:

1.3.1 Positive economics:

Positive statement is a statement about what is and contains no indication of approval or disapproval. A positive statement can be wrong; it can be tested by objective use of evidence.

The tools of positive economics are reason, logic and empiricism

Positive statements are **objective statements** that can be tested, amended or rejected by referring to the available **evidence**. Positive economics deals with **objective explanation** and the testing and rejection of theories. For example:

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- A fall in incomes will lead to a rise in demand for own-label supermarket foods
- If the government raises the tax on beer, this will lead to a fall in profits of the brewers.
- The rising price of crude oil on world markets will lead to an increase in cycling to work
- A reduction in income tax will improve the incentives of the unemployed to find work.
- A rise in average temperatures will increase the demand for sun screen products.
- Higher interest rates will reduce house prices
- Cut-price alcohol has increased the demand for alcohol among teenagers

1.3.2 Normative economics:

A normative statement: expresses a **value judgment** about whether a situation is desirable or undesirable.

Statements that include indicator words such as: should, ought, or prefer are likely to be normative rather than positive

Normative statements are **subjective statements** rather than objective statements – i.e. they carry **value judgments**. For example:

- Pollution is the most serious economic problem
- Unemployment is more harmful than inflation
- The government should increase the minimum wage per hour to reduce poverty.
- The government is right to introduce a ban on smoking in public places.
- The retirement age should be raised to 70 to combat the effects of our ageing population.

Resources are best allocated by allowing the market mechanism to work freely

Illustrative Example

20. Which of the following is a positive economic statement?
    A. If the government increases its tax on gasoline, the price people pay for gasoline will rise.
    B. The government should not let the price of gasoline rise, because this increase would harm poor People
    C. The government should let the price of gasoline rise, because this would help retired people who own stock in petroleum companies.

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D. The government ought not to control the price of gasoline, because its price should be determined in the free marketplace.

***Explanation*:** Positive economics is concerned on the cause and effect relationship between economic variables and it tries to explain how the world actually works.

Answer: A

21. Economists tend to disagree primarily about
    A. The implications of scarcity for our economy.
    B. Which resources are free?
    C. Topics in positive economics.
    D. Issues of normative economics.

Explanation: Economics do disagree often strongly about normative economics.

Answer: D

22. Which one of the following is a positive economic statement?
    A. An increase in the price of butter causes consumers to buy less butter.
    B. Social conscience demands that we increase the minimum wage.
    C. Taxes should be raised to halt inflation.
    D. The sales tax on food should be repealed.

***Explanation*:**

- Positive economic studies about the cause and effect of relationship between variables.
- The consumer buy less of butter is due to an increase in the price of butter.

Answer: A

23. "More jobs should be made available to teenagers." This is an example of
    A. a normative statement
    B. a positive statement
    C. an assumption
    D. a hypothesis

***Explanation*:** Normative economics involves value judgment on whether what happens is socially desirable or undesirable.

Answer: A

24. An editor of a local newspaper called for the government to abolish the minimum wage because it takes advantage of consumers. This is an example of:
    A. a positive economic statement
    B. a pure economic statement
    C. an exogenous use of economics
    D. a normative economic statement

***Explanation*:**

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- Normative economics involves value judgment what is socially desirable or undesirable.
- In the statement it is the opinion of the newspaper editor to abolish the minimum wage is a normative statement.

Answer: D

25. "A compassionate welfare program should have a higher priority than strong defense." This statement is an example of
    A. the ceteris paribus fallacy
    B. a normative economic statement
    C. a positive economic statement
    D. the fallacy of composition

***Explanation*:** Normative economic statements are about 'what ought to be', given the philosophical views of the advocate.

Answer: B

26. Which of the following topics fall under positive economics and normative economics?
    A. People who earn large incomes ought to pay more income tax than people earn low income.
    B. What polices will prevent inflation
    C. The government should increase its defense expenditure.
    D. Is this bed made of wood?
    E. Is she a beautiful woman?
    F. An increase in the personal income tax will slow the growth rate of the economy
    G. The goal of any country's economic policy should be to increase the well-being of its poorest, most vulnerable citizens

Answer: positive economics are: - B, D, and F
Normative economics are: - A, C, E and G

1.4 The economic problem: scarcity and choice:

The most important fact of economics is the law of scarcity: there will never be enough resources to meet everyone's wants. Economics is the study of how scarce resources are allocated among competing en

Limited Resources → SCARCITY of goods and services ← unlimited wants

Scarcity is the widespread condition of limited resources and unlimited wants and needs. An item is a scarce good if the amount available is less than the amount people want if it would be given away free of charge.

1.4.1 Resources:

Resources are those things we use to produce the things we want.
Resources can be classified as free resources and economic resources

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Free resources:

A free resource is one where the amount available is greater than the amount people want at zero prices.

Economic resource:

Economics is the study of how people, firms, and societies use their scarce productive resources to best satisfy their unlimited material wants.

Economic resource is a resource with an available quantity less than its desired use.

Economic or scarce resources are also called factors of production and generally classified as labor, capital, land, or entrepreneurship. Economic resources are the workers, equipment, raw materials, and organizers that are used to produce economic goods.

1.4.2 Factors of production:

The factors of production can be divided in to four categories- land, labor, and capital and entrepreneur ships.

Land: any resource created by nature. This may be arable land, mineral deposits, oil and gas reserves, forests, water resources, oxygen, and so on. The return to land is rent.

Labor: is the physical and mental talent that human beings contribute to the production process. The return to labor is wages.

Capital: manmade equipment like machinery, buildings, roads, vehicles, computers, tools and other manufactured goods used to produce other goods and services. The return to capital is interest.

An entrepreneur: is a person who organizes, manages, and assumes the risks of a firm, taking a new idea or a new product and turning it into a successful business

Entrepreneurial ability: the effort and know-how to put the other resources together in a productive venture. A special type of human resource that organizes the other three factors of production, makes business decisions, innovates, and bears business risk. Return to entrepreneurship is profit. The entrepreneur does four things:

1. Comes up with new ideas
2. Brings all the other resources together
3. Makes the basic decisions, and
4. Takes the risks to earn profits (or losses).

1.4.3 Choice:

The starting point of any economic analysis is the existence of human wants. Human wants are unlimited, but human capacity to satisfy the wants

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is limited. This is the economic problem unlimited wants, very limited means. We seek to overcome the problem of scarcity by exercising our choice. Choice is the act of selecting among limited alternatives. Economics is basically concerned with how people choose when they are faced with limited alternatives.

Limited Resources → **SCARCITY of goods and services** ← **unlimited wants**

↓

ECONOMIZING PROBLEM
(must make choices)

Illustrative Example

27. The problem of scarcity as understood in economics?
    A. Exists because ownership of resources is dominated by a few individuals
    B. Exists because people's wants, in general, are unlimited compared to the available resources
    C. Exists because past generations have over exploited the available resources
    D. Could be eliminated through appropriate government policies

***Explanation*:** Scarcity is a situation in which unlimited wants exceed the limited resources available to full fill those wants.

Answer: B

28. Which of the following is **not** a scarce resource?
    A. Labor, when the unemployment rate is near 10 percent
    B. Fertile land in Deseret
    C. Sunshine for solar power in the deserts areas
    D. Factories used to produce personal computers

***Explanation*:**

- If the prices of all the other resources listed were set to zero, the amount people would want of them would vastly exceed the amount offered.
- Therefore, by definition, these resources are scarce

Answer: C

29. The fundamental reasons people must choose which goods to buy and consume is because of
    A. Scarcity.
    B. Specialization.
    C. People engaging in exchange.
    D. The fact there is many different economic agents.

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***Explanation*:** If resources were not scarce, everyone could have everything he or she wanted, so there would be no need to choose between alternative goods.

Answer: A

30. Land, labor, capital and entrepreneurial talent are often referred to as:
    A. Production possibilities.
    B. Goods and services.
    C. scarce economic resources
    D. unlimited human wants.

***Explanation*:**

- Economic resources are the goods or services available to individuals and businesses used to produce valuable consumer products.
- The classic economic resources include land, labor, capital and Entrepreneurship.
- Economic resources are scarce resources.

Answer: C

31. Economic choice and competitive behavior are the result of
    A. basic human greed.
    B. poverty.
    C. private ownership of resources
    D. scarcity

***Explanation*:** Scarcity of resources in relation to the wants of the people gives rise to the problem of choice.

Answer: D

32. A free good is not scarce because:
    A. Individuals can have all they desire at zero prices.
    B. Price rations the good so that all individuals willing to pay the market price can buy the good and it is, therefore, not scarce.
    C. Human desires for the good exceed the amounts available at a zero money price.
    D. it is an abundant natural resource

***Explanation*:** A free good is one where the amount available is greater than the amount people want at zero prices.

Answer: A

33. Factors of production (resources) include
    A. land, capital, and interest
    B. land, interest, and rent
    C. labor, land, and capital
    D. labor, capital, and profits

***Explanation*:**

- Factors of production are: land, labor, capital.
- All three of these factors of production are required in combination at a time to produce a commodity

Answer: C

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34. The difference between a scarce (economic) good and a free good is that
    A. for free goods at a price of zero enough of the good is available to completely satisfy consumers' demand for the good
    B. free goods no longer exist -- there are only scarce goods
    C. free goods are made with natural resources
    D. free goods are provided by the government

***Explanation*:**
- Economic resources are the goods or services available to individuals and businesses used to produce valuable consumer products. It a resource with an available quantity less than its desired use
- A free resource is one that is plentiful enough to satisfy all desires uses, often with some left over

Answer: A

35. For an economist, a good is scarce when:
    A. the amount of the good available is less than the amount that people want when the good's price equals zero
    B. the amount of the good available is greater than the amount that people want when the good's price equals zero
    C. resources must be used to produce the good
    D. the economic questions of what, how, and for whom to produce, cannot be answered by the economy involved

***Explanation*:**
- Scarcity is the widespread condition of limited resources and unlimited wants and needs.
- An economic resource is more specifically a resource with limited availability relative to desired use.

Answer: A

36. We have to make choices because:
    A. we have unlimited income.
    B. Resources are scarce.
    C. Choices involve a trade-off
    D. both B and C.

***Explanation*:** We seek to overcome the problem of scarcity by exercising our choice.

Answer: B

37. In economics, capital is best defined as:
    A. produced goods which are used as productive resources
    B. private property
    C. money needed to run a business
    D. the primary factor in productivity

***Explanation*:** Capital refers to man –made resources which help in the production of further goods and services.

Answer: A

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38. List economic Resources and their factor payments:

***Explanation*:**

Economic resources are:

- **Land** - includes space (i.e., location), natural resources, and what is commonly thought of as land. land is paid rent
- **Capital** - are the physical assets used in production - i.e., plant and equipment. Capital is paid interest.
- **Labor** - is the skills, abilities, knowledge (called human capital) and the effort exerted by people in production. labor is paid wages
- **Entrepreneurial talent** - (risk taker) the economic agent who creates the enterprise. entrepreneurial talent is paid profits

Opportunity cost:

Every choice involving the allocation of scarce resources involves opportunity costs.

Opportunity cost is the value of what is foregone in order to have something else. The opportunity cost of an action is the loss of the next best alternatives.

1.5 Production possibility curve:

In economics, a production–possibility frontier (PPF) is a graph that shows the various combinations of amounts of two commodities that could be produced using the same fixed total amount of each of the factors of production

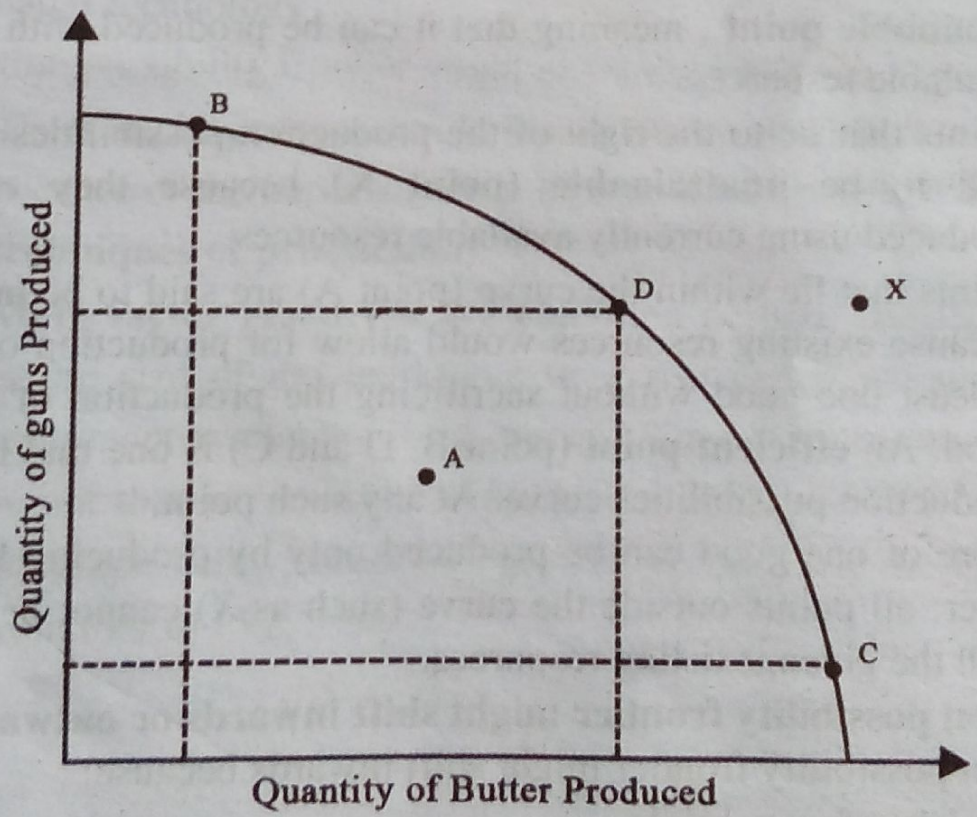

Figure 1.1: production possibility curve

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A PPF typically takes the form of the curve on the right. An economy that is operating on the PPF is said to be efficient, meaning that it would be impossible to produce more of one good without decreasing production of the other good. In contrast, if the economy is operating below the curve(point A), it is said to be operating inefficiently because it could reallocate resources in order to produce more of both goods.

For example, assuming that the economy's available quantity of factors of production does not change over time and that technological progress does not occur, then if the economy is operating on the PPF production of guns would need to be sacrificed in order to produce more butter. If production is efficient, the economy can choose between combinations (i.e., points) on the PPF:

Point *B* if guns are of interest, point *C* if more butter is needed, point *D* if an equal mix of butter and guns is required.

- In the PPF, all points *on* the curve are points of maximum productive efficiency (i.e., no more output of any good can be achieved from the given inputs without sacrificing output of some good)
- All points inside the frontier (such as *A*) can be produced but are productively *inefficient.*
- Any point that lies either on the production possibilities curve (point B, D and C) or to the left of it(point A) is said to be an **attainable point** , meaning that it can be produced with currently available resources.
- Points that lie to the right of the production possibilities curve are said to be **unattainable (point X)** because they cannot be produced using currently available resources.
- Points that lie within the curve (point A) are said to be **inefficient**, because existing resources would allow for production of more of at least one good without sacrificing the production of any other good. An **efficient point** (point B, D and C) is one that lies on the production possibilities curve. At any such point,

  More of one good can be produced only by producing less of the other; all points outside the curve (such as *X*) cannot be produced with the given, existing resources.

A production possibility frontier might shift inwards or outwards:

A production possibility frontier might shift inwards because:

- the labor force shrinks,
- the supply of raw materials is depletes
- a natural disaster decreases the stock of physical capital.

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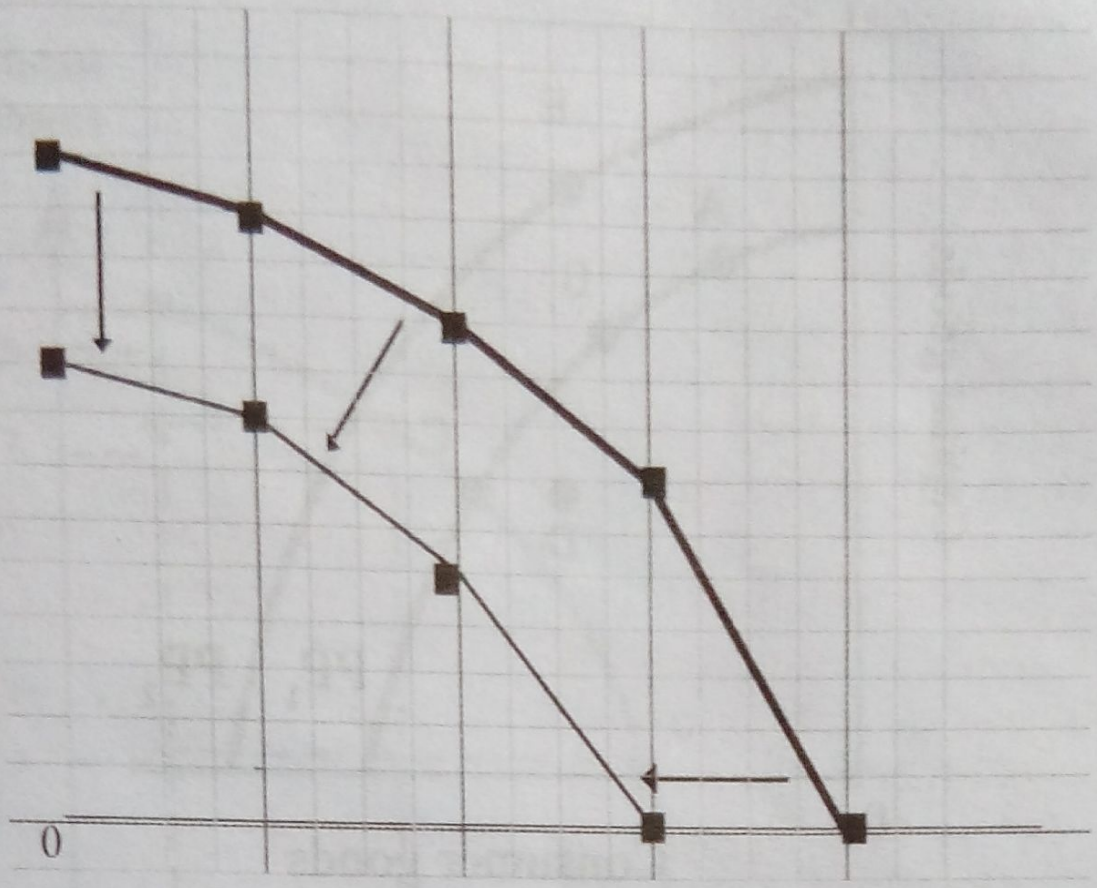

Figure 1.2: production possibility curve move inwards.

We can increase our ability to produce goods and services (or increase our POTENTIAL GDP) if we get:

- more resources
- better resources, and
- better technology

A production possibility frontier might move outwards if:

- There are **improvements in productivity and efficiency** perhaps because of the introduction of **new technology** or **advances in the techniques of production**)
- **More factor resources are exploited** perhaps due to an increase in the size of the workforce or a rise in the amount of capital equipment available for businesses. Since these increase maximum output that we are able to produce it shifts the PPC outward. On the graph below, economic growth would cause the PPC to move from PP1 to PP2.

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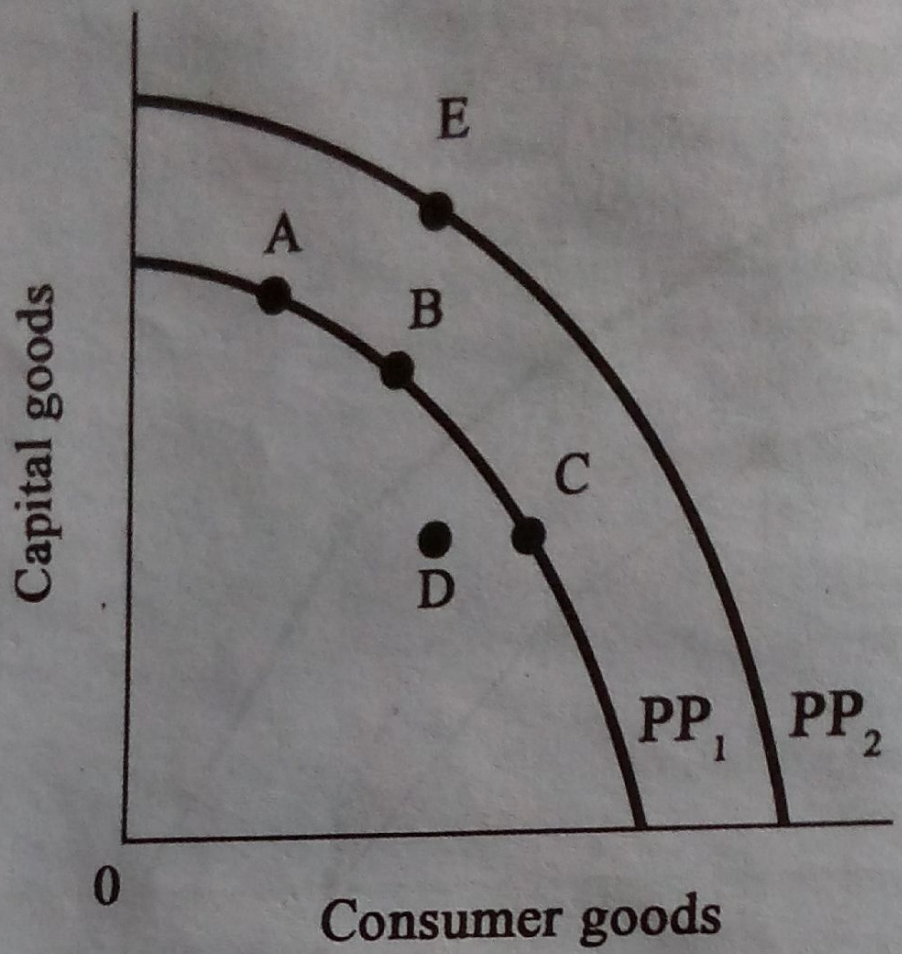

Figure 1.3: production possibility curve moving out.

1.6 Opportunity cost and production possibility frontier:

An opportunity cost will usually arise whenever an economic agent chooses between alternative ways of allocating scarce resources. The opportunity cost of such a decision is the value of the next best alternative use of scarce resources.

1.6.1 Law of Increasing Costs:

Law of Increasing Costs tells us that the more of a good that is produced, the greater its opportunity cost. This reality gives us a production possibility curve that is concave, or *bowed outward*

Opportunity cost can be thought of in terms of how decisions to increase the production of an extra, marginal, unit of one good leads to a decrease in the production of another good.

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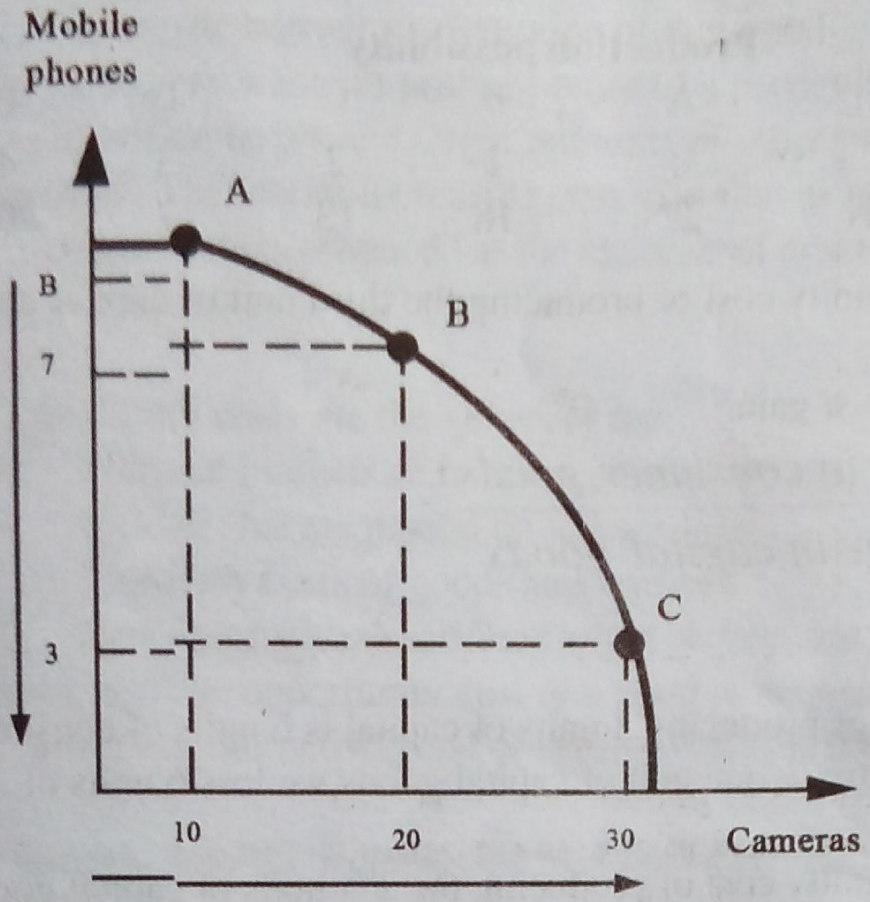

Figure 1.4: increasing opportunity costs.

According to economic theory, successive increases in the production of one good will lead to an increasing sacrifice in terms of a reduction in the other good. For example, as an economy tries to increase the production of good X, such as cameras, it must sacrifice more of the other good, Y, such as mobile phones.

This explains why the PPF is concave to the origin, meaning it is *bowed outwards*. For example, if an economy initially produces at A, with 8 phones and 10 cameras (to 20), and then increases output of cameras by 10, it must sacrifice 1 phones, and it moves to point B.

If it now wishes to increase output of cameras by a further 10 (to 30) it must sacrifice 2 phones, rather than 1, and it moves to point C; hence, opportunity cost increases the more a good is produced.

The gradient of the PPF gets steeper as more cameras are produced, indicating a greater sacrifice in terms of mobile phones foregone.

$$\text{Opportunity cost of a good} = \frac{\text{the amout of the next best alternative given up}}{\text{The amout of a good gained}}$$

Example: Use the table below to complete the following questions. The production possibilities curve below shows the hypothetical relationship between the production of capital goods and consumer goods in an economy.

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Production possibility

Products	A	B	C	D	E
Capital goods	0	1	2	3	4
Consumer goods	22	18	13	7	0

What is the opportunity cost of producing the third unit of capital goods?

***Solution*:**

Opportunity cost = loss/ gain

$$= \frac{change\ in\ consumer\ goods}{change\ in\ capital\ goods}$$

= 7-13/3-2

= 6

The opportunity cost of producing 3 units of capital is 6 units of consumer goods. In order to increase one unit of capital goods we loss 6 units of consumer goods.

1. What is the opportunity cost of producing the 2^{nd} units of capital goods?

***Solution*:**

Opportunity cost = change in consumer goods/ change in capital goods

= 13- 18/ 2-1

= 5

The opportunity cost of producing 2 units of capital is 5 units of consumer goods.

PPF and economic growth:

Economic growth is the increase in incomes and production per person. It results from

- The progress in technology,
- The accumulation of ever larger quantities of productive equipment and ever rising standards of education.
- Increase in natural and human resources

Illustrative Example

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39. The law of increasing opportunity costs states that:
    A. if the prices of all the resources devoted to the production of goods increase, the cost of producing any particular good will increase at the same rate
    B. if the sum of the costs of producing a particular good rises by a specified percent, the price of that good must rise by a greater relative amount

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C. the sum of the costs of producing a particular good cannot rise above the current market price of that good

D. if society wants to produce more of a particular good, it must sacrifice larger and larger amounts of other goods to do so

***Explanation*:** The law of increasing cost says that as more of one commodity is produced at the expense of others, it opportunity cost increases.

Answer: D

40. Opportunity costs are the values of the:
    A. Minimal budgets of families on welfare
    B. Hidden charges passed on to consumers
    C. Monetary costs of goods and services
    D. Best alternatives sacrificed when choices are made

***Explanation*:** The opportunity cost of a good or service is the value of the next best alternative good or service that is given up for it.

Answer: D

41. When one decision is made, the next best alter- native not selected is called
    A. Economic resource.
    B. Opportunity cost.
    C. scarcity.
    D. comparative disadvantage.

***Explanation*:** The opportunity cost of a particular action is the loss of the next best alternative actions.

Answer: B

42. Free goods have an opportunity of:
    A. Positive
    B. Negative
    C. Zero
    D. Cannot be determined

***Explanation*:**

- Free goods have zero opportunity cost because their total supply exceeds total demand. Therefore, no sacrifice has to be made to obtain them.

Answer: C

43. Opportunity cost is best defined as:
    A. marginal cost minus marginal benefit.
    B. the time spent on an economic activity.
    C. the value of the best foregone alternative.
    D. the money cost of an economic decision

***Explanation*:** Opportunity cost is the loss of the next best alternatives. It is the value of the best forgone alternative.

Answer: C

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44. When an Ethiopian government chooses to build more roads, the required resources are no longer available for spending on public education. This dilemma illustrates the concept of:
    A. marginal analysis.
    B. full employment.
    C. full production.
    D. opportunity cost

***Explanation*:** The government chooses to use the available resources to build more roads, to do so the government give up or forgo the use of resources on public education. This is the concept of opportunity cost.

Answer: D

45. The opportunity cost of a new national park is the:
    A. alternative uses for the land and funding for the park.
    B. cost of constructing park buildings and highways to get to it.
    C. cost of hiring staff and park rangers to provide services for visitors.
    D. increased pollution to the wildlife habitat at the park

***Explanation*:** The opportunity cost of a new national park is the alternative use of the land for other purposes such as building office, opening a restaurant.

Answer: A

46. Which of the following is "not" illustrated by a production possibility boundary?
    A. scarcity
    B. opportunity cost
    C. necessity for choice
    D. all of the above are illustrated

***Explanation*:** The PPF describes three important concepts:

- the concept of scarcity
- the concept of choice
- the concept of opportunity cost

Answer: D

47. The production possibilities frontier is bowed-out from the origin, because:
    A. Resources are not equally suited for the production of both goods.
    B. Resources are scarce.
    C. Economic growth leads to inefficiency.
    D. Resources are inefficiently used.

***Explanation*:**

- Typically production possibilities frontiers are bowed outward (i.e. concave) because resources are not equally well suited to the production of all goods and services

Answer: A

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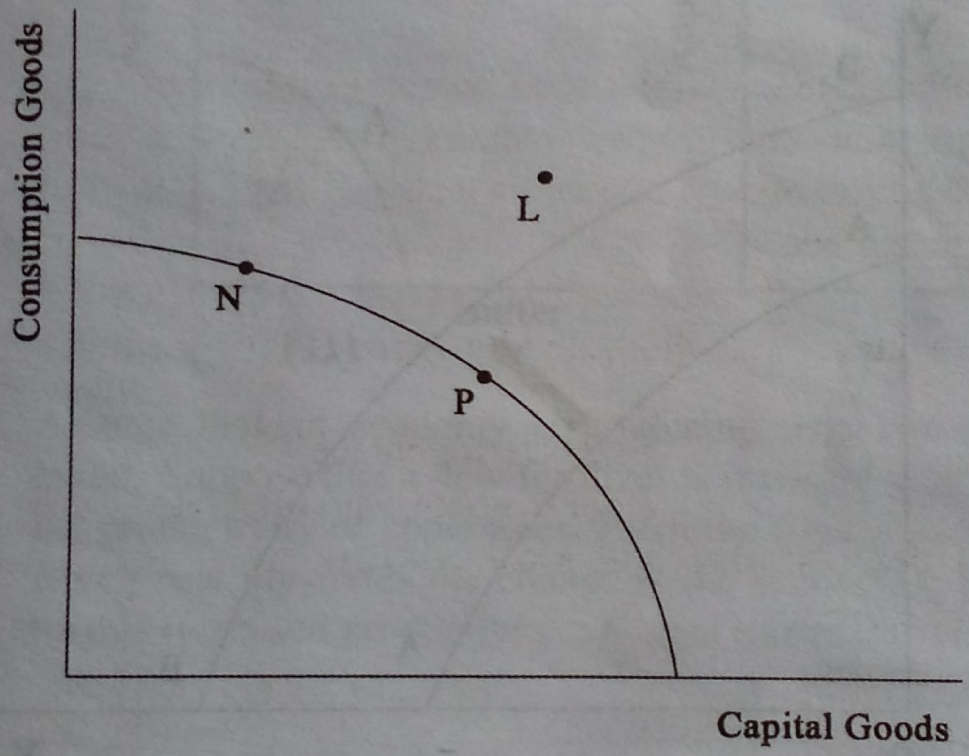
48. The production possibilities frontier will shift outward for all of the following reasons except:
    A. an increase in the unemployment rate.
    B. an increase in the labor force.
    C. an improvement in technology.
    D. an increase in worker productivity.

Explanation: The PPF will shift outward when:

- There is an increase in the amount of capital, natural and human resources
- Progress in technology.

Answer: A

Use the graph to answer question 49:



49. Other things being equal, society's current choice of point P on the curve will:
    A. allow it to achieve more rapid economic growth than would the choice of point N
    B. entail a slower rate of economic growth than would the choice of point N
    C. entail the same rate of growth as would the choice of point N
    D. be unobtainable because it exceeds the productive capacity of the economy

Explanation:

- At point N the economy is producing more of consumer goods and less of capital goods.

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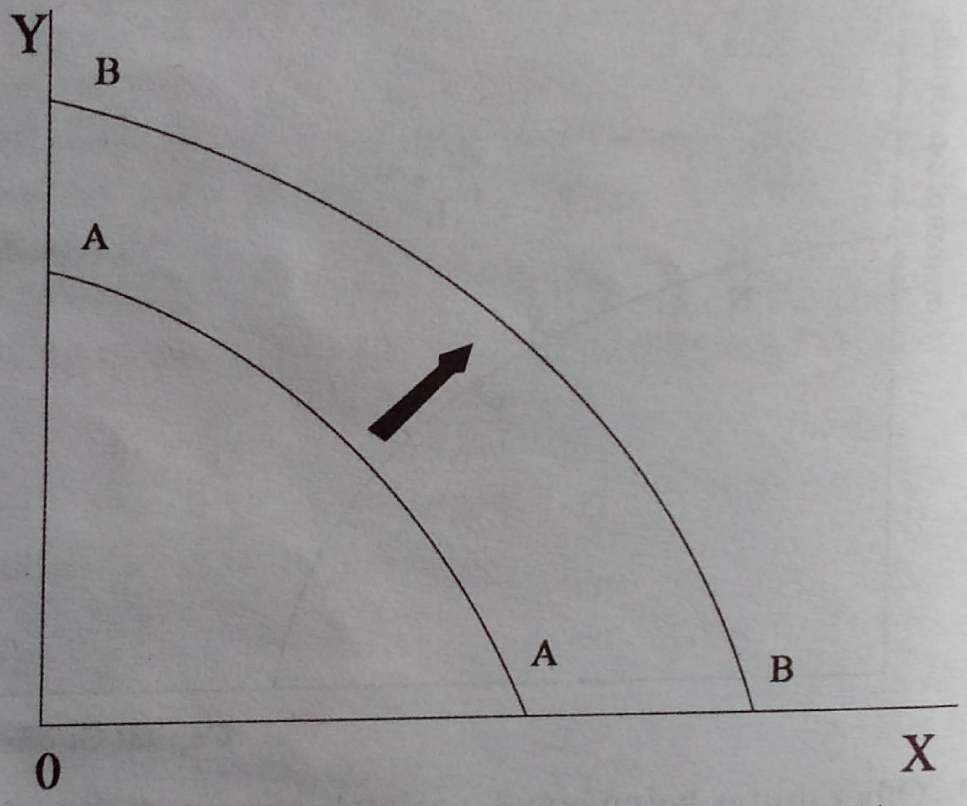
- At point P the economy produces equal combination of both consumer goods and capital goods.
  At point P there is more economic growth than point N.

Answer: A

50. Operating inside a society's production possibilities frontier is a:
    A. drawback of capitalism relative to socialism
    B. symptom of inefficiency or idle resources
    C. way to build reserves to stimulate investment and growth
    D. result whenever the capital stock depreciates rapidly

***Explanation*:** The economy works below the production possibility curve when some of labor, capital and other resources are unemployed or lying idle.

Answer: B



51. According to the graph above, a shift in the production-possibilities frontier from A-A to B-B could result from:
    A. improved technology in the production of both goods
    B. changes in the combination of goods produced
    C. unemployment
    D. inflation

***Explanation*:** PPF shift out ward when there is

- an increase in quality and quantity of resources,
- improved technology
- capital accumulation

Answer: A

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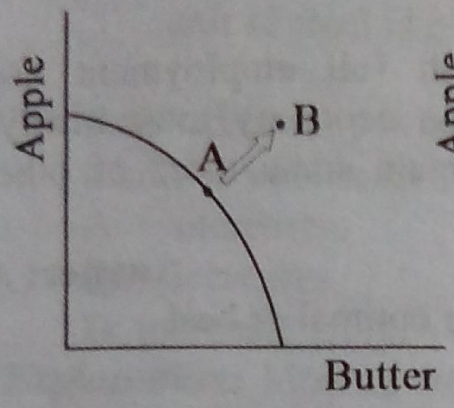
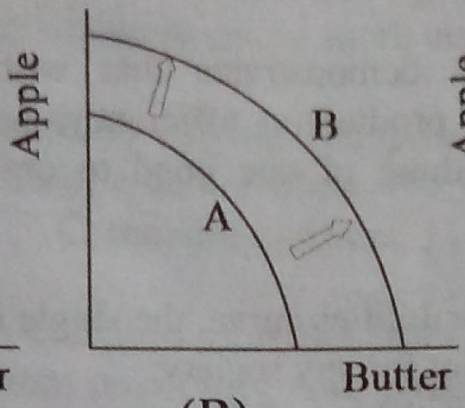
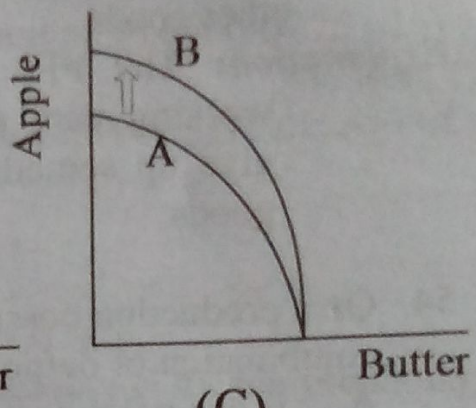
Use the following graph to answer question 52:

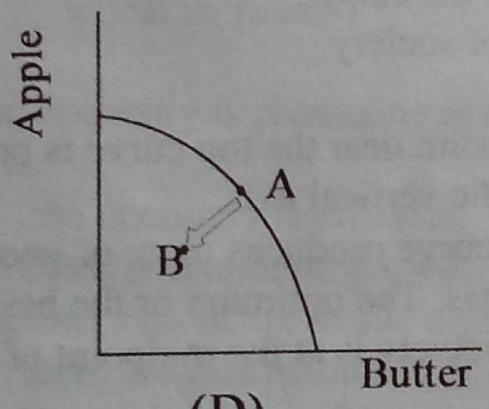




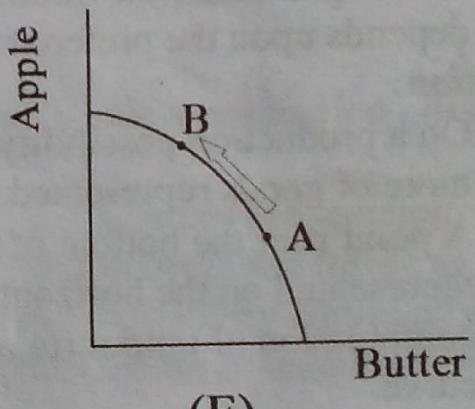
(B)



(C)



(D)



(E)

52. Assume that an economy is producing only two goods: apples and butter. Suppose that a new fertilizer is invented which greatly increases the productivity of apple trees. From the figures above, choose the one which best illustrates the change in the production possibilities caused by this increased productivity, all other things unchanged.

A. (A)
B. (B)
C. (C)
D. (D)

***Explanation*:**

- In choice –B improvement in technology is applied in both industries, so the PPF shift outward
- In choice – C there is an effect of technological advancement in apple industry only. The production of apple increases.

Answer: C

53. The production possibilities curve demonstrates the basic principle that
A. given full employment of all of a nation's resources, producing more of one good necessarily entails producing less of another
B. in a mixed capitalistic system with free markets, the economy will automatically employ all of its resources
C. exchange is a necessary corollary of specialization

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D. given full employment, to produce more of one good a nation will eventually, but not immediately, be forced to forgo production of other goods

***Explanation*:** The PPF demonstrates that with full employment and working with production efficiency, an economy must always give up something of one good to obtain some more of other goods.

Answer: A

54. On a production possibilities curve, the single optimal or best combination of output for any society:
    A. is at a point near the top of the curve.
    B. is at the precise midpoint of the curve.
    C. is at a point near the bottom of the curve.
    D. depends upon the preferences of society

***Explanation*:**

- On a production possibility, a point near the top curve is producing more of goods represented on the vertical axis.
- A point near the bottom of the curve produces more of goods represented on the horizontal axis. The optimum or the best combination of output for any society is at the midpoint of the curve.

Answer: B

Use the table to complete the following question(s). The following economy produces two products.

Production possibilities

Products	A	B	C	D	E	F
Steel	0	1	2	3	4	5
Wheat	100	90	75	55	30	0

55. Refer to the above table. The total opportunity cost of the third units of steel is:
    A. 15 units of wheat.
    B. 20 units of wheat.
    C. 45 units of wheat.
    D. 55 units of wheat.

***Solution*:** Opportunity cost = loss/ gain
= change in wheat /change in steel
= 55 – 75/3-2 = 20 units of wheat

Answer: B

56. Refer to the above table. A change from possibility C to B means that:
    A. 1 unit of steel is given up to get 75 units of wheat.
    B. 2 units of steel are given up to get 75 units of wheat.
    C. 1 unit of steel is given up to get 15 more units of wheat.
    D. 2 units of steel are given up to get 15 more units of wheat.

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***Explanation*:** A change from possibility C to B means decreasing the production of steel and increasing the production of wheat. One unit of steel is given up to get 15 more unit of wheat.(90- 75 =15)

Answer: C

57. Refer to the above table. In moving from possibility A to F, the cost of a unit of steel in terms of a unit of wheat:
    A. increases.
    B. decreases.
    C. remains constant.
    D. increases from A to B, and decreases from B to F.

***Explanation*:** Moving from possibility A to possibility F means producing more and more of steel meaning less and less of wheat. The cost of producing more of steel is increases (the forgone value of wheat increases)

Answer: A

58. If an economy is producing at a point inside a production possibilities curve:
    A. the economy is efficient.
    B. there is economic growth.
    C. resources are unemployed.
    D. resources are fully employed

***Explanation*:** Producing inside a production possibility curve indicates that there are idle resources or underutilization of resources. It indicates Unemployment of resources.

Answer: C

59. The production possibilities curve is a simple model that can be used to show choices: what are the assumptions of PPF

***Explanation*:** The assumptions necessary to represent production possibilities in a simple production possibilities curve model are:

- Efficiency
- fixed resources
- fixed technology

1.7 Basic economic problems of the society:

Any individual, organization, or nation has to make three fundamental types of choices about how to allocate the scarce resources available to it. It has to decide:

What to produce, the problem 'what to produce' the problem of choice between commodities. What commodities are being produced and in what quantities? This question directly springs from the scarcity of resources. It is largely concerned with the allocation of scarce resources among competing uses. Should society devote its limited resources to produce civilian or military goods, luxuries or necessities, goods for immediate consumption or goods that increase the wealth of society (capital goods?)

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How to produce, the problem 'how to produce' is the problem of choice of technique. By what methods are these commodities produced? This question arises because there are various technically possible ways in which goods and services can be produced. It is always technically possible to produce a give quantity of goods with more of labor and less of capital (with labor intensive technology) and with more of capital and less of labor (with capital- intensive technology).

For whom it will produce, this question arises how is the national output shared among the households or what determines the share of each household? How society's output of goods and services divided among its members. Why do some individuals and groups earn large incomes while others earn only small incomes? This question is concerned with the division of the total national product among individuals and groups.

Illustrative Example

60. An example of the "for whom" part of the economic problem is
    A. Should pharmacists be paid more than economists?
    B. Who should decide what goods are produced?
    C. How does the nation decide who should be a doctor and who should be a writer?
    D. Should more food and fewer diamonds be produced?

Explanation:
The "for whom" question asks for whom the goods should be produced. If pharmacists are paid more than economists, pharmacists will be able to buy more goods and services so that more of society's products will be produced for them.

Answer: A

61. Three basic decisions must be made by all economies. What are they?
    A. how much will be produced; when it will be produced; how much it will cost
    B. what the price of each good will be; who will produce each good; who will consume each good
    C. what will be produced; how goods will be produced; for whom goods will be produced
    D. how the opportunity cost principle will be applied; if and how the law of comparative advantage will be utilized; whether the production possibilities constraint will apply

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***Explanation*:** The basic microeconomics problems are: what to produce and how much to produce, how to produce and for whom to produce.

Answer: C

62. Which of the following is the primary incentive in determining WHAT to produce in a free market price system? Produce those products that
    A. enjoy maximum freedom from government controls
    B. are needed by the masses of people
    C. are easiest to produce
    D. will provide maximum profits for the producer

***Explanation*:** The problem of what to produce is the problem of choice between commodities that will provide maximum profit for the producer.

Answer: D

1.8 Economic systems:

An **economic system** is the system of producing and distributing of goods and services and allocating resources in a society. It includes the combination of the various institutions, agencies, entities and consumers that comprise the economic structure of a given community.

Every economic system represents an attempt to solve three fundamental and interdependent problems:

- What goods and services shall be produced and in what quantities?
- How shall goods and services be produced? That is, by whom and with what resources and technologies?
- For whom shall goods and services be produced? That is, who is to enjoy the benefits of the goods and services and how is the total product to be distributed among individuals and groups in the society?

1.8.1 Types of economic system:

a) Pure capitalism
b) Command economy
c) Mixed economy

Capitalism (free market) - refers to the private ownership of the means of production. The market system is based on the demand and supply of products. Demand and supply determine prices and prices act as signals to both producers (suppliers) and consumers (who create demand).

The heart of the market system is the profit motive. Entrepreneurs (those who organize resources into production) are stimulated to take risks and produce goods and services by the prospect of seeing some return on their efforts. The extent to which they achieve this success is dependent on identifying the level of demand for the good or service, as well as their ability to manage production efficiently. If the costs of providing the good

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or service is less than the revenue that they receive from selling those goods and services, they will make a profit. If they can maximize revenue whilst reducing costs to a minimum, the level of profit they make will be higher.

In a capitalist economic system (capitalism):

- production is carried out to maximize private profit, decisions regarding investment and the use of the means of production are determined by competing business owners in the marketplace; production takes place within the process of capital accumulation.
- The means of production are owned primarily by private enterprises and decisions regarding production and investment determined by private owners in capital markets.
- Capitalist systems range from laissez-faire, with minimal government regulation and state enterprise, to regulated and social market systems, with the stated aim of ensuring social justice and a more equitable distribution of wealth

Collectivism, command and planned economic system:

In a planned system, the three basic questions are answered through some form of central planning.

In socialist economic system (socialism)**:**

- Production is carried out to directly satisfy economic demand by producing goods and services for use
- Decisions regarding the use of the means of production are adjusted to satisfy economic demand; investment (control over the surplus value) is carried out through a mechanism of inclusive collective decision-making.
- The means of production are either publicly owned, or are owned by the workers cooperatively.
- A socialist economic system that is based on the process of capital accumulation, but seeks to control or direct that process through state ownership or cooperative control to ensure stability, equality or expand decision-making power, are market socialist systems.

Mixed economy:

- Mixed economy is usually applied to economies where there is a significant component of both command and free market enterprises.
- Economic decisions are taken by the market mechanism and command system.
- All real economies have a mixture of such state ownership and private ownership.

The vast majority of countries around the world have some form of mixed economy where some resources are allocated through the price or market

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mechanism and others are allocated by the state. In theory, such a system is able to combine the best elements of both a planned economy and a market economy. In reality, the proportion of planned and market varies, with some countries placing more emphasis on market solutions to resource allocations and others favoring a greater role for state planning.

Whatever the proportions, there is never a 'best' combination; it will depend to a large extent on the aims and objectives of the government concerned. Even with the existence of both a planned and a market element to an economy, there can still be problems and there will be aspects of market failure and government failure. The latter is where the government attempts to intervene in the market, to try to solve problems that might be caused by the market mechanism. In so doing, this creates other problems that do not ensure an efficient allocation of resources.

Today the dominant form of economic organization at the global level is based on capitalist market-oriented mixed economies.

Illustrative Example

63. Economic planning by central government agencies is primarily associated with
    A. command economies.
    B. market economies.
    C. laissez faire economies.
    D. mixed economies.

Explanation: Government planning is the hallmark of a command economy; in other economies, individuals and firms play the central role in decision making

Answer: A

64. In market economies, the incentive that draws entrepreneurs into industry is
    A. government bonuses for meeting production quotas.
    B. profit.
    C. government assumption of the risk of failure.
    D. government assistance with making output and pricing decisions.

Explanation: The profit motive draws new firms into industry in market economies, because private property rights allow firms to keep those profits.

Answer: B

65. In a mixed economy, the economic decisions of what to produce, how to produce, and who will receive products are made by:
    A. the government.
    B. consumers and firms.
    C. banks and stock markets.
    D. firms, consumers, and government.

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***Explanation*:** Mixed economies provide for some government intervention, such as regulation and provision of public goods, but primarily rely on the decision making of firms and consumers in markets

Answer: D

66. Which of the following might be considered to be a characteristic feature of a planned economy
    A. all income completely evenly distributed
    B. there is no incentive for people to work hard
    C. goods and services produced reflect consumer sovereignty
    D. price is relatively unimportant as a means of allocating resources

***Explanation*:** In command economic system there is :

- relative equality of income
- absence of incentive for hard work and efficiency
- maximum social welfare
- complete control over all economic activities

Answer: B

67. Which one is **not** the feature of capitalist economic system?
    A. high rate of capital formation
    B. the right to private property
    C. clear social and economic objectives
    D. profit motive

***Explanation*:**

- Clear social and economic objectives are features of command economic system.
- The right to private property, high rate of capital formation and profit motive are the feature of capitalist economic system.

Answer: C

68. Which of the following is characteristic of a centrally planned economic system?
    A. Resources are allocated based on relative prices.
    B. Private ownership of resources is fundamental to economic growth.
    C. Government planners decide how best to produce goods and services.
    D. Efficiency is superior to the market economic system.

***Explanation*:** A centrally planned economy decides which goods are needed and how best to provide them to the population. Resources are allocated and goods are distributed by the government, not the price system.

Answer: C

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69. Which of the following is the main feature of capitalism?
    A. Collective owner ship
    B. Clear social and economic objectives
    C. Maximum social welfare
    D. Profit motive

***Explanation*:** The main features of capitalism are:

- The right to private property
- Freedom of enterprises
- Freedom of choice by consumers
- Profit motive
- Competition
- Price mechanism
- Minor role of government

Collective ownership, clear social and economic objectives and maximum social welfare are the features of command economy.

Answer: D

70. A n economy in which a central government authority dominates in decisions of what to produce, how to produce, and for whom to produce using national resources is:
    A. Market economy
    B. Mixed economy
    C. Capitalist economy
    D. command economy

***Explanation*:**

- In a command economic system, the governmental appropriation of property other than land, transferring it from the domain of private property to national control.
- In a command economic system, the central government authority decides what to produce, how to produce and for whom to produce using a national resource.

Answer: D

71. Which of the following statements are true of a free market economy?
    A. Economic resources are publicly owned.
    B. Freedom of enterprise is critical.
    C. The price system allocates resources in the most efficient way.
    D. Resources are allocated by the government.

***Explanation*:** Free- market enterprise economies consider state ownership of productive facilities and government interference in price setting as deplorable exceptions to the rule of private ownership and price determination through the mediation of competitive markets.

Answer: B

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72. In socialist economic system resource allocation is managed is primarily by

A. Large corporation C. Markets
B. Workers committees D. Centrally planning

Explanation: Socialist system utilizes a central planning process to determine the most efficient use of resources and manage the resources accordingly.

Answer: D

1.9 Decision making units:

Household sector:

- They are sellers in the market for resources.
- Sell land, labor, capital, and entrepreneurial activity in exchange for money, which in this case is called income.
- They are buyers in the market for goods and services
- Exchange income for goods and services.

Businesses:

- They are sellers in the market for goods and services.
- Sell goods and services in exchange for money, which in this case is called revenue.
- They are buyers in the markets for resources.
- Exchange the revenue earned in the market for goods and services to buy land, labor and capital in the market for resources.

Government:

Government Just like households and firms the government also earns incomes and makes expenses. Two major functions played are:

a. Earns revenue either from tax or non-tax sources both from households and firms.
b. Provides essential public services such as maintenance of law and order, defense services, judiciary etc.

Foreign sector:

It consists of two kinds of international inflow ¬ export and import of goods and services ¬ economic transactions i.e. inflow and outflow of capital

1.10 The circular flow of economic activities

Economic agents interact in two markets:

A. **Resource markets**: *Input markets* are the markets in which resources—labor, capital, and land—used to produce products, are exchanged.
B. **Product markets:** Output, or product markets are the markets in which goods and services are exchanged.

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Circular flows are classified as: Real Flow and Money Flow.

Real Flow: the flow of factor services from households to firms and corresponding flow of goods and services from firms to households is known to be as real flow.

Money Flow- money acts as a medium of exchange between goods and factor services. Money flow of income refers to a monetary payment from firms to households for their factor services and in return monetary payments from households to firms against their goods and services.

Household sector gets monetary reward for their services in the form of rent, wages, interest, and profit form firm sector and spends it for obtaining various types of goods to satisfy their wants.

The two sector circular flow model

In a two sector circular flow model there are firms and households.

Business firms: produce goods and services

Households: supply the economic resources to the firms and receive payment in terms of money

Three Sectors circular flow model

In this model there are:

- **Household Sector**: This sector is responsible for consumption expenditures. It also owns all productive resources.
- **Business Sector**: This includes the undertake the task of combining resources to produce goods and services.
- **Government sector**: This includes the ruling bodies of the federal, state, and local governments. Regulation is the prime function of the government sector, especially passing laws, collecting taxes, and forcing the other sectors to do what they would not do voluntary.

Four Sectors circular flow model.

The circular flow model illustrates the interaction among the four sectors--household, business, government, and foreign.

- **Household sector**: This includes everyone, all people, seeking to satisfy unlimited wants and needs.
- **Business sector:** This includes the institutions that undertake the task of combining resources to produce goods and services. This sector does the production.
- **Government sector**: This includes the ruling bodies of the federal, state, and local governments. Regulation is the prime function of the government sector, especially passing laws, collecting taxes, and forcing the other sectors to do what they would not do voluntarily.

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- **Foreign sector**: This includes everyone and everything (households, businesses, and governments) beyond the boundaries of the domestic economy. It buys exports produced by the domestic economy and produces imports purchased by the domestic economy, which are commonly combined into net exports (exports minus imports).

Illustrative Example

73. In the product market of the circular flow model,
    A. firms buy finished products from households.
    B. consumers buy factors of production from firms.
    C. firms sell factors of production to the government.
    D. consumers buy finished products from firms.

***Explanation*:**
- In the product market, firms sell products to consumers.
- Households or consumers buy finished products from firms.

Answer: D

74. Which of the following activities would occur in the resource (factor) market?
    A. A teacher buys a new truck from a car dealership.
    B. The department of defense buys a tank from a weapons manufacturer.
    C. A farmer buys farmland from a retiring farmer.
    D. A retirement fund buys stock in a major corporation.

Explanation In the resource (factor) market, firms buy factors of production (land, labor, capital, and entrepreneurial ability) from resource providers.

Answer: C

75. While resources and products flow in one direction of the circular flow model, what flows the other direction?
    A. money
    B. services
    C. Public goods
    D. imported goods

***Explanation*:**
- There are two types of flow. Real flow and Money flow.
- Flow of resources and products is real flow
- The flow of payments for resources and products is money flow Money, in the form of paychecks, profit, and other payments, serves as an incentive for households to provide resources and firms to produce products in a market economy.

Answer: A

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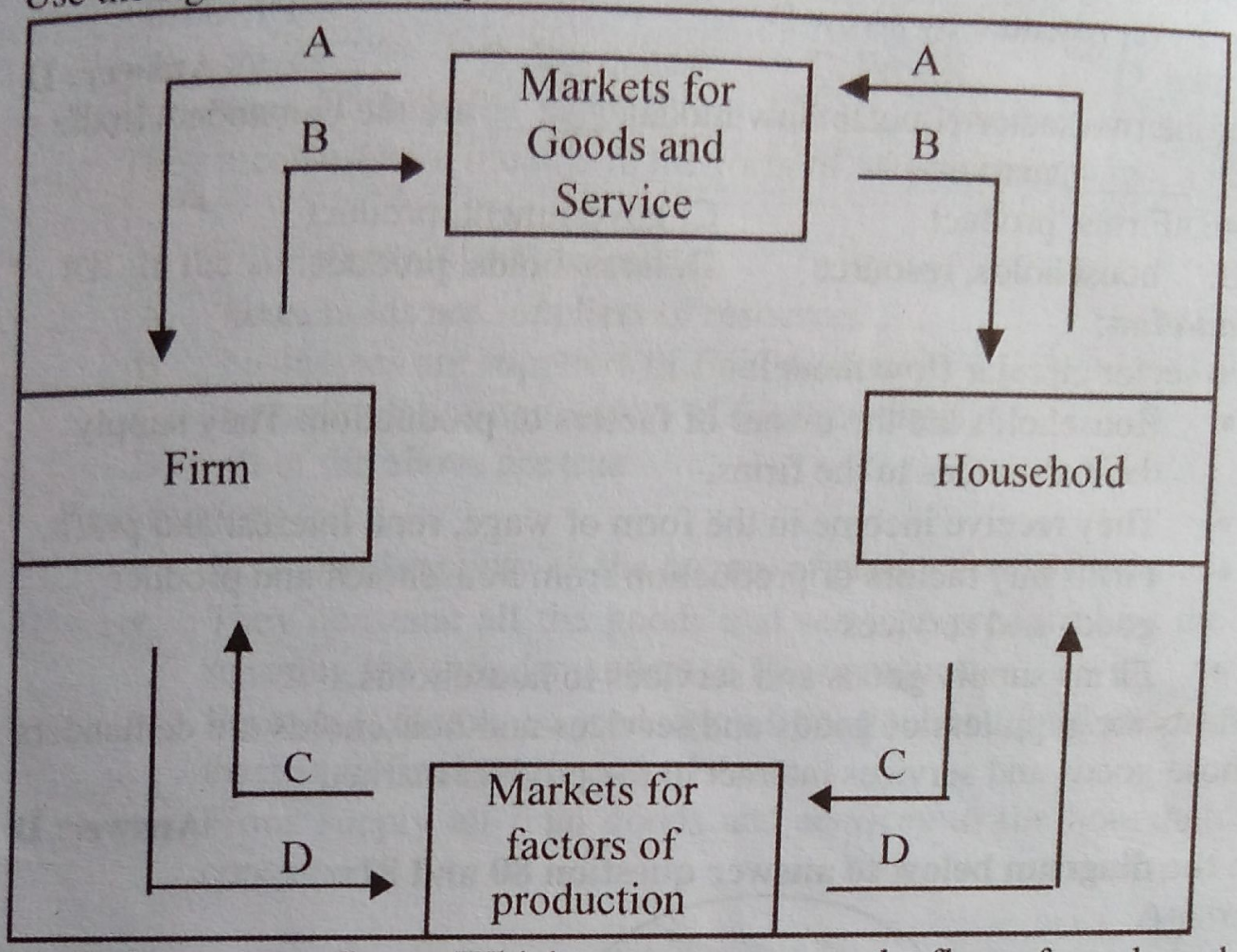
Use the figure to answer question 76-78



76. Refer to above figure. Which arrow represents the flow of goods and services?

A. A B. B C. C D. D

***Explanation*:** Goods and services produce in the firm sector and supply to the house hold for consumption purpose. The flow of goods and services from the firms to the household represent by arrow B

Answer: B

77. Refer to above figure. Which arrow represents the flow of factors of production?

A. A B. B C. C D. D

***Explanation*:** Households are owner of factors of production. Households supply their factors of production to the firm. The flow of factors of production from the household to the firm represent by arrow C

Answer: C

78. Refer to the above Figure. Hana completes her first week of employment working as a hairdresser at Sara beautify salon. On Friday of that week, she receives her first salary. To which of the arrows does this transaction directly contribute?

A. B only B. A and B C. C only D. C and D

***Explanation*:** Hana supplies her factor of resource i.e her skill, knowledge and talent to the firm (at sara beautify salon)which is represented

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by arrow C. The reward of Hana is wage (salary)which represented by arrow D.

Answer: D

79. In the two sector circular flow model,_____ are the demanders in the _____ markets.

A. Firms, product
B. households, resource
C. government, product
D. households, product

Explanation:

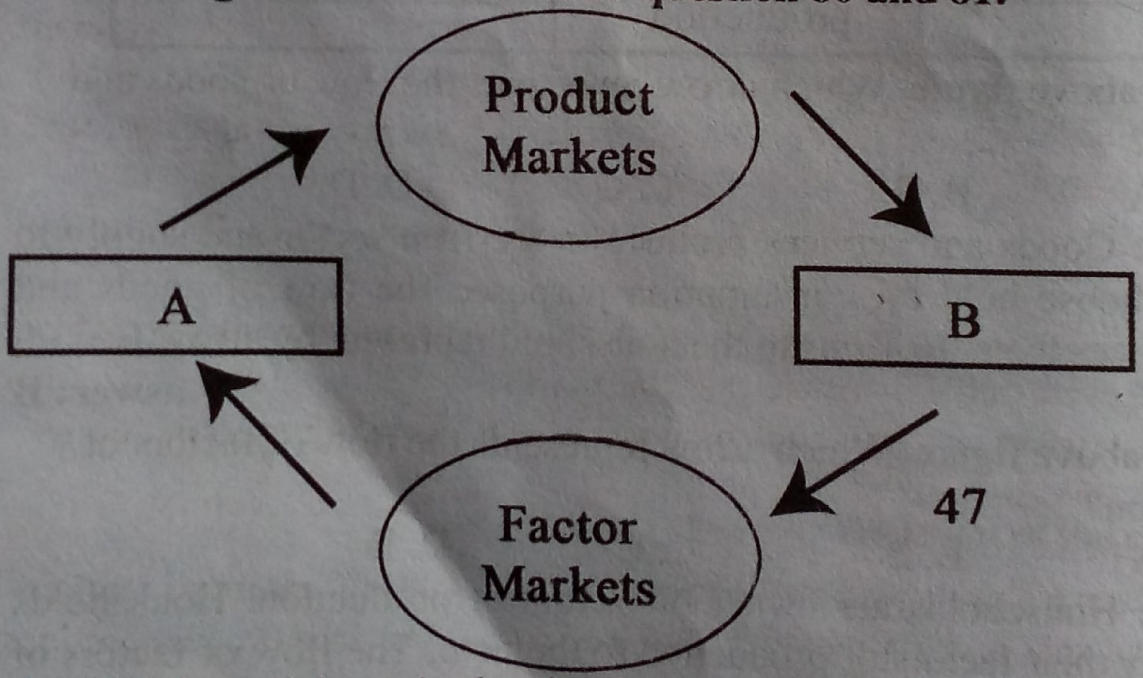
In two sector circular flow model:

- Households are the owner of factors of production. They supply their resources to the firms.
- They receive income in the form of wage, rent, interest and profit.
- Firms buy factors of production from households and produce goods and services
- Firms supply goods and services to households.

So firms are suppliers of goods and services and households are demanders of those goods and services interact in the product market.

Answer: D

Use the diagram below to answer question 80 and 81:



80. Which participant in the circular flow does' A' represent?

A. Firms
B. Government
C. International Community
D. Households

***Explanation*:**

- Factors of production such as land, labor, and capital flow from resource (factor) market to firms.
- 'A' represent firms
- 'B' represent households

Answer: A

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81. All of the following are transferred from B to the factor markets EXCEPT

A. wages B. Revenues C. Profits D. interest

Explanation: Households supply their factors of production to the firms. They receive factor income in the form of wage, rent, interest and profit.

Answer: B

82. In the simple circular flow model:
    A. households are suppliers of resources
    B. businesses are suppliers of final products
    C. households are demanders of final products
    D. all of the above are true

Explanation:

- Households supply all the factors of production to the firms
- They consume all the goods and services produced by the firms, meaning they are demanders of final products
- Firms hire factors of production from the household and transform them in to final goods and services.
- Firms supply all final goods and services to the households, the consumers.

Answer: D

83. The simple circular flow model shows that:
    A. households are on the demand side of both product and resource markets
    B. Businesses are on the supply side of both product and resource markets
    C. households are on the supply side of the resource market and on the demand side of the product market
    D. businesses are on the demand side of the product market and on the supply side of the

Explanation:

- The sale and purchase of inputs creates factor (resource) market and sale and purchase of final goods and services creates product market.
- Households supply inputs to firms. They are on the supply side of resource market.
- Households buy final goods and services from the firms and they are on the demand side of the product market.

Answer: C

84. Which of the following behavioral assumptions is true about basic decision making units in economics?
    A. firms are utility maximizers

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B. firms are welfare maximizers
C. Household is utility maximizers
D. Households are profit maximizers

***Explanation*:**

- Households are basic consuming units who demand goods and services to satisfy their wants. So households are utility maximizers.
- Firms are production units which are setup to produce goods and services for satisfying wants of the households. Firms are profit maximizers.

Answer: C

85. Which of the following is true of the complete circular flow model of an open economy?
    A. All goods and services flow through the government in exchange for resource payments.
    B. There is no role for the foreign sector.
    C. Households supply resources to producers in exchange for goods and services.
    D. Producers provide goods and services to households in exchange for the costs of production.
    E. The government collects taxes from firms and households in exchange for goods and services.

***Explanation*:**

- In the full circular flow model, the role of government is to collect taxes from firms and households in exchange for goods and services.
- Households supply resources in exchange for wages, which they then use to purchase goods and services.

Answer: E

86. In circular flow of income and expenditure, real flows consists of
    A. Money incomes from factor services such as rent
    B. The money expenditure incurred for the purchase of goods and services
    C. Goods and services from the producers to the buyers
    D. Factor services from the producers to the owners of a factor

***Explanation*:**

Real flows consist of:

- Factor services from the owners of factor services to the producers and
- Goods and services from the producers to the buyers.

Answer: C

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Supplementary Exercise on Unit 1

1. The general concern of economics is with the study of the:
   A. degree of competition in stock and bond markets in the economy.
   B. efficient use of limited productive resources to satisfy economic wants.
   C. issue of equality in the distribution of income and wealth among households.
   D. budget deficits in the domestic economy and trade deficits in the international economy
2. As a student of economics, when you speak of scarcity, you are referring to the ability of society to:
   A. employ all of its resources.
   B. consumes all that is produced.
   C. satisfies economic wants given limited resources.
   D. continually makes technological breakthroughs and increase production.
3. Economics is the study of:
   A. increasing the level of productive resources so there is maximum output in society.
   B. increasing the level of productive resources so there is a minimum level of income.
   C. how people, institutions, and society make choices under conditions of scarcity.
   D. the efficient use of scarce resources paid for at the minimum level of cost to consumers and businesses.
4. The primary focus of the study of economics is with:
   A. expanding the production of goods and services.
   B. equalizing the distribution of consumer income and wealth.
   C. making the most efficient use of scarce productive resources.
   D. reallocating resources from consumption to production in the economy.
5. The general concern of economics is with the study of the:
   A. degree of competition in stock and bond markets in the economy.
   B. efficient use of limited productive resources to satisfy economic wants.
   C. issue of equality in the distribution of income and wealth among households.
   D. budget deficits in the domestic economy and trade deficits in the international economy.
6. Which of the following is not a central focus of the "economic perspective"?

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A. Scarcity and choice.
B. The scientific method.
C. Purposeful behavior.
D. Marginal analysis.

7. The term scarcity in economics refers to the fact that:
   A. economic wants are limited and resources are abused.
   B. even in the richest country some people goes hungry.
   C. no country can produce enough products to satisfy everybody's economic wants.
   D. it is impossible to produce too much of any particular good or service in a market economy.
8. The basic economic problem is essentially one of deciding how to make the best use of:
   A. limited resources to satisfy limited economic wants.
   B. limited resources to satisfy unlimited economic wants.
   C. unlimited resources to satisfy unlimited economic wants.
   D. unlimited resources to satisfy limited economic wants.
9. The key economic concept that serves as the basis for the study of economics is:
   A. inflation.
   B. unemployment.
   C. money.
   D. scarcity
10. Which is an illustration of a microeconomic question?
   A. What is the current national rate of unemployment?
   B. Is the economy experiencing a decline in the rate of inflation?
   C. Will a new type of television set increase the number of buyers?
   D. Is the production of goods and services in the economy greater this year than last year?
11. Which question is an example of a microeconomic question?
   A. What should the Federal government do to reduce the trade deficit with Japan?
   B. Will the merger of two airlines likely lead to higher airline ticket prices?
   C. What factors are contributing to the rise of unemployment in the economy?
   D. Will the inflation rate remain relatively stable this year?
12. Microeconomics focuses on:
   A. the workings of the whole economy or large sectors of it.
   B. the individual units that make up the whole of the economy.
   C. issues such as unemployment and inflation.
   D. total output and the general level of prices.

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13. Macroeconomics focuses on:
    A. the individual units that make up the whole of the economy.
    B. studies of how individual markets and industries are organized.
    C. total output and the general level of prices in the economy.
    D. how a business determines how much of a product to produce.
14. The study of how one business firm sets its prices would fall under the study of:
    A. macroeconomics.
    B. microeconomics.
    C. income distribution.
    D. economic growth
15. The branch of economics that focuses primarily on aggregates is:
    A. macroeconomics.
    B. microeconomics.
    C. scientific economics.
    D. consumer economics
16. A point inside the production possibilities curve is:
    A. attainable and the economy are efficient.
    B. attainable, but the economy is inefficient.
    C. unattainable, but the economy is inefficient.
    D. unattainable and the economy are efficient.
17. A point on the frontier of the production possibilities curve is:
    A. attainable and the economy are efficient.
    B. attainable, but the economy is inefficient.
    C. unattainable, but the economy is inefficient.
    D. unattainable and the economy is efficient
18. Which of the following statements would you consider to be a normative one?
    A. Faster economic growth should result if an economy has a higher level of investment
    B. Changing the level of interest rates is a better way of managing the economy than taxation and government expenditure
    C. Higher levels of unemployment will lead to higher levels of inflation
    D. The average level of growth in the economy was faster in the 1990s than the 1980s
19. Which of the following goods or services would not be provided by a pure free-market economy?
    A. Public transport system
    B. pensions
    C. education
    D. National Defense
20. Which of the following are considered to be essential characteristics of a public good?
    A. It must be provided by the government
    B. Consumption by one person does not diminish the quantity available for others

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C. Once the good is provided, it is not possible to exclude people from consuming it
D. It must benefit the whole economy

21. Which of the following might be considered to be a characteristic of a planned economy?
A. All income is completely evenly distributed
B. Price is relatively unimportant as a means of allocating resources
C. Goods and services produced reflect consumer sovereignty
D. There is no incentive for people to work hard

22. Economic growth is depicted by:
A. a shift in the production possibilities frontier outward
B. a movement from inside the curve toward the curve
C. a shift in the production possibilities frontier inward
D. a movement along a production possibilities frontier toward capital goods

23. Which of the following will not shift a country's production possibilities frontier outward?
A. an advance in technology
B. an increase in labor force
C. an increase in the capital stock
D. a reduction in unemployment

24. Which of the following is part of the opportunity cost of going on holiday?
A. the money you spent on a theatre show
B. the money you could have made if you had stayed at home and worked
C. the money you spent on airline tickets
D. the money you spent on food

25. All of the following are examples of opportunity cost except:
A. The leisure time sacrificed to study for an exam.
B. The tuition fees paid to a university.
C. The income which could have been earned by a college student had he or she worked full time instead of attending college.
D. The building which could have been built with the construction materials and labor used to build a new university library.

26. Which of the following is the best definition of opportunity costs?
A. The amount of one good that must be given up in order to produce one more unit of another good.
B. The amount of money that must be paid in order to purchase one more unit of a good.
C. The amount of an input that must be used in order to produce one more unit of a good.

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D. The price of a good that must be charged in order for a merchant to sell one more unit.

27. A free good is not scarce because:
    A. Individuals can have all they desire at zero price.
    B. Price rations the good so that all individuals willing to pay the market price can buy the good and it is, therefore, not scarce.
    C. Human desires for the good exceed the amounts available at a zero money price.
    D. It is an abundant natural resource.
28. Microeconomics is not concerned with
    A. The quantities of each good produced.
    B. The size of aggregate money flows in the economy.
    C. The distribution of goods among households.
    D. The allocate function of changes in relative prices.
29. All of the following basic economic questions deal with microeconomics except:
    A. How is the supply of goods allocated among the members of the society?
    B. What goods and services are being produced and in what quantities?
    C. By what methods are goods and services being produced?
    D. Is the economy's capacity to produce goods growing over time?
30. Which is not considered to be an economic resource?
    A. land (or natural resources
    B. money
    C. labor
    D. tools and machinery
31. Name the four (4) factors of production and the income earned from each factor
32. What are the basic economic questions all economic systems must answer when dealing with the problem of scarcity
33. Who owns and controls the factors of production in a
    A. pure market economy? B. pure command economy?
    C. mixed economy?
34. What is society's basic economic problem?

Answer to Supplementary Exercise on Unit 1

1. B	2. C	3. C	4. C	5. B	6. C	7. D	8. B	9. D	10. C
11. B	12. A	13. C	14. B	15. A	16. B	17. A	18. B	19. D	20. B
21. D	22. A	23. D	24. B	25. C	26. A	27. A	28. B	29. D	30. B

31. Land – rent , Capital – interest , Labor – wage, Entrepreneur – profit
32. What to produce, how to produce, for whom to produce
33. A. Private B. Government C. private and public
34. Unlimited human wants and very limited means to satisfy these wants

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UNIT TWO

Theory of demand and supply

The supply and demand model is designed to explain how prices are determined in perfectly competitive markets. Supply and demand are like two blades of a scissors: The demand blade tells us how much of something buyers want to buy, and the supply blade tells us how much sellers want to sell.

In economic terminology the term demand conveys a wider and definite meaning than in the ordinary usage. Ordinary demand means a desire, whereas in economic sense it is something more than a mere desire. It is interpreted as a want backed by the- purchasing power.

Demand refers to the different quantities of a product that a consumer is willing and able to buy at various prices in a given period of time.

2.2 Demand and quantity demand:

Demand is the amount of a product that people are willing and able to purchase at each possible price during a given period of time.

The **quantity demand** is the amount of a product that people are willing and able to purchase at one, specific price. Demand schedule and demand curve

A ***demand schedule*** is a table showing how much of a given product a household would be willing to buy at different prices.

Demand curve: Graphically shows how much of good consumers are willing to buy (holding their incomes, preferences, and other things constant) at different prices.

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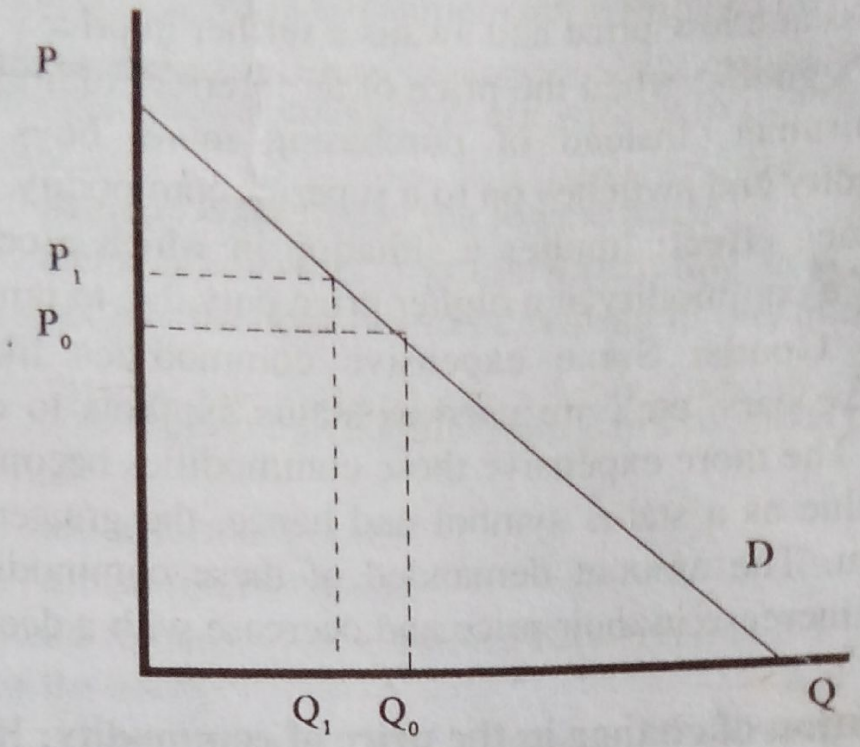

Figure 2.1: demand curve.

Here, as price increases to P1, quantity demanded falls to Q1.

An individual's **quantity demanded** of any good is the total amount that individual would choose to buy at a particular *price*.

Market demand is the sum of all the quantities of a good or service demanded per period by all the households buying in the market for that good or service.

Law of demand:

The **Law of Demand** is commonly described as: "Holding all else equal, when the price of a good rises, consumers decrease their quantity demanded for that good." In other words, there is an inverse, or negative, relationship between the price and the quantity demanded of a good.

$$\uparrow P \rightarrow \downarrow Qd$$

Assumption of the Law of Demand:

The assumptions are:

- Income of the consumer is given and constant.
- No change in tastes, preference, habits, etc,
- Constancy of the price of other goods.
- No change in the size and composition of population.

Exception of the law of demand:

There are certain situations in commodities do not follow the law of demand. These are known as exception of the law of demand. These are:

- **Continuous changes in the price lead to the exceptional behavior**: if the price shows a rising trend a buyer is likely to buy more at a high price for protecting himself against a further rise.

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As against it when the price starts falling continuously, a consumer buys less at a low price and awaits a further in price.

- **Giffen's goods:** when the price of an inferior commodity declines, the consumer, instead of purchasing more, buys less of that commodity and switches on to a superior commodity.
- **Ignorance effect:** implies a situation in which a consumer buys more of a commodity at a higher price only due to ignorance.
- **Veblen Goods:** Some expensive commodities like diamonds, expensive cars, etc., are used as status symbols to display one's wealth. The more expensive these commodities become, the higher their value as a status symbol and hence, the greater the demand for them. The amount demanded of these commodities increase with an increase in their price and decrease with a decrease in their price.
- **Expectation of change in the price of commodity:** If a household expects the price of a commodity to increase, it may start purchasing a greater amount of the commodity even at the presently increased price. Similarly, if the household expects the price of the commodity to decrease, it may postpone its purchases. Thus, some argue that the law of demand is violated in such cases.

Why the demand curve slopes downward?

Income effect: the change in quantity demanded resulting from a change in the consumer's purchasing power (or real income).

As price of good decreases, consumer's income *appears to increase*, so consumers demand a greater quantity of a good as the price goes down (If PA falls, DA increases as the purchasing power of individual's incomes increases).

Substitution effect: the change in quantity demanded resulting from a change in the price of one good relative to the price of other goods.

As the price of one good decrease, other goods *appear to become more expensive,* so consumers demand a greater quantity of the good whose price has decreased (If PA increases relative to PB, then DA falls).

Diminishing marginal Utility: The more one consumes of a good, the less additional utility that last unit consumed provides, and therefore consumers are only willing to buy additional units of a good if the price decreases.

Illustrative Examples

1. The demand curve shows
   A. the maximum amount consumers are willing to pay for particular units of a good

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B. the minimum amount consumers are willing to pay for particular units of a good
C. the average amount consumers are willing to pay for particular units of a good
D. that consumers want to pay the lowest price

Explanation: Demand curve is a graph illustrating how much of a given product a household would be willing to buy at different prices.

Answer: A

2. The more money people make the more pairs of shoes they buy. We can conclude that
   A. Shoes are a normal good.
   B. Shoes are an inferior good.
   C. Demand for shoes has elasticity between 0 and 1
   D. All of the above.

Explanation:

- A good for which demand increases when income rises and decreases when income falls is a normal good.
- As people makes more money they buy more of shoe, then shoe is a normal good.

Answer: A

3. Demand shows the relationship between
   A. income and quantity needed per unit of time
   B. price of a good and the available quantity of that good per unit of time
   C. the price of a good and the quantity consumers are willing and able to buy in a given time period
   D. income and quantity demanded per unit of time

Explanation: Demand is the different quantities of a product that consumers are willing and able to buy at various prices in a given period of time other things remain unchanged.

Answer: C

4. When the price of apples goes up:
   A. The demand for apples will decrease, *ceteris paribus*
   B. The demand for apples will decrease, *ceteris paribus*
   C. The quantity demanded for apples decreases, *ceteris paribus*
   D. The quantity demanded for apples will increase, *ceteris paribus*

Explanation:

- Law of demand states that the tendencies for the quantity demand of a good in a market is decline when its prices rise.
- When the price of apples increases, the quantity demand for apple decreases.

Answer: C

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5. If the economic policy of Ethiopia was causing people to lose jobs; in the market for agricultural product, this would cause:
   A. An increase in demand
   B. A decrease in demand due to higher price
   C. A decrease in demand due to lower income
   D. An increase in prices and quantity

***Explanation*:**

- The policy causes to decrease the consumer income.
- An increase in income increases the demand for goods.
- A decline in income reduces the demand for the goods.
- So the policy decreases the demand for the good due to a lower in consumer income.

Answer: C

6. What is distinction between demand and the quantity demanded?

***Explanation*:**

- Demand refers to a schedule of quantities that will be bought per unit of time at various prices. It refers to the entire demand curve.
- Quantity demanded refers to a specific amount that will be demanded per unit of time at a specific price. It refers to a point on a demand curve.

7. Explain the difference between the slope of a straight-line demand curve and its elasticity

***Explanation*:**

- The slope is simply the rise over the run. i.e. the change in price over the change in quantity.
- The elasticity is the *percentage* change in quantity over the *percentage* change in price. i.e. the average price divided by the average quantity multiplied by the reciprocal of the slope

8. Why the demand curves slope downward?

***Explanation*:** The demand curve slopes downward because in general, the higher the price of the good, the fewer people will want to buy it.

9. Groups of factors give rise to a down ward slopping individual demand curve?
   A. Change in the number of consumers of a commodity and change in the number of use of the commodity
   B. Substitution effect of price change, income effect of price change, diminishing marginal utility
   C. Change in the preference and change in technology
   D. The taste of the commodity being Giffen or prestige good

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***Explanation*:** Demand curve slopes downwards to the right. There is an inverse relationship between the price of the commodity and its demand. There are several reasons for this inverse relationship:

- Operation of the Law of diminishing marginal utility
- Income effect
- Substitution effect
- New consumers creating demand.

Answer: B

10. Which of the following describes the theory behind the demand curve?
    A. Decreasing marginal utility as consumption rises.
    B. Increasing marginal cost as consumption rises.
    C. Decreasing marginal cost as consumption rises.
    D. Increasing total utility at an increasing rate as consumption rises.

***Explanation*:** One of the foundations of the Law of Demand is falling marginal utility as more of a good is consumed. You can eliminate any choices that refer to marginal cost and a downward sloping demand curve would not be the result of total utility that increases at an increasing rate.

Answer: A

11. A buyer is said to have a demand for a good only when:
    A. The buyer wants to own the good
    B. The buyer is both willing and able to purchase the good
    C. The price of the good is low
    D. There is enough supply of the good available for purchase.

***Explanation*:** Demand refers to the different quantities of a product that a consumer is willing and able to buy at various prices in a given period of time.

Answer: B

2.3 Determinant of demand:

Some of the factors that determine demand for a commodity are:-

Price - The law of demand states that when prices rise, the quantity demanded falls. This also means that, when prices drop, demand will rise.

Income of the consumer- Demand represents the consumer's willingness and ability to pay for a good. Income is a major factor in that "ability" to pay component. For most goods, when income increases, demand for the good increases. Thus, for these **normal goods**, increased income results in a graphical rightward shift in the entire demand curve. There are other **inferior goods**, fewer in number, where higher levels of income produce a decrease in the demand curve.

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Tastes and Preferences:
We have different internal tastes and preferences. Collectively, consumer tastes and preferences change with the seasons (more flowers in January, fewer wedding cloths in December); with fashion trends; or with advertising. A stronger preference for a good is an increase in the willingness to pay for the good, which increases demand.

Expectation of future price and income:
The future expectation of a price change or an income change can cause demand to shift today.
Demand can also respond to an expectation of the future availability of a good.

- **Future price:** consumers' current demand will increase if they expect higher future prices; their demand will decrease if they expect lower future prices.
- **Future income**: consumers' current demand will increase if they expect higher future income; their demand decreases if they expect lower future income.

Population and Number of Buyers: the more buyers lead to an increase in demand; fewer buyers lead to decrease.

Advertisement: when a product is aggressively advertised through all the possible media, the consumers buy the advertized commodity even at a high price and many times even if they don't need it

Illustrative Examples

12. If the price of product L increases, the demand curve for a close substitute product M will:
    A. remain unchanged
    B. shift to the right
    C. shift to the left
    D. none of the above

***Explanation*:**

- If two goods L and M are substitute, an increase in the price of good L will increase the demand for good M.
- The demand curve for good M shift to the right.

Answer: B

13. If buyers expect the price of a good to rise in the future, the result is:
    A. A decrease in supply today
    B. An increase in supply today
    C. A decrease in quantity demand today
    D. An increase in demand today.

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Explanation:
If consumer expect that in the near future price of the goods will rise, then in the present they will buy more of the commodity. This leads to an increase in the demand of a good today.

Answer: D

14. If an increase in the price of product A, results in an increase in the demand for product B, one may conclude that products A and B are
    A. substitute goods
    B. complementary goods
    C. unrelated goods
    D. inferior goods

Explanation:
- The demand for a good will increase if the price of substitute for the good rises and the demand for a good will decrease if the price of substitute falls.
- Product A and product B are substitute goods.

Answer: A

15. If the real income of a consumer decreases and, as a result, his demand for product X increases, it can be concluded that product X is a/an
    A. complementary good
    B. normal good
    C. inferior good
    D. substitute good

Explanation: Inferior good is a good for which demand decreases when income rises and increases when income falls. Product X is an inferior good.

Answer: C

16. If cars and petrol are complementary goods, then an increase in the price of petrol will result in:
    A. an increase in the demand for cars
    B. an increase in the demand for petrol
    C. a decrease in the demand for cars
    D. a decrease in the demand for petrol

Explanation:
- The demand for a good increase if the price of a complement decreases and the demand for a good decrease if the price of a complement increases.
- An increase in the price of petrol decreases the demand for cars.

Answer: C

17. If the price of product 'X' decreases, the demand for a close substitute product 'Y':
    A. is inelastic
    B. will shift to the left
    C. will not be affected
    D. Is elastic

Explanation: If product X and Y are substitute goods, an increase in the price of product X will increase the demand for product Y, and a

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decrease in the price of product X will decrease the demand for product Y(product Y shift to the left).

Answer: B

18. An inferior good is a product
    A. that is not expensive
    B. for which there is no demand
    C. for which demand increases as income increases
    D. for which demand falls as income increases

***Explanation*:** Inferior good is a good that consumers buy a smaller quantity with the increase in consumer income.

Answer: D

19. If coffee and tea are substitutes, an increase in the price of coffee will cause:
    A. a decrease in the demand for tea
    B. the consumer's tastes to change so that she now prefers coffee more than before its price rose
    C. an increase in the demand for tea
    D. a movement along the demand curve for tea

***Explanation*:**

- Substitute goods are goods that can serve as replacement for another.
- When the price of one good increases, demand for the other good increases.
- An increase in the price of coffee will increase the demand for tea.

Answer: C

20. To say that' shero wot' is an inferior good, as economists use the term
    A. means that as the price of 'shero wot' falls, the quantity demanded of shero wot falls
    B. means that there is no real income effect when the price of changes
    C. means that as the average level of income falls, the demand for shero wot rises
    D. means that the supply of shero wot is perfectly inelastic

***Explanation*:**

- **Inferior goods** are Goods for which demand decreases as income increases.
- When consumer income falls the demand for shero wot increases.

Answer: C

21. Suppose the price of good 'X' has decreased; which in turn, leads to an increase in the demand for good 'Y'. Economic analysis states that the goods (X and Y) must be:
    A. substitute goods
    C. normal goods

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B. complementary goods D. inferior goods

***Explanation*:** Two goods are complement if increasing the price of one good lowers the demand for the other. Good X and Y are complements.

Answer: B

22. Which of the following would increase the amount of an inferior good that buyers would like to purchase?
    A. an increase in buyers' incomes
    B. an increase in the price of complement
    C. a decrease in the price of a substitute
    D. a decrease in buyers' income

***Explanation*:** The demand for inferior goods increases when the income of the consumer decreases.

Answer: D

23. Which of the following is not a determinant of demand?
    A. tastes of a consumer
    B. price of related goods
    C. income of the consumer
    D. the price of the inputs to produce the good.

***Explanation*:**

- Income, taste and price of related goods are determinant of demand.
- The price of the input is the determinant of supply.

Answer: D

24. If an increase in demand quantity for a commodity as a result of in income of consumers is to be analyzed, which of the following best express the assumption that must have been held by the analyst?
    A. Price of the goods and price of the other goods are held constant
    B. Price of the goods and price of other goods are changing slowly
    C. All factors, other than income, are changing at a reasonable rate
    D. All factors, other than income, are held constant.

***Explanation*:**

- To express the relationship between consumer income and quantity demand we hold other things being constant.
- Every laws and principles are develop under consideration of Ceteris Paribus which is meant 'Other things remain constant'
- To know an increase in demand due to an increase in income, we held all factors other than income held constant.

Answer: D

25. If average household income rises and we observe that the demand for raw meat increases, raw meat must be:
    A. An inferior good. C. a surplus good.

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B. A normal good. D. a necessity.

***Explanation*:** When income increases and demand increases, the good is a normal good. Had the demand for raw meat decreased, they would be an inferior good.

Answer: B

2.4 Movement along a demand curve and shift in the demand curve

2.4.1Movement along a demand curve:

A movement refers to a change along a curve.

- On the demand curve, a movement denotes a change in both price and quantity demanded from one point to another on the curve.
- The movement implies that the demand relationship remains consistent. Therefore, a movement along the demand curve will occur when the price of the good changes and the quantity demanded changes in accordance to the original demand relationship
- In other words, a movement occurs when a change in the quantity demanded is caused only by a change in price, and vice versa.

2.4.2 Shift in the demand curve:

A change in any determinant of demand except for the good's price causes the demand curve to shift. We call this a **change in demand**.

- If buyers choose to purchase more at any price, the demand curve shifts rightward an increase in demand.
- If buyers choose to purchase less at any price, the demand curve shifts left warda decrease in demand

Illustrative Examples

26. If the price of gasoline fell, the market demand curve for automobile tires, a complement, would

A. shift to the right. C. remains stationary.

B. shift to the left. D. becomes more elastic.

***Explanation*:**

- Complementary goods are those goods which are used in conjunction with another good.
- If the price of one good increases the demand for the good is decreases.
- If the price of gasoline fell the demand for automobile tires increase meaning it shift to the right.

Answer: A

27. When economists say the demand for a product has increased, they mean that the

A. Demand curve for the product has shifted to the left.

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B. price of the product has fallen, and consequently consumers are buying more of the product.
C. cost of producing the product has increased.
D. amount of the product that consumers are willing to purchase at Various prices have increased.

***Explanation*:**

- When demand for a product increases its demand curve shift to the right and when it decreases its demand curve shift to the left.
- Shift of a demand curve occurs when there is a change other than price.

Answer: D

28. If the price of good B decreased, you would expect the
A. demand of good B to increase
B. quantity demanded of good B to decrease
C. quantity demanded of good B to increase
D. demand curve of good B to shift to the left

***Explanation*:**

- A change in price would change the quantity demand for a product.
- When factors other than price changes would change demand for a product.
- If the price of good B decreased quantity demand of good B is increased.

Answer: C

29. Which of the following will NOT change the demand for a good? A change in
A. the number of consumers in the market
B. the prices of related goods
C. the price of the good itself
D. expectations about future prices

***Explanation*:** Change in demand occurs when there is a change in

- An increase in the number of consumers.
- The price of related goods
- Expectation of future price
- Change in the price of the good itself causes a change in quantity demand.

Answer: C

30. Which one of the following will cause the demand curve for gasoline to shift to the right?
A. a fall in the price of cars
B. an increase in the supply of gasoline

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C. a fall in the price of gasoline
D. a rise in the price of cars

***Explanation*:**

- Cars and gasoline are complementary goods.
- If the price of on good increase, the demand for the good decrease.
- A fall in the price of cars will cause the demand for gasoline to increase(shift to the right)

Answer: A

31. Suppose that a demand curve for a product is negatively sloped and that the price of the product increases from 4.50 Birr to Birr 5.00. Which of the following will result?
    A. the supply of the product will decrease
    B. quantity demanded of the product will increase
    C. the demand for the product will increase
    D. quantity demanded of the product will decrease

***Explanation*:**

- The law of demand states that there is a negative relationship b/n the price of a good and quantity demand.
- At higher price quantity demand will decreases and vice-versa.
- The price of a product increase from Birr 4.50 to Birr 5.00, then quantity demand of the product will decrease.

Answer: D

32. Which of the following will cause a change in quantity demand for good X?
    A. a change in the price of substitute good Y
    B. a change in the price of complementary good Z
    C. a change in the price of X
    D. a change in consumer incomes

***Explanation*:**

- If the price of a good changes but everything else remains the same, there is a movement along the demand curve.
- A movement along the demand curve shows a change in the quantity demanded. It is caused by a change in price of a product.

Answer: C

33. Which of the following would NOT shift the demand curve for a good or service?
    A. a change in income
    B. a change in the price of a related good
    C. a change in expectations about the price of the good or service
    D. a change in the price of the goods or service

***Explanation*:** Shift in the demand curve occurs when there is a change in:-

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- tastes and preferences
- prices of related goods and services
- consumers income
- number of consumers
- expectations of future prices and income

A change in the price of a good causes movement along on a demand curve.

Answer: D

34. Which of the following will NOT cause a shift to the right in the demand curve for beer?
    A. A change in the price of beer
    B. A health study indicating positive health benefits of moderate beer consumption.
    C. An increase in the price of a substitute
    D. A decrease in the price of a complement.

Explanation: A change in the price of beer would induce a movement along the demand curve for beer, rather than a shift in the demand curve for beer.

Answer: A

35. A student says "an increase in price causes a decrease in demand", true/ false

Explanation:

- An increase in price only causes a movement along the demand curve.
- The phrase decrease in demand means that the demand curve shifts.
- The demand curve NEVER shifts because of a price change.

Answer: False

36. Carefully explain the difference between *quantity demanded* and *demand* for a good/product

Explanation:

- When the demand for a good is related to its price, it is called the quantity demanded.
- Ceteris Paribus, the quantity demanded for a good increase when the price of the good decreases, and vice versa. For a change in price of a good we move along the demand curve
- On the other hand, when the demand for a good is related to other factors, e.g., income, taste, substitutes, etc., it is called **demand**. For instance, when income increases, at a given price level, a consumer can purchase more; hence the **demand** increases and demand function shifts to the right

37. For each of the following markets indicate whether the stated change causes a shift in the demand curve or a movement in the demand curve.

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A. The housing market: consumers' income increases
B. The orange juice market: the price of orange decreases.
C. The food and restaurant market: the number consumers in the area decreases
D. The coffee market: the price of coffee increases.
E. The beef market: a change in information about the effect of cholesterol.
F. The travel market: an increase in the number of Taxis in a country.

***Explanation*:**

- The statements that cause a shift in the demand curve are: A, C, E and F.
- The statements that cause a movement of demand curve are: B and D

2.5 Supply:

Supply indicates the various quantities of a product that sellers are willing and able to provide at various prices in a given period of time.

A supply schedule is a table showing how much of a product firms will supply at different prices

Supply curve - Shows the relationship between the quantity supplied and the price of a good, everything else remaining the same. It is a graph illustrating how much of a product a firm will supply at different prices.

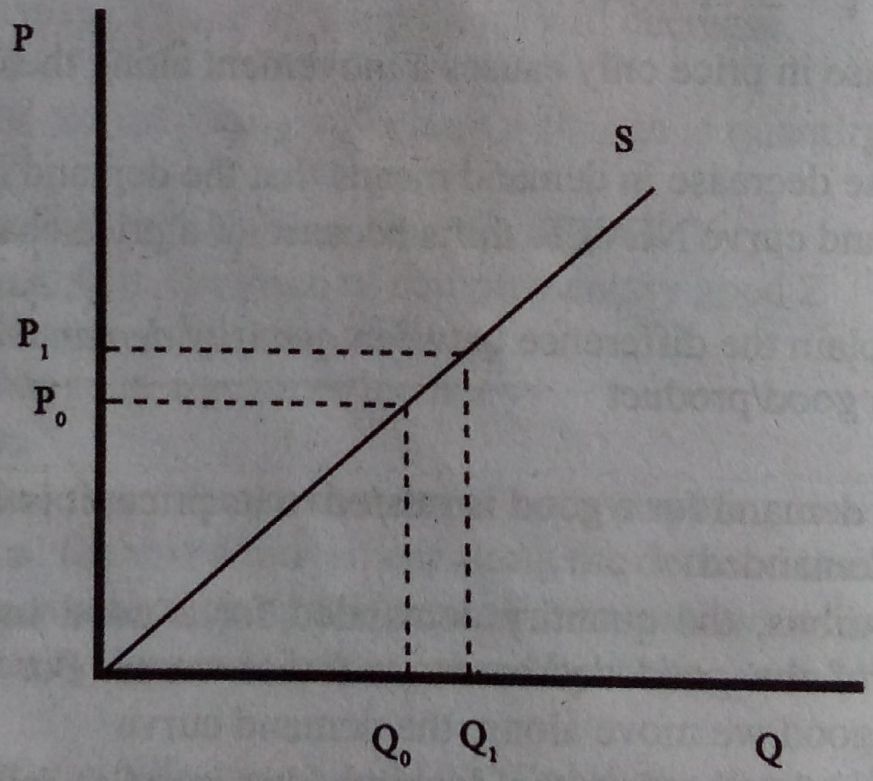

Figure 2.6: supply curve.

Here, as price increase from P0 to P1, quantity supplied increase from Q0 to Q1.

2.5.1 Law of supply:

The Law of Supply is commonly described as: "*Holding all else equal, when the price of a good rises, suppliers increase their quantity supplied*

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for that good." In other words, there is a direct, or positive, relationship between the price and the quantity supplied of a good.

$$\uparrow P \longrightarrow \uparrow Q_s$$

This means that supply curves typically have a positive slope.

2.5.2 Determinant of supply:

Good's own price: an increase in price will induce an increase in the quantity supplied.

Prices of related goods: firms often react to change in the price of related products. If beef prices rise, production may respond by raising more cattle. But leather comes from cowhide. Thus, an increase in beef prices may actually increase the supply of leather.

The state of technology: If there is a technological advancement in one good's production, the supply increases.

Expectations: Sellers' expectations concerning future market conditions can directly affect supply. If the seller believes that the demand for his product will sharply increase in the foreseeable future the firm owner may immediately increase production in anticipation of future price increases. The supply curve would shift out.

Price of inputs: Inputs include land, labor, energy and raw materials.[7] If the price of inputs increases the supply curve will shift left as sellers are less willing or able to sell goods at any given price.

Number of suppliers: As more firms enter the industry the market supply curve will shift out driving down prices.

Taxes and Subsidies: A per unit tax is treated by the firm as an additional cost of production and would therefore decrease the supply curve, or shift it leftward. A subsidy is essentially the anti-tax, or a per unit gift from the government because it lowers the per unit cost of production and would increase the supply curve or shift to the right.

Illustrative Examples

38. Which of the following is correct with respect to a firm's supply of a given product? The supply curve shows
    A. the amount of profit that will be earned for various output levels
    B. the amount of a good that will be available for sale at various prices
    C. an inverse relationship between price and quantity supplied
    D. the amounts of a good that will be sold at various prices

Explanation: Supply is the relationship between the amount of a good or services offered for sale at a given price.

Answer: B

39. Which of the following is true with regard to an increase in the supply of a good?

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A. it is most likely caused by an increase in the price of the good
B. it is represented by an upward shift of the supply curve
C. it is represented by a rightward shift of the supply curve
D. it is represented by a movement upward along the supply curve

***Explanation*:** If the price of a good remains the same but other non-price determinants of supply changes, supply changes and there is a shift of the supply curve.

Answer: C

40. Which of the following is *not* an example of a "non-price determinant of supply" of Wheat?
A. Available technology for making Wheat.
B. The price inputs for making wheat.
C. Number of Wheat producers.
D. Price of Wheat.

***Explanation*:** Non-price determinants of supply are those determinants of supply other than price. These are:-

- Improved technology
- The price of inputs or factors of production
- Number of suppliers

Price of Wheat is own price determinant of supply.

Answer: D

41. A leftward shift of a product supply curve depends upon:
A. an improvement in the relevant technique of production
B. a decline in the prices of needed inputs
C. an increase in consumer incomes
D. some firms leaving an industry

***Explanation*:** Supply of a product decreases:

- Backward technology
- An increase in the price of inputs or factors of production
- Decrease the number of firms producing a product
- Some firms leaving an industry decrease the number of producers and the supply of the product. The supply curve shift to the left.

Answer: D

42. Why does the supply curve slopes upward?

***Explanation*:**

- Supply is a positive relationship between the price of the good and the quantity supplied of the good by firms.
- The supply curve slopes upward because higher prices give firms more incentive to produce and sell more

43. Suppose that aluminium is a key production input in the production of bicycles. If the price of aluminium falls and all other variables are held constant, we expect

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A. The demand for aluminium to rise.
B. The supply of bicycles to rise.
C. The supply of bicycles to fall.
D. The demand for bicycles to rise.

***Explanation*:** This is a determinant of supply. If the raw material becomes less costly to acquire, the marginal cost of producing bicycles falls. Producers increase the supply of bicycles. Recognizing this as a supply determinant allows you to quickly eliminate any reference to a demand shift.

Answer: B

2.5.3 Movements versus shifts of supply:

Movements along the supply curve occur only if there is a change in quantity supplied caused by a change in the good's own price.

A shift in the supply curve, referred to as a change in supply, occurs only if a non-price determinant of supply changes.

Illustrative Examples

44. For each of the following markets indicate whether the stated change causes a shift or movement along a supply curve:
    A. The Teff market: there is a drought in the Teff cultivation area
    B. The Automobile market: the price of cars decreases
    C. The banana market: an increase in the price of an input
    D. The egg market: there is a disease that reduces the number of hens.
    E. The fish market: the price of fish increases.

***Explanation*:**

- The statements that cause shift of supply curve are: A, C and D
- The statements that cause movement along the supply curve are: B and E

45. Which of the following will not cause supply to change?
    A. an increase in cost of production
    B. an increase in the number of firms
    C. an increase in the price of another product
    D. an increase in demand

***Explanation*:** Supply of a good depends on:

- The price of the good;
- The prices of factors of production;
- The price of other goods produced;
- Expected future prices;
- The number of suppliers;
- Technology

An increase in demand is not the factor that supplies to change.

Answer: D

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46. If technology improves, that will be reflected on the supply curve as:
    A. an increase (shift to the right)
    B. move along the supply curve
    C. no change
    D. a decrease (shift to the left)

***Explanation*:**

- If there is an improvement in production technology the production efficiency increases which reduce the unit of cost of production and the firm would supply more than before.

Answer: A

46. A leftward shift in the supply curve indicates
    A. that more suppliers have entered the industry
    B. decrease in the quantity supplied at each price
    C. that more is demanded at each price
    D. an increase in the quantity supplied at each price

***Explanation*:** A left ward shit of the supply curve indicates that there is a decrease in the quantity supplied at each price.

Answer: B

47. Which of the following statements is true, regarding the supply of a particular good, and that good's own price?
    A. A price increase shifts the supply curve to the right.
    B. A price decrease shifts the supply curve to the right.
    C. A price increase shifts the supply curve downward.
    D. A price change alone does not shift the supply curve.

***Explanation*:** A shift in the supply curve, referred to as a change in supply, occurs only if a non-price determinant of supply changes. Such as:

- Price of related goods
- Prices of inputs
- Technology
- Weather condition

Answer: D

48. Which of the following is an example of movement *along* a supply curve?
    A. The quantity of apples offered for sale increases as the price of apples rises.
    B. Thanks to good weather conditions, apple growers enjoy a bumper crop this year.
    C. The price of orange doubles, increasing demand for apples.
    D. The price of fertilizer increases, making it more expensive to produce apples.

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***Explanation*:**

- Weather condition
- Price of related goods (the price of orange)
- The price of inputs or factors of production (price of fertilizers). These factors cause a change in supply curve(shift the supply curve)
- A change in the price of a product (price of apples) cause a movement along on a supply curve.

Answer: A

49. Which of the following is an example of movement *along* a supply curve?
    A. The quantity of apples offered for sale increases as the price of apples rises.
    B. An apple orchard burns down in an accidental fire, decreasing the number of suppliers on the market.
    C. Thanks to good weather conditions, apple growers enjoy a bumper crop this year.
    D. The price of fertilizer increases, making it more expensive to produce apples.

***Explanation*:**

- If the price of a good changes but everything else influencing suppliers' planned sales remains constant, there is a movement along the supply curve.
- Good weather conditions increases supply(shift supply to the right)
- Decrease the number of suppliers and an increase in the price of fertilizers decrease supply(shift supply to the left)

So movement along a supply curve occurs when there is a change in the price of apples.

Answer: A

50. Which one of the following is more likely to happen to the supply curve of bread following an increase in the cost of flour mills and a reduction in the production of wheat, if all other things remain the same?
    A. A left ward shift of the supply curve
    B. A right ward shift of the supply curve
    C. Both left ward and right ward shifts in the supply curve
    D. Cannot be determined from the given information

***Explanation*:** The most important variables that shift supply curve are:

- Price of inputs
- Technological change
- Price of substitute in production
- Expected future prices
- Number of firms in the market

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An increase in the price of inputs shifts the supply curve to the left. Price of input is the factor most likely to cause the supply curve for a product to shift is a change in the price of inputs.
When the cost of flour mills increases and reduces the production of wheat, the supply of bread decreases and the supply curve shift to the left.

Answer: A

2.6 Market Equilibrium:
Equilibrium refers to a situation in which the price has reached the level where quantity supplied equals quantity demanded

P	QD	QS
1.10	0	1000
1.00	100	900
90	200	800
80	300	700
70	400	600
60	500	500
50	600	400
40	700	300
30	800	200

As can be seen, this market will be in equilibrium at a price of 60. At this price the demand equals the supply, and the market will clear. 500 quantities will be offered for sale at 60p and 500 will be bought - there will be no excess demand or supply at 60p.

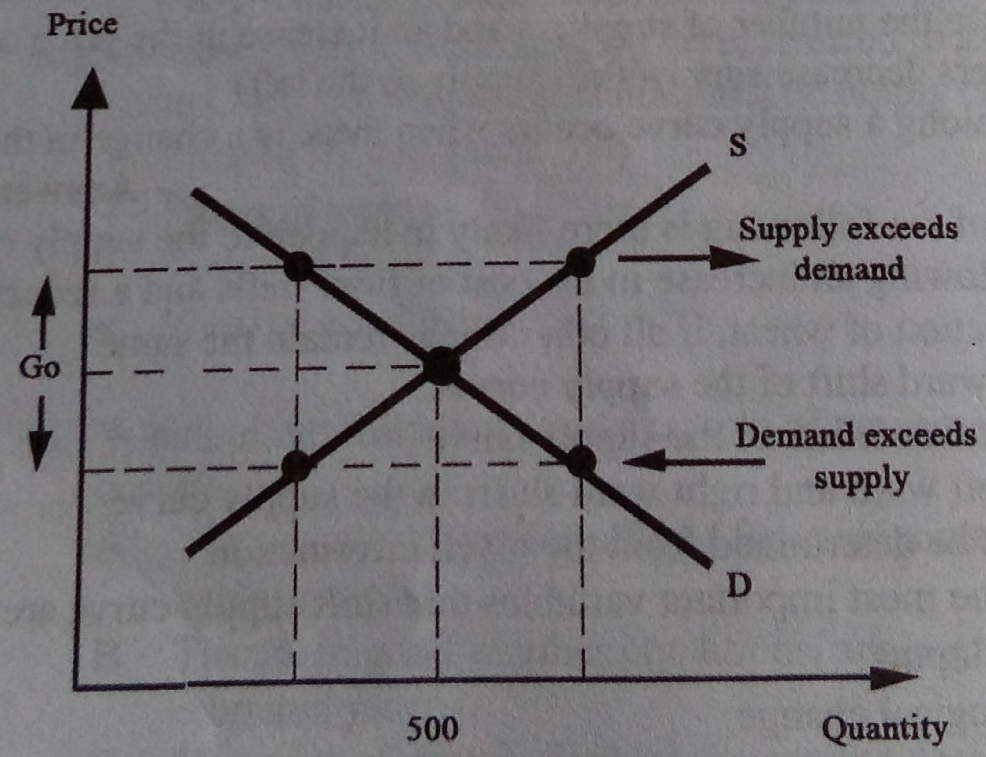

Figure 2.7: market equilibrium.

Equilibrium Price:
The price that balances quantity supplied and quantity demanded.

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On a graph, it is the price at which the supply and demand curves intersect (at p = 60)

Equilibrium Quantity

The quantity supplied and the quantity demanded at the equilibrium price. On a graph it is the quantity at which the supply and demand curves intersect(at Q = 500)

- At low prices, producers supply less and consumers want to buy more than at the equilibrium price.
- This creates an **excess demand**, and causes a shortage of the product.

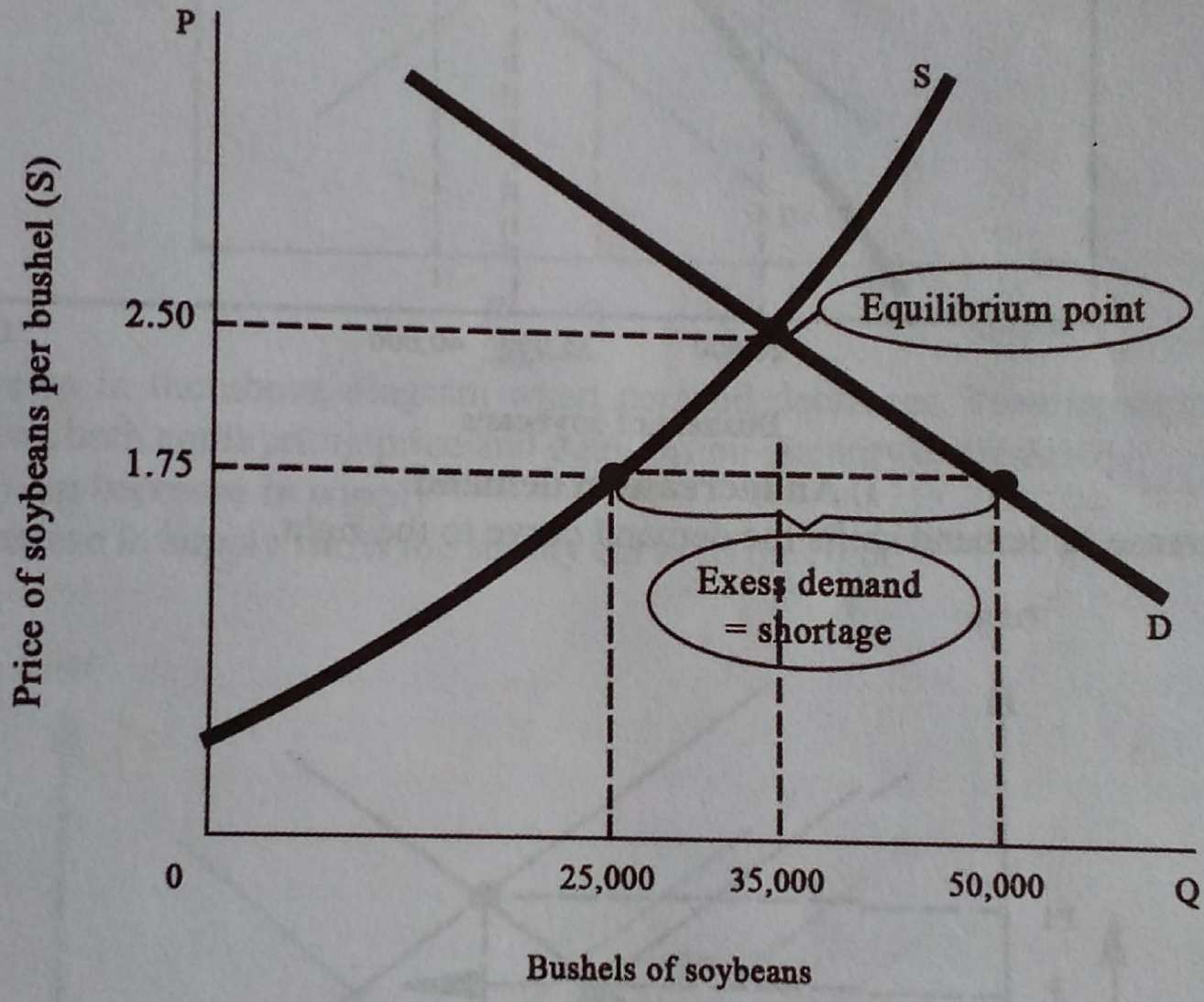

Figure 2.8: market equilibrium.

- At high prices, producers are willing to produce more of the product, but consumers are willing to buy less than at the equilibrium price.
- Excess supply, the condition where quantity supplied exceeds quantity demanded at the current price, will result. This creates surplus.

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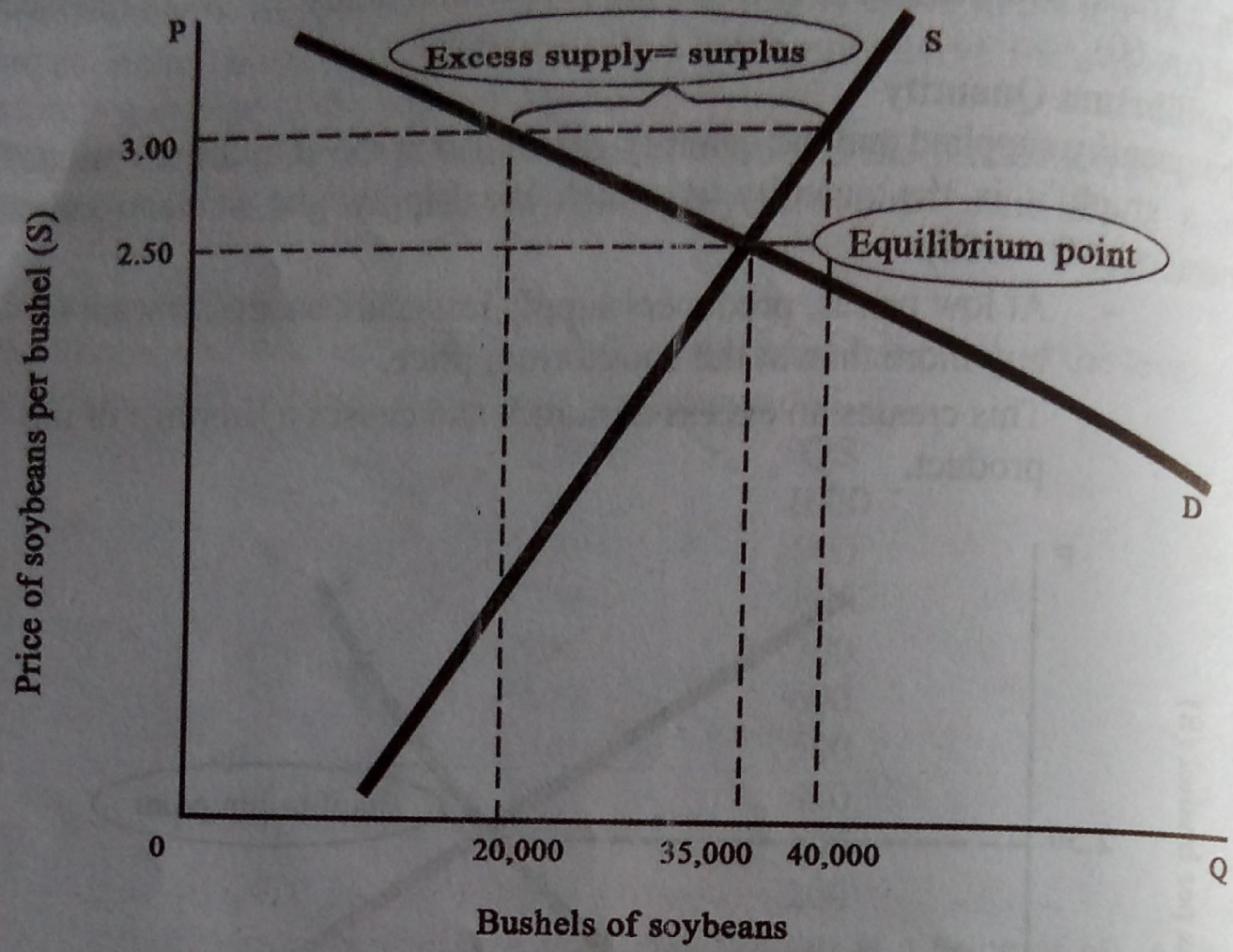
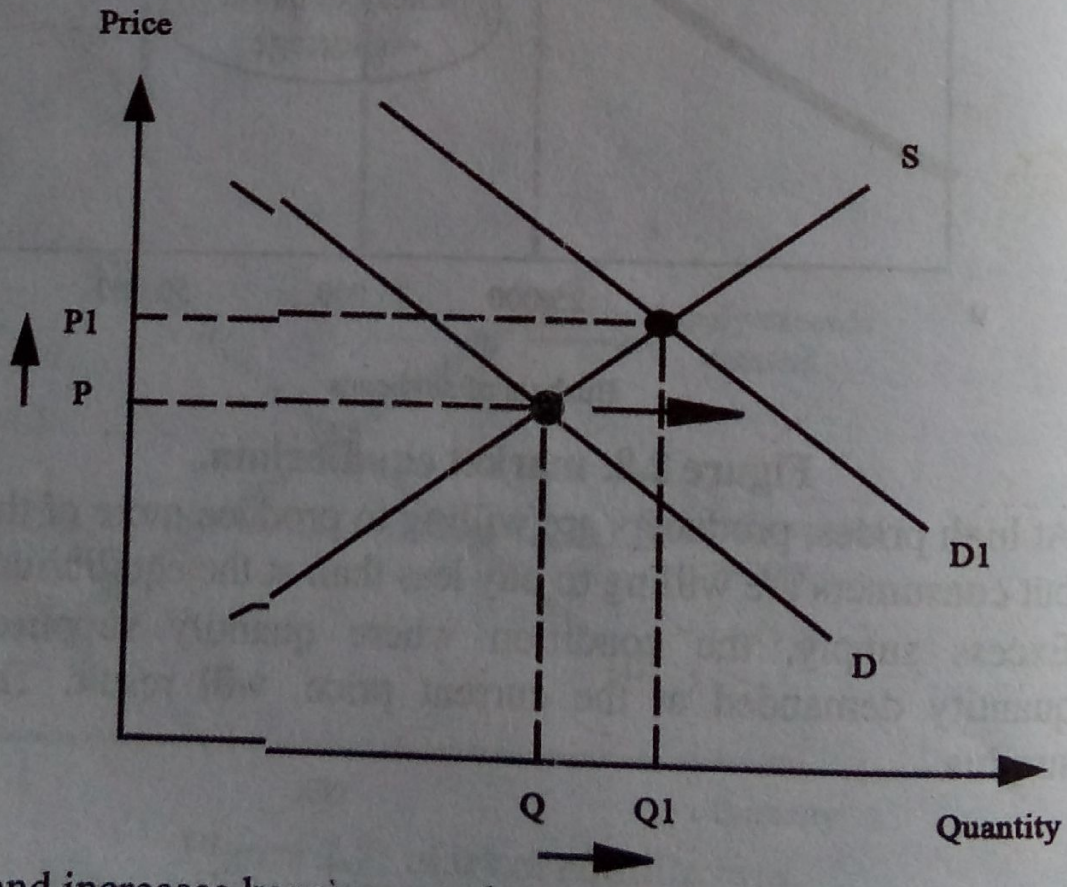


i) An increase in demand

An increase in demand shifts the demand curve to the right.



When demand increases keeping supply constant, both equilibrium price and equilibrium quantity increases.

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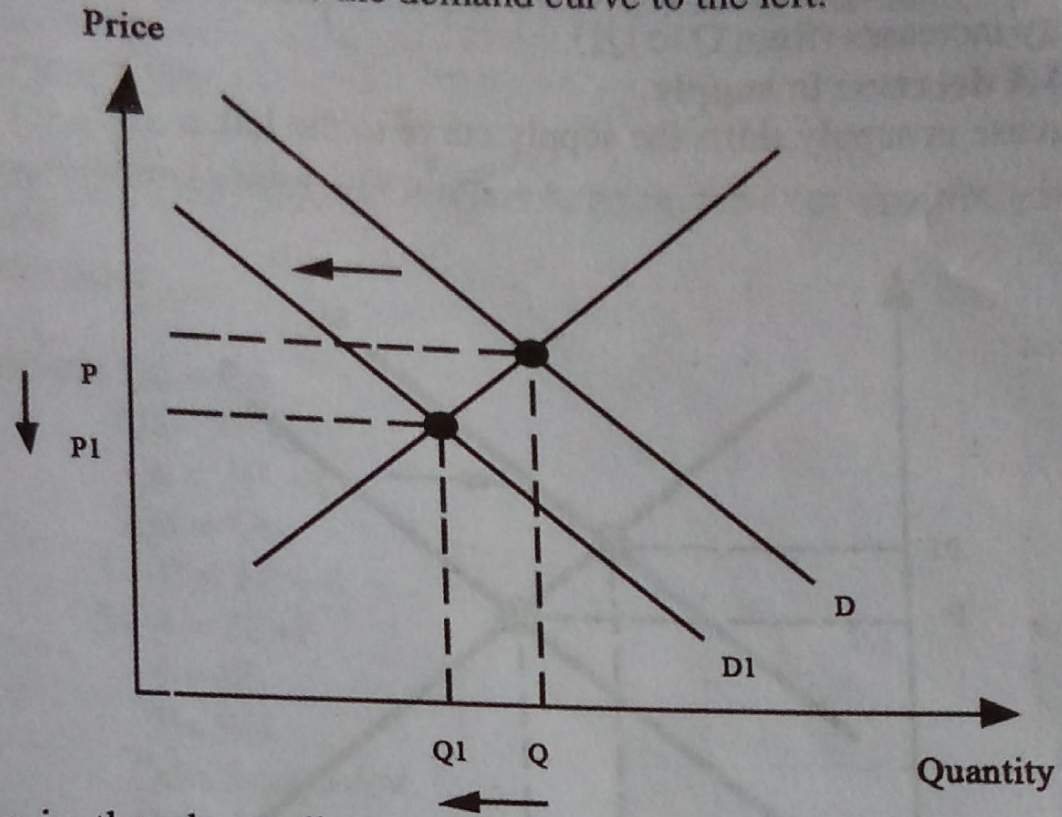
ii) A decrease in demand

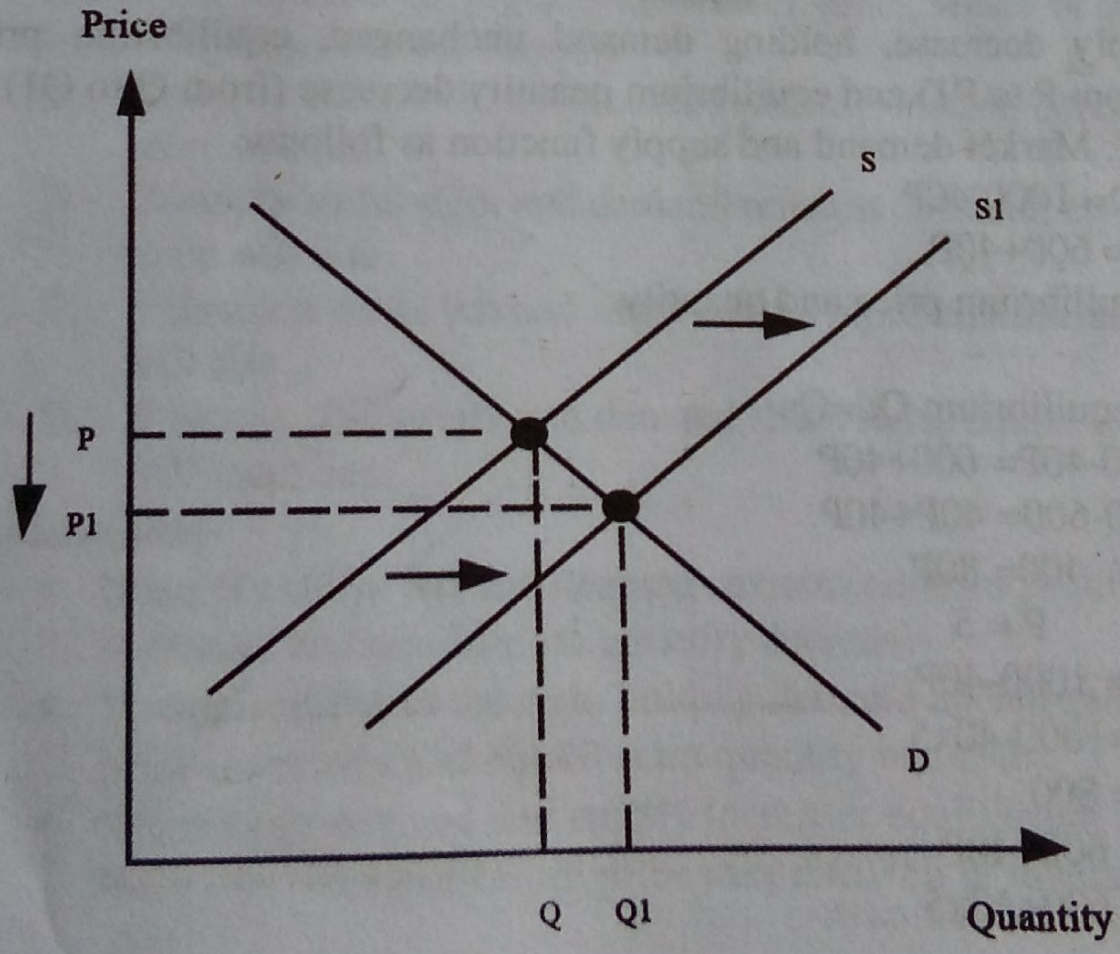
A decrease in demand shifts the demand curve to the left.



As shown in the above diagram when demand decreases, keeping supply constant, both equilibrium price and equilibrium quantity decreases.

iii) An increase in supply

An increase in supply shifts the supply curve to the right.



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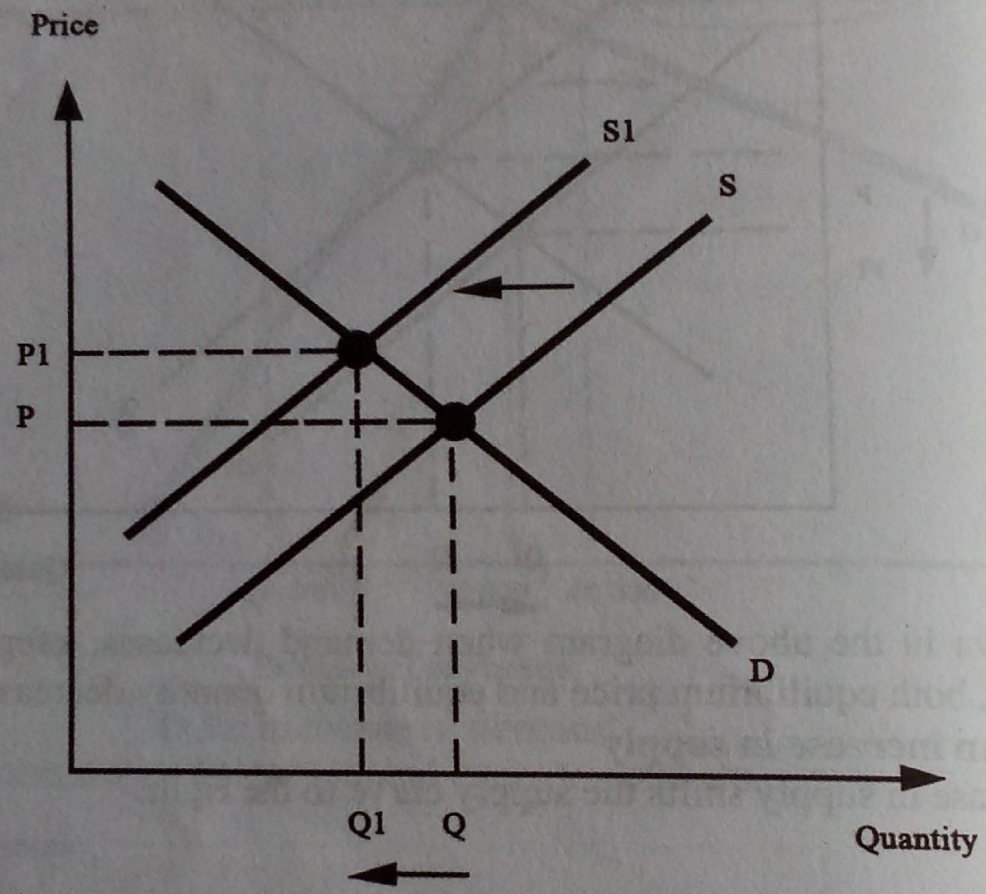
As shown in the above diagram when supply increases, holding demand unchanged equilibrium price decrease (from P to P1) and equilibrium quantity increases (from Q to Q1)

iv) A decrease in supply

A decrease in supply shifts the supply curve to the left.



When supply decrease, holding demand unchanged, equilibrium price increase (from P to P1) and equilibrium quantity decrease (from Q to Q1)

Example: 1. Market demand and supply function as follows:

DD= 1000-40P

SS= 600+40P

Calculate equilibrium price and quantity.

Solution:

At equilibrium Qd=Qs

1000-40P= 600+40P

1000-600= 40P+40P

400= 80P

P = 5

Qd= 1000-40P

Qd=1000-40x5

Qd= 800

Qs= 600+40P

Qs= 600+40x5

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Qs= 800

Example: 2. Individual demand and supply functions for a commodity are given respectively as:

Qd = 5 - P

Qs = 3P- 4

Where Q refers to quantity and P refers to price. Find out Equilibrium quantity and

Equilibrium price

Solution:

At equilibrium Qd = Qs

Qd= 5 - P

Qs = 3P - 4

Qd = Qs

5 – P = 3P – 4

5+ 4 = 3P+P

9= 4P,

P= 9/4

Qd= 5-p= 5-9/4

Qd= 11/4

Qs= 3P-4= 3x9/4 - 4

Qs= 11/4

Illustrative Examples

51. Given the standard supply-and-demand diagram, which of the following statements is correct?
    A. If supply shifts left and demand remains constant, equilibrium price will fall
    B. If supply shifts right and demand remains constant, equilibrium price will rise
    C. If demand shifts left and supply shifts right, equilibrium quantity will rise
    D. If supply shifts right and demand shifts right, equilibrium quantity will increases.

***Explanation*:**

- If supply shifts left and demand remains constant, equilibrium price increases and equilibrium quantity decreases.
- If supply shifts to the right holding demand unchanged, equilibrium price decreases and equilibrium quantity increases.
- When both demand and supply increases equilibrium quantity increases but equilibrium price may rise, fall or unchanged.

Answer: D

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52. If supply and demand both decreases, we can say that equilibrium quantity:
    A. and equilibrium price must both decline
    B. must decline, but equilibrium price may rise, fall, or remain unchanged
    C. price must fall, but equilibrium quantity may rise, fall, or remain unchanged
    D. and equilibrium price must both increase

***Explanation*:**
If both demand and supply decreases, the equilibrium quantity would fall. But the equilibrium price may rise, fall or remain unchanged.

Answer: B

53. Excess demand occurs when:
    A. quantity demanded is less than quantity supplied
    B. goods are scarce
    C. the actual price is greater than the equilibrium price
    D. the actual price is less than the equilibrium price

***Explanation*:**
- Excess demand, or shortage, is the condition that exists when quantity demanded exceeds quantity supplied at the current price.
- It occurs when the actual price is set below the equilibrium price.

Answer: D

54. When there is a shortage in a market
    A. consumers are willing to buy more of the good at the current price
    B. the equilibrium price is below zero
    C. quantity supplied exceeds quantity demanded
    D. firms are willing to sell more of the good at the current price

***Explanation*:** Shortage occurs when the actual price is set below the equilibrium price. At lower price consumers are willing to buy more of the good at the current price

Answer: A

55. According to basic supply and demand analysis, when the demand for tea caused sugar prices to rise, what happened to the equilibrium price and quantity of tea?
    A. Equilibrium price and quantity both increased
    B. Equilibrium price and quantity both decreased.
    C. Equilibrium price increased and equilibrium quantity decreased
    D. Equilibrium price decreased and equilibrium quantity increased

***Explanation*:**
- Sugar and tea are complements.

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- When the price of sugar increases, the demand for tea decreases. Thus, the equilibrium price and quantity for tea will decrease.

Answer: B

56. If the demand curve for desktop computers shifts rightward and at the same time the supply curve shifts leftward, then
    A. the equilibrium price definitely increases.
    B. the equilibrium price definitely decreases.
    C. the equilibrium quantity definitely increases.
    D. the equilibrium quantity definitely decreases.

***Explanation*:**

- When demand increases and supply decreases, the price rises and the quantity increases, decreases, or remains constant.
- When the demand for desktop computer shifts right ward and its supply curve shifts left ward, then the equilibrium price increases.

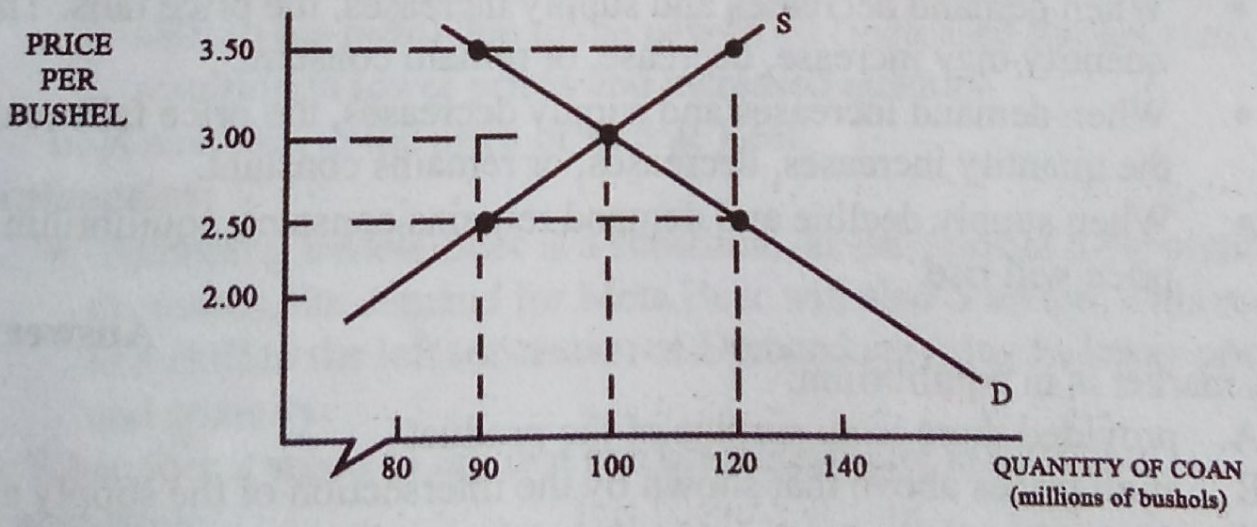
Answer: A

57. What happens if the price of the product is below the equilibrium price?
    A. The buyers will stop purchasing a "cheap" product
    B. The producer will lower the price to sell more products
    C. There will be excess demand for the product
    D. There will be a surplus of the product

***Explanation*:**

- The price below the equilibrium price causes a shortage.
- A shortage results if at the current price the quantity demand exceeds the quantity supplied.
- The price is too low to equate the quantity demanded with the quantity supplied.

Answer: C



58. The equilibrium price is not 2.50 per bushel because
    A. at that price there would be an excess supply of 30 million units and this would tend to drive the price down.

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B. at that price there would be an excess supply of 30 million units and this would tend to drive the price up.
C. at that price there would be an excess demand of 30 million units and this would tend to drive the price down.
D. at that price there would be an excess demand of 30 million units and this would tend to drive the price up.

***Explanation*:**

- At price 2.50 quantity demand is 120 and quantity supply is 90.
- At this price there will be excess demand of 30 units (120-90= 30) and this would tend to drive the price up

Answer: D

59. At the price of 3.50, there would be
A. excess supply of 30 million bushels.
B. excess supply of 80 million bushels.
C. excess demand of 20 million bushels.
D. excess demand of 30 million bushels

***Explanation*:**

- At price 3.50 quantity demand is 90 units and quantity supply is 120 units.
- At this price there will be excess supply of 30 units(120- 90)

Answer: A

60. Which of the following statements is incorrect?
A. If demand increases and supply decreases, equilibrium price will rise
B. If supply increases and demand decreases, equilibrium price will fall
C. If demand decreases and supply increases, equilibrium price will rise
D. If supply declines and demand remains constant, equilibrium price will rise

***Explanation*:**

- When demand decreases and supply increases, the price falls. The quantity may increase, decrease, or remain constant.
- When demand increases and supply decreases, the price falls and the quantity increases, decreases, or remains constant.
- When supply decline and demand remains constant, equilibrium price will rise.

Answer: D

61. A market is in equilibrium:
A. provided there is no surplus of the product
B. at all prices above that shown by the intersection of the supply and demand curves
C. if the amount that producer wants to sell is equal to the amount that consumers want to buy

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D. whenever the demand curve is down-sloping and the supply curve is up-sloping

***Explanation*:**

- Equilibrium is a situation in which opposing forces balance each other.
- Equilibrium in a market occurs when the price is such that the opposing forces of the balance of buyers and sellers balance each other.

Answer: C

62. A surplus of a product will arise when price is:
    A. A above equilibrium with the result that quantity demanded exceeds quantity supplied
    B. above equilibrium with the result that quantity supplied exceeds quantity demanded
    C. below equilibrium with the result that quantity demanded exceeds quantity supplied
    D. below equilibrium with the result that quantity supplied exceeds quantity demanded

***Explanation*:**

- **Excess supply,** or surplus, is the condition that exists when quantity supplied exceeds quantity demanded at the current price.
- It occurs when the price set above equilibrium price.

Answer: B

63. Consider the market for Meta Beer using Supply & Demand. Show the effect of each of the following on equilibrium price and quantity of each of the following: (explain what shifts and why).
    A. Opening new Beer factory.

***Explanation*:**

- opening a new Beer factory increase the supply of Beer (supply shift to the right) due to the new firms (increased market size) resulting in lower prices and increased quantity.

B. A decrease in the price of Bedele Beer.

***Explanation*:**

- Assuming Bedele Beer is a substitute, as the price of a substitute decreases, the demand for Meta Beer will also decrease. This results in a shift to the left (decrease) of Demand resulting in lower prices and quantity

64. What does it mean to say that market equilibrium is stable?

***Explanation*:**

- When the market reaches equilibrium, the natural price and market price are equal.

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- At equilibrium, the quantity supplied equals the quantity demanded meaning the market clears.
- If the market is not in equilibrium, ie has a surplus or shortage, the invisible hand will guide it towards equilibrium

65. Explain how equilibrium price and quantity will change in the following cases:
   A. Demand for a commodity increases, the supply remains the same
   B. Supply for a commodity decreases, the demand reaming the same
   C. Both demand and supply increase by equal amount.

***Explanation*:**

   A. When demand increase keeping supply curve unchanged, both equilibrium price and quantity increase
   B. Under the condition of stable demand curve, if there is a decrease in supply equilibrium price increases and equilibrium quantity decreases.
   C. When both demand and supply increase, the quantity increases. The price remains constant.

66. Given a demand function and supply function

$Qd = 22-p$

$Qs = 10+ p$, where Q is quantity and p is price.

Calculate equilibrium price and quantity

Solution:

$Qd = Qs$

$22-P= 10+P$

$22-10 = P+P$

$12= 2P$, $P= 12/2$

$P= 6$

$Qd= 22- P$

$Qd= 22-6$

$Qd= 16$

$Qs= 10+P$

$Qs= 10+6$

$Qs= 16.$

67. Suppose the following demand and supply function is given:

$Qd = 200 – 20P$

$Qs = 25 +15P$, where Qd stands for quantity demand and Qs is quantity supplied

Find out equilibrium quantity and price.

Solution:

At equilibrium $Qd = Qs$

$200- 20P= 25+15P$

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200-25 = 15P+20P
175= 35P
P= 175/35
P= 5
Qd= 200- 20P
Qd= 200- 20X5
Qd= 100
Qs= 25+15P
Qs= 25+15X5
Qs= 100

68. An individual demand for maize is given by:
Qd = 10 – 0.5P, where
Qd = Quqntity demand of maize per month in kilograms
P= price of maize in Birr/ Kilogram
The market price of maize was 10 Birr and if there are 500 people in a village, what was the total demand for maize.

Solution:
Qd = 10 – 0.5P
Qd = (10 – 0.5P)500
Qd = 5000 – 250P , P=10 Birr
Qd = 5000 – 250x10, Qd = 2500 Kg

69. Consider the market demand and supply function Qd= 21-3P and Qs = -3+5p respectively.
Calculate equilibrium price and quantity.

Solution:
Qd= 21-3p
Qs= -3+5p
Qd=Qs
21-3P= -3+5P
21+3= 5P+3P
24= 8P, P=24/8
P= 3
Qd=21-3P
Qd=21-3X3
Qd= 12
Qs= -3+5P
Qs= -3+5x3
Qs= 12

70. There are 10,000 identical consumers in the market for commodity X each with a demand function given by Qd= 12-2p and 1000 identical

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producers of c0mmodity X, each with supply function given by Qs = 20P, where p is the price and Q is quantity.

Calculate equilibrium price and quantity

Solution:

Qd= (12-2P) 1000= 120,000- 20,000P

Qs= (20P) 1000 = 20,000P

Market equilibrium, Qd = Qs

120,000- 20,000p= 20,000p

120,000= 20,000p+ 20,000p

120,000= 40,000p, p= 120,000/40,000

P= 3

Qd= 120,000- 20,000P

Qd=120,000- 20,000X3

Qd= 120,000- 60,000

Qd= 60,000

Qs= 20,000P

Qs= 20,000X3

Qs= 60,000

71. In the above equation suppose the government decides to collect a sales tax of Birr 2 per unit.

Calculate the new market equilibrium and price.

Solution:

Sales tax= 2

Qs= 20,000p

Qs/ sale tax= 20,000p/2

Qs= 10,000p

At equilibrium Qd=Qs

120,000- 20,000p= 10,000p

120,000= 10,000p + 20,000p

120,000= 30,000p, p= 120,000/30,000

P= 4

Qd= 120,000- 20,000p

Qd= 120,000- 20,000x 4

Qd= 40,000

Qs= 10,000p

Qs= 10,000x4

Qs= 40,000

72. Consider the market for coffee suppose that the market demand for coffee is given by the equation Qd = 200 – 10P and the market supply for coffee is given by the equation Qs = 10P – 40, where Qd = quantity demand, Qs = quantity supplied, P= price.

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A. In equilibrium how much coffee would be sold and at what price.
B. What would happen if suppliers set the price of coffee at 16 Birr? What is the market adjustment process?

Solution:

A. At equilibrium, Qd = QS

$200 - 10P = 10P - 40$

$200 + 40 = 10P + 10P$

$240 = 20P$

$P = 12$

$Qd = 200 - 10P = 200 - 10(12)$

$Qd = 80$

$Qs = 10P - 40 = 10(12) - 40$

$Qs = 80$

At equilibrium 80 units of coffee would be sold at price of 12 Birr.

B. With P= 16, consumers would buy $Qd = 200 - 10(16) = 40$ units of coffee With P= 16, producers will supply $Qs = 10(16) - 40 = 120$ units of coffee. There will be an excess supply of coffee which would bring the price down. As the price decreased quantity supply decreased while quantity demand increase until both were equal to at price of 12 and quantity 80.

73. Only one of the following is true about the effect of change in supply and demand on equilibrium price. Which one is it?
   A. If the increase in supply is greater than the increase in demand, the equilibrium price rises.
   B. If the increase in demand is greater than the increase in supply, the equilibrium price rises.
   C. If the increase in demand is equal to the increase in supply, the equilibrium price rises.
   D. If the increase in demand is less than the increase in supply, the equilibrium prices rises.

***Explanation*:**

- If both demand and supply increases the equilibrium quantity would certainly rise. But the equilibrium price may rise, fall or remain unchanged.
- If the increase in demand is greater than the increase in supply, the equilibrium price rises.
- If the increase in demand is smaller than the increase in supply, the equilibrium price falls.
- If the increase in demand and increase in supply are equal, the equilibrium price remains unchanged.

Answer: B

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74. Individual demand and supply functions for a commodity are given respectively as Qd 100 – 20P and Qs = 40P – 120, where Q refers to the quantity and P price. If there are 20 buyers and 10 suppliers of the commodity, the equilibrium quantity and price are _____ and _____ respectively.

A. (40,40) B. (4,40) C. (400,4) D. (4,20)

***Solution*:**

$Qd = 100 - 20P$

$Qs = 40P - 120$

At equilibrium, $Qd = Qs$

$(100-20P)20 = (40P - 120)10$

$2000 - 40P = 400P - 1200$

$2000 + 1200 = 400P + 400P$

$3200 = 800P$

$P = 4$

$Qd = 2000 - 400p = 2000 - 400(4)$

$Qd = 400$

$Qs = 400P - 1200 = 400(4) - 1200$

$Qs = 400$

Answer: C

Question 76 and 77 re based on the following information:

A firm supply curve for oil is given by $Qs = 30p - 24$, where, Qs = the quantity of oil supplied in kilograms per day and, p = price of oil in Birr per kilogram. There are 10 firms in the oil industry and the price of oil on a specific day of the week was found to be Birr 10 per kilogram.

75. The quantity of oil supplied by the industry per day is_____

A. 2760 B. 276 C. 27600 D. 27.6

***Solution*:**

$Qs = 30p - 24$

$P = 10$ Birr per kilogram

$Qs = (30p - 24)10 = 300p - 240$

$300(10) - 240$

$Qs = 2760$

The quantity of oil supplied by the industry is 2760.

76. The quantity of oil supplied by an individual firm per day is_____

***Solution*:**

$Qs = 30p - 24, p = 10$

$Qs = 30(10) - 24$

$Qs = 300 - 24$

$Qs = 276$

The quantity of oil supplied by an individual firm per day is 276

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2.7 Price Controls:

Price floors and price ceilings are government mandated prices that attempt to control the price of a good or service

2.7.1 A price ceiling:

Price ceiling is usually imposed to keep down the price of something perceived as too expensive. To have any effect, it must be imposed below the market price.

A price ceiling is installed when consumers feel the market equilibrium price is "too high."

- A price ceiling creates a permanent shortage at a price below equilibrium.
- The more price elastic the demand and supply curves, the greater the shortage.
- The price ceiling reduces net benefit by under allocating resources to the production of the good.

2.7.2 Price floor:

Price floor is usually imposed to keep up the price of something perceived as too cheap. To have any effect, it must be set above the market price.

- A price floor is installed when producers feel the market equilibrium price is "too low."
- A price floor creates a permanent surplus at a price above equilibrium.
- If the government purchases the surplus, taxpayers eventually pay the bill.
- The more price elastic the demand and supply curves, the greater the surplus and the greater the government spending to purchase the surplus.
- The price floor reduces net benefit by over allocating resources to the production of the good

Illustrative Examples

77. When a price floor is above the equilibrium price,
    A. quantity demanded will exceed quantity supplied.
    B. quantity supplied will exceed quantity demanded.
    C. the market will be in equilibrium.
    D. quantity demand equals with quantity supply.

Explanation: A price set above equilibrium price causes a surplus. Surplus occurs when quantity supplied exceeds quantity demand (QS > QD)

Answer: B

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78. Suppose the government of Ethiopia sets a ceiling on the price of oil. Which of the following situations could result?
   A. People pay less than the equilibrium price for oil.
   B. The quantity of oil demanded exceeds the quantity of oil supplied.
   C. A shortage exists in the oil market.
   D. All of the above.

***Explanation*:**
- Price ceiling sets below equilibrium price and makes people to pay less than the equilibrium price
- Below equilibrium price quantity demand exceeds quantity supplied
- Price ceiling causes shortage

Answer: D

79. An effective price ceiling in the market for good *X* likely results in
   A. a persistent surplus of good *X*.
   B. a persistent shortage of good *X*.
   C. an increase in the demand for good *Y*, a substitute for good *X*.
   D. a decrease in the demand for good *Z*, a complement with good *X*.

***Explanation*:** Price ceiling cause persistent shortage which in turn:
- cause rationing
- black market
- reduce quality of goods and services

Answer: B

80. A price ceiling is
   A. a maximum legal price
   B. the only legal price
   C. a minimum legal price
   D. the legal price of housing

***Explanation*:**
Price ceiling: a government price control that sets the maximum allowable price for a good.

Answer: A

81. Dairy price supports (floor) which raise milk prices received by farmers are likely to:
   A. lowers the price of milk
   B. result in shortages of milk
   C. helps consumers
   D. cause increased production of milk

***Explanation*:**
- Price floor is the minimum price set by the government to support producers.

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- When the government fixes a price floor for dairy products, milk production increases.

Answer: D

82. Price ceilings and price floors are usually intended to benefit:
    A. government by increasing government revenue
    B. buyers (ceilings) and sellers (floors)
    C. buyer's D. sellers

***Explanation*:**
Price ceiling is applied to support consumers and price floor is set to support producers.

Answer: B

83. A price floor
    A. is set below the equilibrium price for the product
    B. is set above the equilibrium price for the product
    C. c h a n g e s the equilibrium quantity for the product
    D. shifts the supply and demand curves for the product

***Explanation*:** A price floor is the minimum price which is set above the equilibrium price for the product

Answer: B

84. An effective price floor would cause:
    A. a price lower than equilibrium
    B. a surplus
    C. a shortage
    D. a price greater than equilibrium price

***Explanation*:**
- Price floor cause a persistent surplus, with the quantity supplied greater than the quantity demand.
- In the case of price floors on agricultural products, the surplus are bought by the government and put in storage.

Answer: B

85. The normal reason given for a price floor is:
    A. To help low income consumers.
    B. To help low cost producers
    C. To help importers
    D. To help high cost producers

***Explanation*:** Price floor are imposed by the government in order to help the suppliers of goods and services.

Answer: D

86. When government intervenes in a competitive market by imposing an effective price ceiling, we would expect the quantity supplied to and the quantity demanded to.

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A. fall; rise
B. fall; fall
C. rise; rise
D. rise; fall

***Explanation*:**

- The fact that the price ceiling is effective implies that the government is forcing the maximum price to be lower than the equilibrium price. Thus, we would expect the quantity supplied to fall and the quantity demanded to rise

Answer: A

87. Which of the following is true of a price floor?
A. The price floor shifts the demand curve to the left.
B. An effective floor creates a shortage of the good.
C. The price floor shifts the supply curve of the good to the right.
D. To be an effective floor, it must be set above the equilibrium price.

***Explanation*:**

- Price floors are legal minimum prices so they are set above equilibrium.
- A surplus result.

Answer: D

2.8 Elasticity:

Elasticity is a measure of just how much the quantity demanded will be affected by a change in price or income or change in price of related goods.

2.8.1 Price Elasticity of Demand:

***The* price elasticity of demand** measures the magnitude by which consumers change the quantity demanded in response to a change in the price of the product. The more elastic demand is, the more responsive it is to price changes.

$$\textbf{Price elasticity of demand} = \frac{\textit{Percentage change in quantity demanded}}{\textit{Percentage change in price}}$$

Point-price elasticity-it measures elasticity at a point on a demand curve.

$$\textbf{Ed} = \% \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Example: 11. Suppose the price of a certain good fell from \$1.00 to \$0.50 and, as a result, the quantity demanded increased from 500 to 750 units. Calculate price elasticity of demand

***Solution*:**

$$\textbf{Ed} = \% \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\frac{750-500}{0.5-1} \times \frac{1}{500}$$

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$^{e}d = 1$

B. Over this range, the demand curve is:

The demand curve is unit-elastic

Arc elasticity- it measures price elasticity of demand between two points.

$$Ed = \% \frac{\%\Delta Q}{\%\Delta P} = \frac{\Delta Q / Q_0}{\Delta P / P_0} = \frac{(Q_2 - Q_1)P_1 + P_2}{(P_2 - P_1)Q_1 + Q_2}$$

Suppose the price of a product increases from Birr 1000 to 1200. As a result quantity demand decreases from 2000 to 1500 units. Calculate the price elasticity of demand.

***Solution*:**

Q1= 2000, P1= 1000

Q2= 1500 P2= 1200

$$Ed = \% \frac{\%\Delta Q}{\%\Delta P} = \frac{(Q_2 - Q_1)P_1 + P_2}{(P_2 - P_1)Q_1 + Q_2}$$

$$= \frac{1500 - 2000}{1200 - 1000} \times \frac{1200 + 1000}{1500 - 2000}$$

= 500/200X 2200/3500

= 11/7

2.8.2 Types of elasticity:

Relatively inelastic when the percentage change in quantity demanded is *less than* the percentage change in price (so that Ed < 1). If *E*d < 1, demand is said to be **"price inelastic"** for good *X*. The initial change in the price exceeded, in percentage terms, the responsiveness of the consumer

Unitarily elastic demand when the percentage change in quantity demanded is *equal to* the percentage change in price (so that Ed = 1). If demand is said to be **"unit elastic"** for good *X*.

The initial change in the price is exactly equal to, in percentage terms, the responsiveness of the consumer.

Relatively elastic when the percentage change in quantity demanded is *greater than* the percentage change in price (so that Ed > 1). If Ed >1, demand is said to be "**price elastic**" for good *X*. The responsiveness of the consumer exceeded, in percentage terms, the initial change in the price

2.8.3 Determinant of price elasticity of demand:

- **The number of close substitutes** – the more close substitutes there are in the market, the more elastic is demand because consumers find it easy to switch

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- **The degree of necessity or whether the good is a luxury** – necessities tend to have an inelastic demand whereas luxuries tend to have a more elastic demand.
- **The proportion of a consumer's income allocated to spending on the good** – products that take up a high percentage of income will have a more elastic demand
- **The time period allowed following a price change** – It is usually easier to find substitutes for an item in the long run than in the short run. Therefore, demand tends to be more elastic in the long run than in the short run.
- **Whether the good is subject to habitual consumption** – consumers become less sensitive to the price of the good of they buy something out of habit (it has become the default choice
- **Peak and off-peak demand** - demand is price inelastic at peak times and more elastic at off-peak times

Illustrative Examples

88. Suppose a grocery store normally sells 100 cartons of milk per day and the price elasticity of demand for milk is 1.7. If the store lowers the price of milk by 10%, about how many cartons of milk will it then sell per week?

***Solution*:**

ed=$\Delta Q/\Delta P$

$$ed = \frac{Q_2 - Q_1}{\Delta P} \div \frac{Q_2 - 100}{10} = 1.7$$

$Q_2 - 100 = 10 \times 1.7$

$Q_2 = 17 + 100$

$Q_2 = 117$

89. Suppose the demand curve for agricultural products is inelastic, then
    A. the demand curve will be relatively steep
    B. buyers are relatively unresponsive to changes in price
    C. total farm income will decrease if the market price falls
    D. all of the above

***Explanation*:**

- If the Price elasticity of demand is less than 1, demand is inelastic
- A low coefficient implies that changes in price have little influence on demand
- Total revenue decreases when price falls if the price elasticity of demand is inelastic.
- The demand curve for inelastic demand is relatively steep.

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Answer: D

Question 91 based on the following scenario:

A 4% increase in the price of tomatoes leads to a 1% reduction in the quantity of tomatoes demanded.

90. The price elasticity of demand for tomatoes is:

A. 0.5 B. 0.6 C. 0.25 D. 1.25

***Solution*:**

$ed = \Delta Q / \Delta P$

$ed = 1\%/4\%$, $ed = 0.25$

91. Which of the following is not a determinant of the price elasticity of demand?

A. the price elasticity of supply
B. whether the product is a necessity
C. whether the product is a luxury
D. the time period in question

***Explanation*:**

Some of the determinants of price elasticity of demand are:

- The availability of substitutes
- The number of use of a commodity
- The nature of the product- necessity or luxury
- Time – short period or long period of time

So the price elasticity of supply is not the determinant of price elasticity of demand.

Answer: A

92. If the percentage change in quantity demanded for a product is smaller than the percentage change in price, then demand for the good is

A. infinitely elastic C. perfectly inelastic
B. of unitary elasticity D. inelastic

***Explanation*:** If the percentage change in the quantity demanded is less than the percentage change in price, then the magnitude of the elasticity of demand is between zero and 1, and demand is said to be inelastic.

Answer: D

93. Price elasticity of demand measures:

A. the change in quantity supplied to the change in price
B. the extent to which a demand curve shifts from the change in an exogenous variable
C. the response between two goods when the price of one good changes
D. consumer responsiveness of quantity demand to price changes

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***Explanation*:**

- **Price elasticity of demand**: A measure of the responsiveness of the quantity demanded of a good to a change in its price, other things remaining the same.
- It is the percentage change in demand divided by percentage change in price.

Answer: D

94. When the price of a certain good increases by 30%, quantity demanded decreases by 2%. What is the price elasticity of demand?

***Solution*:**

ed= 2%/30%

ed= 1/15

95. The price elasticity of demand for a good is larger when
    A. the good has more substitute goods
    B. the good is relatively inexpensive
    C. the good is more of a necessity
    D. the time period is shortened

***Explanation*:**

- The price elasticity of demand is affected by existence of substitutes.
- The more substitutes available for a product, the more elastic its demand
- The demand for necessity is less elastic
- The shorter the time period, the more inelastic.

Answer: A

96. At a price of Birr 20, a store can sell 24 picture frames a day. If the price falls to Birr 18, the store can sell 36 picture frames a day. Calculate the price elasticity of demand.

***Solution*:**

Given: $Q_1 = 24$, $Q_2 = 36$

$P_1 = 20$, $P_2 = 18$

$$Ed = \frac{36-24}{18-20} \times \frac{20}{24} = \frac{12}{-2} \times \frac{20}{24}$$

ed= 5

97. The XYZ store knows that the price elasticity of demand for X is 2.0. If the price to increase by 10%:
    A. Quantity demanded of X would rise by 20%
    B. Quantity demanded of X would fall by 20%
    C. Quantity demanded of X would rise by 5%
    D. Quantity demanded of X would fall by 5%

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***Solution*:**

ed=*ΔQ/ΔP*

2= ΔQ/ 10%, ΔQ= 2x 10%

ΔQ= 20%

Answer: B

98. Given greater availability of substitutes for coca-cola what will happen to demand elasticity.
    A. It will increase
    B. It will not be affected
    C. It will decrease
    D. Cannot be determined

***Explanation*:**

- If a commodity has close substitute its demand tends to be more elastic.
- The greater the possibility of substitution, the greater the price elasticity of demand

Answer: A

99. If a 10% change in price causes a 5% change in quantity, we would say that demand is:
    A. Elastic
    B. Unit elastic
    C. inelastic
    D. perfectly inelastic

***Solution*:**

ed=*ΔQ/ΔP*

ed= 5%/ 10%

ed= 0.5%

Elasticity of demand between zero and one has an inelastic demand.

Answer: C

100. The quantity of a good demanded rises from 1000 to 1500 units when the price falls from 1.50 to 1.00 per unit. The price elasticity of demand for this product is approximately:

***Solution*:**

$$ed = \Delta Q / \Delta P \times P/Q$$

$$ed = \frac{1500 = 100}{1-15} \times \frac{1.5}{1}$$

ed= 3/2 =

101. A long-run demand curve, as compared to a short-run demand curve for the same commodity, is generally:
    A. more elastic
    B. less elastic
    C. of the same elasticity

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D. steeper if the curves are plotted against the same horizontal scale.

***Explanation*:**

- The longer the time period the greater the elasticity of demand for commodities. This is because consumers can substitute goods in the long run.
- Demand is more elastic in the long run than in the short run.

Answer: A

102. Perfectly inelastic demand indicates:

A. A horizontal curve showing different quantities demanded at a constant price.
B. An upright curve showing quantity changes little with a large movement in price.
C. A vertical curve showing a constant quantity demanded at different prices.
D. a flatter curve showing a large decrease in quantity demanded with a small increase in price.

***Explanation*:**

- If Price elasticity of demand is zero(ed = 0) demand is **perfectly inelastic**
- demand does not change at all when the price changes
- the demand curve will be vertical

Answer: C

103. If demand is price elastic,

A. a 1 percent decrease in the price leads to an increase in the quantity demanded that exceeds 1 percent.
B. a 1 percent increase in the price leads to an increase in the quantity demanded that exceeds 1 percent.
C. the price is very sensitive to any shift of the supply curve.
D. a 1 percent decrease in the price leads to a decrease in the quantity demanded that is less than 1 percent.

***Explanation*:** If Price elasticity of demand is greater than one(**ed > 1**), then demand responds more than proportionately to a change in price.

Answer: A

104. The more substitutes available for a product,

A. the larger is its income elasticity of demand.
B. the smaller is its income elasticity of demand.
C. the smaller is its price elasticity of demand.
D. the larger is it's the price elasticity of demand

***Explanation*:**

- the more close substitutes there are in the market, the more elastic in demand because consumers find it easy to switch

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- The more substitutes are available for a product, the larger the price elasticity of demand.

Answer: D

105. If the price of rose flower and the revenues from flower production increase, we know that the price elasticity of demand for rose flower is:

A. Greater than one
B. less than one
C. equal to one
D. equal to zero

***Explanation*:**

- If a fall in price of a commodity causes total revenue to increase, demand for the good is elastic(ed >1)
- If a fall in price of a commodity causes total expenditure on a commodity to decrease, demand for the commodity is inelastic(ed< 1)

An increase in the price of rose flower increases total revenue if the demand for rose flower is inelastic(less than one)

Answer: B

107. If a demand for a product is said to be "elastic", the value of the elasticity coefficient is:

A. zero
B. greater than one
C. equal to one
D. less than one

***Explanation*:**

- An elastic demand have the value of elasticity coefficient > +1.

Answer: B

108. An elasticity coefficient of -0.1 means that

A. the demand curve is perfectly inelastic.
B. the demand curve is perfectly elastic.
C. the relative changes in price and quantity are equal.
D. expenditures on the good would increase if prices were reduced.

***Explanation*:**

- If the coefficient of elasticity of demand is a number less than one, we say demand is inelastic.
- Demand increase when price reduces and decrease when price rises.

Answer: D

109. Which of the following will **not** be a determinant of the price elasticity of demand for a commodity?

A. The absence of substitute for the good.
B. The presence of substitutes for the good.
C. The length of time period to which the demand curve pertains.
D. The cost of producing the commodity.

***Explanation*:** Determinants price elasticity of demand is:-

- Availability of substitute

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- The amount of time involved
- Whether the product is a luxury or a necessity
- The proportion of price relative to income

The cost of producing the commodity is not the determinant of price elasticity of demand.

Answer: D

110. The price elasticity of demand will increase with the length of the period to which the demand curve pertains because:
   A. consumers' incomes will increase
   B. the demand curve will shift outward.
   C. all prices will increase over time.
   D. consumers will be better able to find substitutes.

***Explanation*:**

- Demand is more price elastic, the longer those consumers have to respond to a price change.
- They have more time to search for cheaper substitutes and switch their spending.

Answer: D

111. How does the nature of a commodity influence its price elasticity of demand?

***Explanation*:**

A commodity for a person may be a necessity, a comfort or a luxury.

- When a commodity is a necessity its demand is generally inelastic.
- When a commodity is a comfort its demand is generally elastic.
- When a commodity is a luxury its demand is generally more elastic that the demand for comforts.

112. Which of the following **cannot** be a characteristic of a product with inelastic demand?
   A. It has a large number of close substitutes
   B. Buyers spend a small percentage of their income on it
   C. Consumers regard the good as a necessity
   D. The period of time for which elasticity is calculated is very short

***Explanation*:**

- If no close substitutes are available, the substitution effect will be small and the demand is inelastic.
- When the goods represent only a negligible portion of the budget the income effect will be insignificant and demand is inelastic.
- The more necessary a good is, the lower the elasticity
- The longer a price change holds, the higher the elasticity of demand.

Answer: A

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113. Suppose the price of beef rises by 10 percent and the quantity demanded of beef falls by 20 percent. We can conclude that

A. Demand for beef is price elastic and consumer spending on beef is falling.
B. Demand for beef is price elastic and consumer spending on beef is rising.
C. Demand for beef is price inelastic and consumer spending on beef is falling.
D. Demand for beef is price inelastic and consumer spending on beef is rising.

***Explanation*:** If the percentage change in *Q*d is greater than the percentage change in price, the good is elastic.
In this situation of rising prices, total spending on beef will fall because the upward effect of prices is outweighed by the downward effect of quantity.

Answer: A

114. Without making an adjustment such as finding the absolute values of the percentage changes, the price elasticity of demand would be negative because

A. price and demand are directly related
B. price and demand are inversely related
C. price and quantity demanded are inversely related
D. price and quantity demanded are directly related

***Explanation*:**

- As the price of good increases, the quantity demanded decreases and vice versa.
- Because the relationship is always negative, we normally concentrate on the absolute values of numbers.

Answer: C

115. If a firm raises the price of its product, its total revenue will

A. always increase
B. increase only if demand is price inelastic
C. increase only if demand is price elastic
D. remain constant, regardless of price

***Explanation*:** If a firm increases the price of its price, its total revenue will increase if only if demand is price inelastic. In this case total revenue increases because the percent change in quantity sold is more than the percent change in price. **Answer: B**

116. Melaku spends exactly the same amount of candy bars each week, regardless of their price. Melaku's demand curve for candy bars is

A. upward-sloping
B. backward-bending
C. perfectly inelastic
D. perfectly inelastic

***Explanation*:** A perfectly inelastic demand curve is perfectly vertical. This means that a change in price has no effect on the quantity demanded.

Answer: C

117. Economists distinguish between normal and inferior goods using:

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A. price elasticity of demand
B. price elasticity of supply
C. income elasticity of demand
D. tax incidence

Explanation:

- A normal good is one that consumers will purchase more of as their incomes increase.
- An inferior good is one that consumers will buy less of as their incomes increase.
- Economists distinguish between normal and inferior goods by using income elasticity of demand.

Answer: C

118. Luxury goods are usually

A. price inelastic
B. income inelastic
C. income elastic
D. goods with negative income elasticity

Explanation: A luxury good is one that consumers can live without. So as the price of luxury goods increase, there is a large effect on the quantity demanded as consumers choose to buy less of the good.

Answer: C

2.8.4 Effect on total revenue

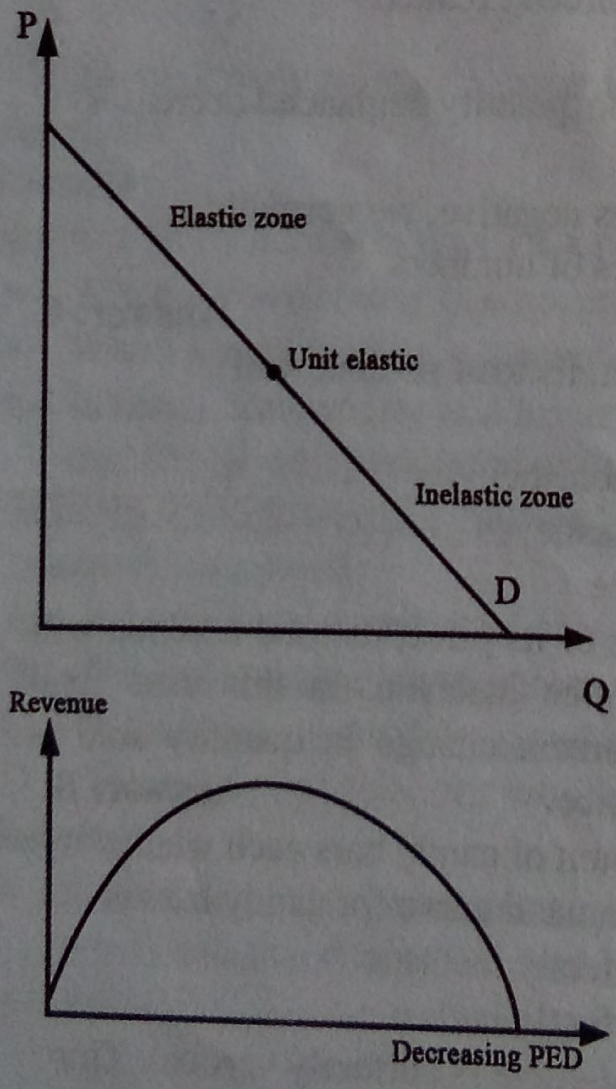

Figure 2.8: effect on total revenue.

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As shown in the above diagram as price decreases in the elastic range, TR increases, but in the inelastic range, TR decreases. TR is maximized at the quantity where PED = 1.

Elastic demand and the total-revenue test: Demand is elastic if a decrease in price results in a rise in total revenue, or if an increase in price results in a decline in total revenue. (Price and revenue move in opposite directions).

Inelastic demand and the total-revenue test: Demand is inelastic if a decrease in price results in a fall in total revenue, or an increase in price results in a rise in total revenue. (Price and revenue move in same direction).

Unit elasticity and the total-revenue test: Demand has unit elasticity if total revenue does not change when the price changes.

Note:

- Inelastic demand $Ed < 1$: $\% \Delta Q < \% \Delta P$ so Total Revenue increases with a price increase.
- Elastic demand $Ed > 1$: $\% \Delta Qd > \% \Delta P$ so Total Revenue decreases with a price increase.
- Unit elastic demand $Ed = 1$: $\% \Delta Qd = \% \Delta P$ so Total Revenue remains the same.

Total Revenue Test:

- **If P↑ and TR↑ => inelastic D**
- **If P↓ and TR↓ => inelastic D**
- **If P↑ and TR↓ => elastic D**
- **If P↓ and TR↑ => elastic D**

Illustrative Examples

119. How does elasticity affect whether a company should raise or lower prices?
   A. Raise prices if demand is elastic
   B. Raise prices if demand is unit elastic
   C. Raise prices if demand is inelastic
   D. Raise prices if demand is perfectly elastic

***Explanation*:** When a demand for a good is inelastic (ep < 1), a raise in price raises total revenue and a fall in price reduces total revenue. Therefore, a company raise price if demand is inelastic.

Answer: C

120. Producers' total revenue will decrease if:
   A. The price rises and demand is inelastic
   B. Income increase and the good is normal good

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C. The price rises and demand is elastic
D. Income falls and the good is inferior good.

***Explanation*:**

- When demand for a good is elastic (ep >1), a raise in price reduces total revenue and a fall in price raises total revenue.
- Producers' revenue will decrease if the price increases and demand is elastic.

Answer: C

121. The demand for a good is elastic if:
A. A decrease in its price results in a decrease in total revenue.
B. The good is a necessity
C. An increase in its price results in an increase in total revenue
D. An increase in its price results in a decrease in total revenue.

***Explanation*:**

- Demand is elastic if a decrease in price results in a rise in total revenue, or if an increase in price results in a decline in total revenue. (Price and revenue move in opposite directions)

Answer: D

122. If the price elasticity of demand for a product equals one, as its price rises the:
A. total revenue increases
B. quantity demand does not change
C. total revenue does not change
D. quantity demanded increases.

***Explanation*:**

- When demand for a commodity is unit-elastic (ep= 1) total revenue does not change when the price changes.

Answer: C

1203. A rise in the price of a product lowers the total revenue from the product if the
A. Good is an inferior product.
B. Demand for the product is inelastic.
C. Demand for the product is elastic.
D. Income elasticity of demand exceeds 1.

***Explanation*:** If demand is elastic, a rise in price will cause quantity demand to decrease by a greater percentage than the percentage increase in price, and total revenue will fall.

Answer: C

124. What happens to the total revenue of a firm if the firm increases the price of its product that has price inelastic demand?
A. Total revenue remains elastic C. Total revenue increases

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B. Total revenue remains unchanged D. Total revenue decreases

***Explanation*:** When demand is inelastic, a cut in price reduces total revenue and an increase in price raises total revenue.

Answer: C

125. Suppose price of a good falls from Birr 12 to Birr 10 per unit. As a result its quantity demand increases from 100 t0 120 units. Tell whether the demand for a good is elastic, inelastic or unit elastic.

***Solution*:**

At price 12, quantity demand is 100
Total expenditure = PxQ= 12x100
TR= 1200
At rice 10, quantity demand is 120
TR= PxQ= 10x120
TR= 1200

- Total expenditure (TR) on the commodity remains the same.
- Price elasticity of demand is unit elastic.

126. Suppose price of a commodity rises from Birr 18 to Birr 19 per unit. As a result, its quantity demand falls from 80 units to 60 units. Find out the price elasticity of demand.

***Solution*:**

TR at price 18 is
TR= PxQ= 18x 80
TR= 1440
TR at price 19 is
TR= PxQ= 19x60
TR= 1140

- Thus at the rise in price total revenue decreases
- The price elasticity of demand is elastic (ep >1)

127. A small business estimates price elasticity for the product to be 3. To raise total revenue, owners should

A. Decrease price as demand is elastic.
B. Decrease price as demand is inelastic.
C. Increase price as demand is elastic.
D. Increase price as demand is inelastic.

***Explanation*:** If *E*d > 1, the firm increases total revenue by decreasing the price because the quantity demanded rises by a greater percentage than the fall in price. **Answer: A**

2.8.5 Income elasticity of demand:

Income elasticity of demand tells us the relationship a product's quantity demanded and income.

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It measures the sensitivity of quantity demand change of product X to a change in income

$$EY = \frac{\%\Delta Qd}{\%\Delta Y} = \frac{Y}{Q} =$$

Example: Q12. Suppose consumer's income rises from Birr 500 to Birr 550, his purchase of a good A increase from 25 units per month to 50 units. Calculate the income elasticity of demand for good A.

***Solution*:**

$$EY = \frac{\%\Delta Qd}{\%\Delta Y} = \frac{Y}{Q}$$

$$\frac{50-25}{550-500} \times \frac{550+500}{50+25}$$

$$\frac{25}{50} \times \frac{1050}{75} = 75$$

Types of Income elasticity of demand:

- A negative income elasticity of demand (Ey < 0) is associated with inferior goods; an increase in income will lead to a fall in the demand and may lead to changes to more luxurious substitutes.
- A positive income elasticity of demand is associated with normal goods; an increase in income will lead to a rise in demand.
- If income elasticity of demand of a commodity is less than 1(0< Ey<1), it is a necessity good.
- If the elasticity of demand is greater than 1(Ey > 1), it is a luxury good or a superior good.
- A zero income elasticity of demand (Ey=0) occurs when an increase in income is not associated with a change in the demand of a good. These would be sticky goods.

Note:

- ***If E I > 1, the good is normal and income elastic (a luxury)***
- ***If 1> EI > 0, the good is normal but income in elastic (a necessity)***
- ***If EI < 0, the good is inferior.***

Illustrative Examples

128. A negative income elasticity of demand indicates that
   A. the good is inferior
   B. t h e good is a luxury item
   C. t h e good has an inelastic demand curve

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D. the good is normal

***Explanation*:**

- A negative income elasticity of demand is associated with inferior goods.
- an increase in income will lead to a fall in the demand and may lead to changes to more luxurious substitutes

Answer: A

129. If the income elasticity of demand for a good is greater than 0 and less than, one then the good is:

A. An economic luxury
B. An economic necessity
C. One with relatively little demand
D. Inferior good

***Explanation*:**

- Necessities have an income elasticity of demand between zero and one.
- Expenditure on these goods may increase with income, but not as fast as income does
- So the proportion of expenditure on these goods falls as income rises.

Answer: B

130. For which of the following goods would income elasticity of demand be positive and relatively large

A. Inferior
B. normal
C. luxury
D. necessities

***Explanation*:**

- Luxury goods have an income elasticity of demand > + 1 i.e., demand rises more than proportionate to a change in income. In this case increase in income is accompanied by relatively larger increase in quantity demanded.

Answer: C

131. The income elasticity of demand is high for

A. Shelter. B. Luxuries. C. Clothing. D. Food

***Explanation*:**

- Shelter, clothing and food are necessities.
- Luxury items usually have higher income elasticity because when people have a higher income, they don't have to forfeit as much to buy these luxury items.
- A good having income elasticity more than one and which therefore bulks larger in consumer's income as become richer is luxury.

Answer: B

132. If the income elasticity of demand is +4

A. the good is an inferior good.
B. the good is an inelastic normal good.
C. the good is an elastic normal good.

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D. the good is an elastic inferior good.

***Explanation*:**

- A positive income elasticity of demand is associated with normal goods; an increase in income will lead to a rise in demand.
- Income elasticity of a good greater than one is elastic. So the good is normal elastic.

Answer: C

133. Luxuries are distinguished from necessities by

A. the number of substitutes available for each.
B. the fact that luxuries have high prices and necessities have low ones.
C. the high price elasticity of demand for luxuries and the low price elasticity of demand for necessities.
D. the high income elasticity of demand for luxuries and the low income elasticity of demand for necessities.

***Explanation*:**

- Luxury goods have high income elasticity of demand. In this case increase in income is accompanied by relatively larger increase in quantity demanded.
- Necessity goods have low income elasticity of demand. In this case proportionate increase in income is accompanied by less than increase in quantity demanded.

Answer: C

134. If a good's income-elasticity-of-demand estimate equal.

A. 3.12, an economist would call the good a necessity.
B. 0.25, an economist would call the good a luxury.
C. -0.66, an economist would call the good an inferior one.
D. -0.50, an economist would call the good a complementary one.

***Explanation*:**

- **Necessities** have an income elasticity of demand of between 0 and +1.
- **Luxuries** on the other hand are said to have an income elasticity of demand > +1.
- **Inferior goods** have a negative income elasticity of demand.

Complementary and substitute goods are refer to with the effect of cross elasticity.

Answer: C

135. If the income elasticity is negative what does this tell us? Explain.

***Explanation*:**

- If the income elasticity of demand is negative it tells us that consumers' purchases go in the opposite direction of an change in income.
- For example if income goes up, purchases go down.

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- Therefore this must be an inferior good.

136. When Sara income is Birr 2000, she purchases 100 Kg of a good. When her income increase to Birr 2400, she purchases 120 Kg of a good.

A. What is her income elasticity?

Solution:

$$eI = \frac{change\ in\ quantity\ demnd}{change\ income} = \frac{income}{quantity}\ \text{change in}$$

$$EY = \frac{\%\Delta Qd}{\%\Delta Y} = \frac{Y}{d}$$

120-100/2400-2000

20/400

eI= 0.05

B. What type of good is?

Explanation:

Necessities have an income elasticity of demand of between 0 and +1. Demand rises with income, but less than proportionately. So the good is necessary.

137. What is zero income elasticity of demand mean?

Explanation:

- Zero income elasticity of demand for a good implies that a given increase in income does not at all lead to an increase in quantity demand of a good.
- Zero income elasticity signifies the quantity demand of the good is quite unresponsive to change in income.

2.8.6 Cross elasticity of demand:

Cross elasticity of demand is a measure of how much the quantity demanded of one good responds to a change in the price of another good.

$$E_{A,B} = \frac{\%\text{change in quantity demand of product A}}{\%\text{change in price of product B}}$$

$$E_{A,B} = \frac{PB_1 + PB_2}{QA_1 + QA_2} \times \frac{\Delta QA}{\Delta QB}$$

- If cross elasticity is positive, then A and B are substitutes.
- If cross elasticity is negative, then A and B are complements.
- If cross elasticity is zero, then A and B are unrelated, independent products.

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Example: 1. If the price of coffee rises from Birr 35 per kg to Birr 55 per kg as a result the consumers demand for tea increase from 400 units to 600 units. Calculate the cross elasticity of tea for coffee.

***Solution*:**

Change in quantity demand= Qt2-Qt1 = 600-400 of tea

Change in price of coffee= Pc2- Pc1= 55-35

$$Etc = \frac{600-400}{55-35} \times \frac{35}{400}$$

Example.2: If the cross price elasticity of demand is -0.65, what would the percentage change in quantity if the price of the other good decreased by three percent? Are these goods substitutes or complements

***Solution*:**

$= \Delta Qx/\Delta Py$

$0.65 = \Delta Q/3\%$, $\Delta Q = 0.65x3\%$

$\Delta Q = 2\%$

Illustrative Examples

138. If goods R and K have a cross elasticity of -5 and goods R and S have a cross elasticity of 5,
   A. R and K are substitutes; R and S are complements
   B. R and K are complements; R and S are substitutes
   C. K is price inelastic
   D. S is price inelastic

***Explanation*:**

- If the cross elasticity of demand is positive, then the two goods are substitute
- If the cross elasticity is negative, then the two goods are complements
- If the cross elasticity is zero, the two goods are unrelated.

So Goods R and K are complements and Goods R and S are substitutes.

Answer: B

139. If a rise in the price of good 1, decreases the quantity of good 2 demanded,
   A. The cross elasticity of demand is negative.
   B. Good 1 is an inferior good.
   C. Good 2 is an inferior good.
   D. The cross elasticity of demand is positive

***Explanation*:**

- Complement good is a good that tends to be consumed together with another good.

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- The demand for good increases if the price of a complement decreases and the demand for good decreases if the price of a complement increases.
- Complementary goods have negative cross elasticity

Answer: A

140. Cross-price elasticity measures whether:
   A. goods are normal or inferior.
   B. Two goods are substitutes or complements.
   C. demand is elastic or inelastic.
   D. supply is steeper than demand or vice versa.

***Explanation*:**

- The concept of cross elasticity of demand is important to determine whether the good is substitute or complement.
- The good which cross elasticity is positive is substitute good
- The good which cross elasticity of demand is negative is complementary good.

Answer: B

141. Some goods are not closely related to each other and are neither substitutes nor complements. For such goods, the cross-price elasticity of demand would be
   A. positive.
   B. negative.
   C. zero
   D. Cannot tell without more information

***Explanation*:**

Unrelated products have a zero cross elasticity.

Answer: C

142. What does Cross elasticity of demand mean?

***Explanation*:**

- If the cross-elasticity of demand is negative that means that consumers purchases of some good, say good A, go in the opposite direction on a price change in some other good, say good B.
- If the price of good B goes up consumers will purchase less of it , and also *less of its complement.*
- Therefore, if the cross-elasticity of demand is negative, the two goods must be complements

142. What will happen to the demand for sugar when the price of cinema admissions changes, ceteris paribus?

***Explanation*:**

- Nothing happens between the two goods, because sugar and cinema are unrelated goods.
- If the cross price elasticity of demand is zero then the two goods in question will be totally unrelated (or independent)

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143. Suppose that you know that elasticity of your product is 1.5 and the cross price elasticity with a competitor is +1.25; and the income elasticity is 1.8.

A. Interpret each of these values in terms of elasticity, and percentage effects

***Explanation*:**

- Elasticity of 1.5 means that the product is elastic, a 1% change in price will cause the quantity demanded to decrease by 1.5%
- Cross elasticity of +1.25 is positive, the products are considered substitutes. This values means that 1% change in our competitors price would cause our sales to decrease by 1.25%
- Income elasticity of 1.8 means that the product is a normal, elastic good and probably viewed as a luxury.

A value of 1.8 means that as consumer income goes up by 1%, our sales will increase by1.8%.

B. The company is considering raising prices to earn more money. What is your advice?

- Since the price elasticity is 1.5, they should NOT raise their price. As noted above, if they raise prices by 1%, they will lose 1.5% of their sales and their total revenue will decline.

C. Does the cross price elasticity indicate that your customers are loyal or just seeking the best price (explain).

- The cross price elasticity of +1.25% indicates that if I change my prices by 1%, then my competitor's sales may increase by +1.25% (and vice versa). Therefore, it appears that my customers are very willing to switch if the price is right.

144. Briefly discuss the practical significance of cross price elasticity of demand

***Explanation*:**

- The cross price elasticity shows the responsiveness of quantity demanded for a good to the change in the price of another good.
- Using cross price elasticity we can easily identify what type of goods we are dealing with.
- When the sign of the calculated cross price elasticity is positive, two goods are substitutes; when the sign is negative two goods are complements.

145. If the price of firm A's long-distance service rises by 5 percent and the quantity demanded for firm B's long-distance service increases by 10 percent, we can say that

A. Demand for firm B is price elastic.

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B. Supply for firm B is price elastic.
C. Firms A and B are substitutes because the cross-price elasticity is greater than zero.
D. Firms A and B are complements because the cross-price elasticity is less than zero.

Explanation: The cross-price elasticity measures how sensitive the *Q*d of good *X* is to a change in the price of good *Y*. If this elasticity is greater than zero, the two goods are substitutes and if it is negative, the two goods are complements. **Answer: C**

2.9 Price Elasticity of Supply:

Price elasticity of supply measures the relationship between change in quantity supplied and a change in price.

The value of elasticity of supply is **positive**, because an increase in price is likely to increase the quantity supplied to the market and vice versa.

Price elasticity of supply measures the responsiveness of quantity supplied to changes in price

The formula for measuring price elasticity of supply is:

$$es = \frac{\%\ \Delta\ In\ quanity\ \text{sup}\ plied}{\%\Delta in\ \text{Pr}\ ice}$$

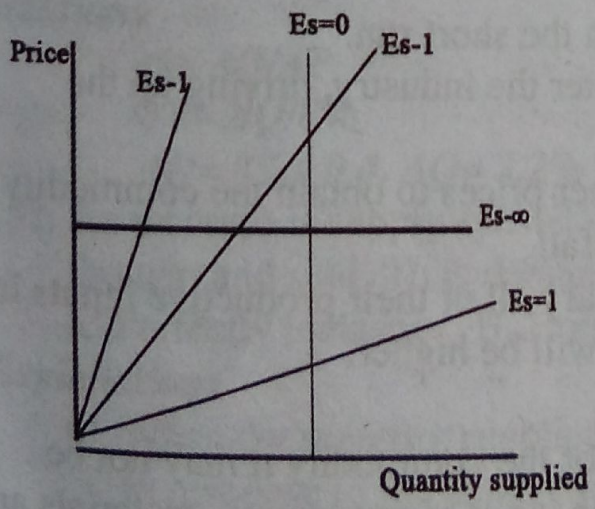
Example: when the price of commodity changes from Birr 6 to Birr 8. Quantity supplied changes from 60 to 80. Calculate the price elasticity of supply.

Solution:

$$Es = \frac{\%\ \Delta Q}{\%\Delta P} \times \frac{P}{Q}$$

Es = 80-60/8-6X6/60

Es = 6



1. Es=0 - perfectly inelastic supply - no response in supply to the change in price - quantity supplied of a commidity remains the same whatevet its price
2. Es<1 - inelastic supply
3. Es-1 unitary elastic supply % change in quantity supplied is the same as the % change in price
4. Es>1 - elastic supply
5. Es=∞ - perfectly elastic supply - producers are willing to supply as much as they can at one particular price and supply nothing at any other price

Figure 2.10: Types of elasticity of supply.

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2.9.1 Determinants of elasticity of supply:

1. The availability of resources (substitutes)

- If a product has many substitutes, then producers can easily alter the pattern of production if its price rises or fall. Its elasticity will be relatively high.
- If producer have no alternative to substitute goods and it is difficult to change the pattern of production if the prices changed, the price elasticity of supply will be low.

2. Time: The more time a producer has to respond to price changes the more elastic the supply.

3. Inventories: A producer who has a supply of goods or available storage capacity can quickly respond to price changes

4. Excess capacity: A producer who has unused capacity can quickly respond to price changes in his market assuming that variable factors are readily available

Illustrative Examples

146. If the elasticity of supply is zero, then

A. supply is very elastic.
B. the supply curve is horizontal.
C. the quantity supplied is the same regardless of price.
D. Both b and c are correct.

***Explanation*:**

- When the price elasticity of supply for a product is *perfectly inelastic* (es= 0), changes in the price do not affect the quantity supply for the product.
- The supply curve is vertical.
- The quantity supplied is the same even if the price rises. **Answer:** C

147. The short-run supply curve is more inelastic than the long-run supply curve because

A. resources are generally cheaper in the short run.
B. in the short run new firms will enter the industry, driving up the prices of resources.
C. consumers are willing to pay higher prices to obtain the commodity now, rather than wait until prices fall.
D. firms have less time to plan and vary all of their productive inputs in the short run; thus, per-unit costs will be higher.

***Explanation*:**

- In the short run a rise in the price of the commodity it may not be possible for the firm to increase the input of production, materials and capital.

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- Therefore, the supply is price inelastic. **Answer: D**

148. The elasticity of supply is typically greater when
   A. producers have less time to respond to price changes.
   B. producers are operating near the limits to their production.
   C. producers have less access to raw materials necessary for production.
   D. producers have more time to respond to price changes

Explanation: The longer the time producers get to make adjustments for changing the level of output in response to the change in price, the greater the response of the output, that is, the greater the elasticity of supply. **Answer: D**

149. If a price decrease from Birr 50 to Birr 40 results in a decrease in quantity supplied from 14 units to 10 units, calculate the price elasticity of supply.

Solution:

$es = \Delta Q/\Delta P$

$$es = \frac{10-14}{40-50} \times \frac{50}{14}$$

$es = 1.4$

150. Price elasticity of supply is the
   A. change in quantity supplied due to a change in quantity demanded
   B. responsiveness of supply to change in price
   C. responsiveness of supply to changes in costs
   D. responsiveness of the price to changes in supply

Explanation: The price elasticity of supply is a unit – free measure of how sensitive the quantity supplied to price. **Answer: B**

151. The price elasticity of supply for gasoline is 0.4. If the price of gasoline rises by 8%, what is the expected change in the quantity of gasoline supplied?

Solution:

$es = \Delta Q/\Delta P$

$0.4 = \Delta Q/8\%$

$\Delta Q = 8\% x\ 0.4,\ \Delta Q = 3.2\%$

152. An increase is supply caused no change in the equilibrium quantity bought and sold. Thus demand must be:

A. Perfectly inelastic B. Horizontal C. Elastic D. Unitary elastic

Explanation:

- When the quantity supplied of the commodity does not change at all in response to the change in its price, the elasticity of supply is zero.
- The supply curve is vertical.

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- It is said to be perfectly inelastic

Answer: A

153. If the price elasticity of supply for a good is 0.5, then
   A. an increase in the price boosts the quantity supplied by a larger percentage.
   B. the supply is elastic.
   C. the percentage change in the quantity supplied is less than the percentage change in price.
   D. None of the above answers are correct.

***Explanation*:**

- es= 0.5 means price elasticity of supply is inelastic.
- If the percentage change in the quantity supply is less than the percentage
- change in price, then the magnitude of the elasticity of supply is between zero and 1

Answer: C

154. The value of perfectly elastic supply:
   A. is equal to zero
   B. ranges between 0 and 1
   C. reports infinity
   D. is equal to 1.

***Explanation*:**

- In perfectly elastic supply a slight reduction ion price will cause the quantity supplied to fall to zero
- A slight rise in price will bring about indefinitely large amount increase in the quantity supplied.
- The value of perfectly elastic supply will be equal to infinity
- Its supply curve will be horizontal

Answer: C

155. The elasticity of supply measures the sensitivity of
   A. supply to changes in costs.
   B. quantity supplied to a change in price.
   C. price to changes in supply.
   D. quantity supplied to quantity demanded.

***Explanation*:**

- The elasticity of supply measures the degree of responsiveness of quantity supplied of a good to change in its own price

Answer: B

Supplementary Exercise on Unit 2

1. The demand for chicken increases as the result of higher beef prices. This indicates that
   A. chicken and beef are substitutes.
   B. chicken and beef are complements.
   C. the market demand for beef is inelastic.
   D. the market demand for chicken is elastic.

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2. If goods A and B are substitutes, a decrease in the price of good B will
   A. increase the demand for good A
   B. increase the quantity demanded of good B
   C. decrease the demand for good A
   D. both B and C
3. If the demand for product J shifts to the left as the price of product K increases, then
   A. J and K are complementary goods
   B. J and K are not related goods
   C. the number of consumers of product K has increased
   D. J and K are substitute goods
4. If the price of a substitute to good X increase,
   A. the demand for good X will increase.
   B. the market price of good X will decrease.
   C. the demand for good X will decrease.
   D. demand for good X will not change
5. If the price elasticity of demand for tires is 0.8, the demand is inelastic.
   A. True B. False
6. A 10 percent increase in the quantity of orange demanded results from a 20 percent decline in its price. The price elasticity of demand for orange is
   A. 0.5. B. 20.0. C. 2.0. D. 10.0.
7. Suppose a rise in the price of banana from Birr 5.50 to Birr 6.50 per kg decreases the quantity demanded from 12,500 to 11,500 units. The price elasticity of demand is
   A. 0.5. B. 1000.0. C. 2.0. D. 1.0.
8. The price elasticity of demand is 5.0 if a 10 percent increase in the price results in decrease in the quantity demanded by:
   A. 10 percent B. 50 percent C. 2 percent D. 5 percent
9. Two goods are complements in consumption if:
   A. both are inferior goods
   B. the cross elasticity of demand between them is negative
   C. both has negative price elasticity
   D. one has a positive elasticity and the other has negative price elasticity
10. If goods are complements, definitely their
    A. Income elasticity is negative. B. Income elasticity is positive.
    C. Cross elasticity is positive. D. Cross elasticity is negative.
11. A 10 percent increase in the price of sugar reduces sugar consumption by about 5 percent. The increase causes households to
    A. spend more on sugar.
    B. spends less on sugar.

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C. spends the same amount on sugar.
D. consumes more goods like coffee and tea that are complements of sugar.

12. A fall in the price of Pepsi that causes a household to shift its purchasing pattern AWAY from Substitutes and toward Pepsi is the:
A. income effect of a price change.
B. substitution effect of a price change
C. diminishing marginal utility effect of a price change.
D. complementary effect of a price change.

13. If beans are inferior goods, then
A. decreased bean prices result in people buying less beans.
B. there is no relationship between bean purchases and prices of complements.
C. the demand for beans falls as income rises.
D. people will substitute other goods for beans as the prices of substitutes rise

14. The degree to which the quantity supplied changes according to a change in any determinant of supply is identifying as:
A. elasticity of demand
B. elasticity of supply
C. price elasticity of demand
D. elasticity of income

15. If the real income of a consumer decreases and, as a result, his demand for product X increases, it can be concluded that product X is a/an
A. complementary good
B. normal good
C. inferior good
D. elasticity of income.

16. The most important determinant of price elasticity of demand is
A. the slope of the demand curve.
B. the availability of substitutes.
C. the price of other goods.
D. the income of the consumer.

17. Price ceilings and price floors
A. are desirable because they make markets more efficient as well as equitable.
B. cause surpluses and shortages to persist since price cannot adjust to the market equilibrium price.
C. can be enacted to restore a market to equilibrium.
D. are imposed because they can make the poor in the economy better off without causing adverse effects.

18. Which of the following statements is true, regarding the supply of a particular good, and that good's own price?
A. A price increase shifts the supply curve to the right.
B. A price decrease shifts the supply curve to the right.
C. A price increase shifts the supply curve downward.
D. A price changes alone does not shift the supply curve

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19. If demand for a good is perfectly inelastic, then the demand curve will be
    A. a horizontal line.
    B. a vertical line.
    C. a straight line with a constant negative slope.
    D. flatter and flatter as the price of the good falls.
20. Which of the following statements is *not* true when price is above the equilibrium price?
    A. There is a shortage of the commodity
    B. The quantity demand exceeds the quantity supplied of the commodity.
    C. The pressure on the commodity price is downward.
    D. There is a surplus of the commodity.
21. The intersection of a market demand curve and a market supply curve for a commodity determines:
    A. the equilibrium price for the commodity.
    B. the equilibrium quantity for the commodity.
    C. the point of neither surplus nor shortage for the commodity.
    D. all of the above.
22. Ceteris paribus, the fewer substitutes there are for a good the more the demand for the good, and the longer period of time people have to adjust to a price change, the more the good_____.
    A. elastic; elastic
    B. elastic; inelastic
    C. inelastic; elastic
    D. inelastic; inelastic
23. The demand curve is downward sloping due to:
    A. income effect.
    B. income and substitution effect.
    C. substitution effect.
    D. consumption effect.
24. If both demand and supply of bottled water decrease, what will be the effect on the equilibrium price and quantity?
    A. Both price and quantity will decrease
    B. Both price and quantity will increase
    C. Price will decrease and quantity will increase
    D. Quantity will decrease and price will increase, decrease, or remain the same
25. An increase in demand can be represented graphically as:
    A. A left ward shift
    B. A right ward shift
    C. Shift down
    D. Move to the left

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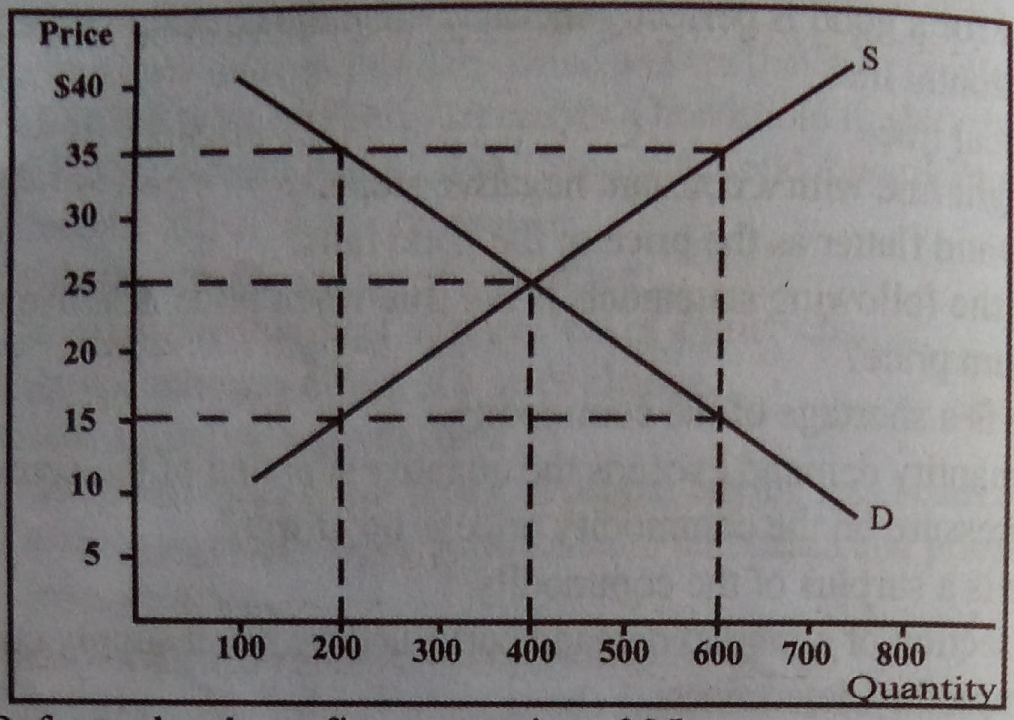

26. Refer to the above figure, at price of 35
    A. There would be a shortage of 400 units
    B. There would be a shortage of 200 units
    C. There would be a surplus of 200 units
    D. There would be a surplus of 400 units
27. Refer to the above figure equilibrium price and quantity are respectively
    A. 35 and 200 B. 35 and 600 C. 25 and 400 D. 15 and 200
28. If the price of good B decreased, you would expect the
    A. demand of good B to increase
    B. quantity demanded of good B to decrease
    C. quantity demanded of good B to increase
    D. demand curve of good B to shift to the left
29. Two goods are substitutes when a decrease in the price of one good
    A. decreases the demand for the other good.
    B. decreases the quantity demanded of the other good.
    C. increases the demand for the other good.
    D. increases the quantity demanded of the other good.
30. Which of the following might cause the demand curve for an inferior good to shift to the left?
    A. a decrease in income
    B. an increase in the price of a substitute
    C. an increase in the price of a complement
    D. None of the above is correct.
31. Which type of good is a competing product that consumers can use in place of another?
    A. complementary good C. disposable good
    B. substitute good D. exchange good

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32. Which term refers to the desire, willingness, and ability to buy a good or service?

A. demand C. utility B. supply D. price

33. Which of the following changes is *less likely* to have an effect on demand?

A. a change in the number of consumers B. a change in profit

C. a change in consumers' income D. a change in consumers' tastes

34. Which statement about the effects of the law of demand is most accurate?

A. When prices decrease, quantity demanded increases

B. When prices decrease, quantity demanded stays the same

C. When prices decrease, quantity demanded decreases

D. When prices decrease, quantity demanded causes prices to decrease further

35. The price at which there is neither surplus nor shortage is called:

A. the adjustment price. C. the fair price.

B. the equal quantity. D. the market-clearing price.

36. If shoe-repair services are an inferior good, then:

A. if people's incomes go up, they spend less for shoe repair.

B. if the price of shoe repair falls, people want less of it.

C. people will only want shoe repair services if they are deceived by advertising.

D. people do not base their decisions to buy shoe repairs on costs-benefit reasoning

37. A fall in the price of cabbage from Birr10.50 to Birr9.50 increases the quantity demanded from 18,800 to 21,200. Calculate the price elasticity of demand

38. Suppose the market has demand and supply function as follows:
Determine equilibrium price and quantity.

39. The demand curve for product X is given by P=300-6Qd while the supply curve is represented by Qs= 10+0.5P

A. Calculate equilibrium price and quantity

B. What is the price elasticity of demand at equilibrium?

C. What is the price elasticity of supply at equilibrium?

Answer to Supplementary Exercise on Unit 2

1. A 2. A 3. A 4. A 5. A 6. A 7. B 8. B 9. C 10. D 11. B
12. B 13. C 14. B 15. C 16. B 17. B 18. D 19. B 20. D 21. D
22. C 23. B 24. A 25. B 26. D 27. C 28. C 29. C 30. D 31. B
32. A 33. B 34. A 35. D 36. A 37. 1.3 38. P= 7.15, Q = 243.5
39. A. P= 60, Q= 40 B. 0.25 C. 0.75

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Theory of Consumer Behavior

The value a consumer places on a unit of a good or service depends on the pleasure or satisfaction he or she expects to derive from having or consuming it at the point of making a consumption (consumer) choice.

In economics the satisfaction or pleasure consumers derive from the consumption of consumer goods is called "**utility**".

Consumers, however, cannot have everything they wish to have.

Consumers' choices are constrained by their incomes

Within the limits of their incomes, consumers make their consumption choices by evaluating and comparing consumer goods with regard to their *utilities*.

What does utility mean in economics:

All economic activities aim at fulfilling unlimited human wants. Utility can be regarded as the potentiality of goods and services to satisfy a human wants. Since utility refers to the inherent quality of a commodity or service, it does not come under the purview of precise quantitative measurement. A person can only realize how much utility he or she derives from a commodity. Therefore, utility has no physical or material existence and the concept is purely subjective introspective.

Total utility (*TU*) is the total amount of happiness received from the consumption of a certain amount of a good. Total utility generally increases as the quantity consumed of a good increases. Example: The satisfaction of consuming 8 units.

Marginal utility (*MU*) is the extra utility received from consuming one additional unit of the good per unit of time while holding constant the quantity consumed of all other commodities. Example: the satisfaction of consuming the 8th unit.

Relationship between total utility and marginal utility

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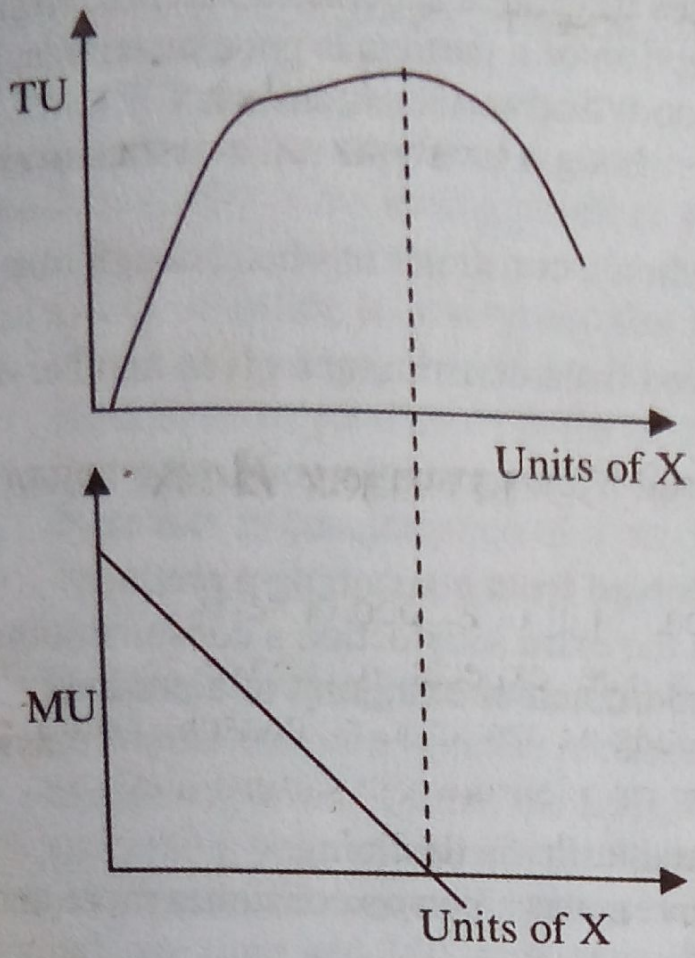

Figure 3.1: relationship between total utility and marginal utility.

As shown in the above figure:

- When Total utility increases by smaller and smaller amount, marginal utility decreases.
- When total utility reaches maximum, (at 5th units), marginal utility is zero.
- When total utility declines, marginal utility is negative.

Illustrative Example

1. According to the theory of consumer behavior, more is always better

   A. True B. False

***Explanation*:**

- Consumer behavior theory rests upon the more is better assumption regarding the utility tied to consumption.
- All goods and services are desirable in the sense of being able to satisfy consumer wants.
- As a result, consumers will always prefer more to less of any good or service.

Answer: True

2. Marginal utility measures the consumer's overall level of satisfaction derived from consumption activities

   A. True B. False

Explanation:

- Total utility measures the consumer's overall level of satisfaction derived from consumption activities.

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- Marginal utility measures the added satisfaction derived from a one-unit increase in consumption of a particular good or service, holding consumption of other goods and services constant.

Answer: False

3. Marginal utility is the
   A. satisfaction achieved when a consumer has had enough of a product.
   B. total satisfaction received from consuming a given number of units of a product.
   C. extra satisfaction received from consuming one more unit of a product.
   D. average satisfaction received from consuming a product.

***Explanation*:** Marginal utility is the extra satisfaction a consumer obtains from consuming an additional or extra unit of a product

Answer: C

4. Total utility
   A. is negative when marginal utility is declining.
   B. has a constant rate of increase as a person consumes more and more of a good.
   C. cannot decrease as a person consumes more and more of a good.
   D. is equal to the sum of the marginal utilities of all units consumed.

***Explanation*:** Total utility refers to the overall sense of satisfaction that someone will drive from the consumption of goods and services.

Answer: D

5. If, when you consume another piece of candy, your marginal utility is zero, then
   A. you should consume less candy.
   B. you want more candy.
   C. you have maximized your total utility from consuming candy.
   D. you have not yet reached the point of diminishing marginal utility.

***Explanation*:**

- Once marginal utility assumes a value of zero, it is no more possible to increase total utility by consuming more of the product.
- Total utility reaches maximum

Answer: C

6. If total utility increases at a decreasing rate as a consumer consumes more coffee, then marginal Utility must
   A. be negative
   B. remains constant.
   C. decrease.
   D. increase

***Explanation*:** When total utility increases at a decreasing rate, marginal utility decreases but remain positive

Answer: C

7. Utility refers to

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A. the satisfaction a consumer receives from consuming a good
B. the usefulness of a good
C. a shift in the demand curve for a good
D. a decline in the supply of a good

***Explanation*:** Utility is the total amount of satisfaction a consumer obtains from consuming a product. **Answer: A**

8. In the theory of utility, it is assumed that marginal utility
   A. is zero as consumption of a product increases
   B. diminishes beyond some point as consumption of a product increases
   C. increases as consumption of a product increases
   D. increases as consumption of a product remains constant

***Explanation*:**

- Marginal utility refers to the extra utility a consumer gets from one additional unit of a specific product.
- In a short period of time, the marginal utility derived from successive units of a given product will decline. **Answer: B**

9. If Ayele gets 100 units of utility from watching the show "Betoch Drama" one time and 150 units of utility from watching the same show twice, his marginal utility from watching the second time is

   A. 250 B. -50 C. 50 D. 150

***Explanation*:**

- Marginal utility refers to the extra utility a consumer gets from one additional unit of a specific product
- In first watch Ayele gets 100 units of satisfaction, in the second watch he gets 150 units of satisfaction. So he gets additional 50 units of satisfaction(150 units minus 100 units= 50 units satisfaction) **Answer: C**

10. The utility of a specific product:
    A. is determined by consumer income.
    B. is determined by the price of the product.
    C. varies from person to person using the product.
    D. is constant from person to person using the product.

***Explanation*:**

- Utility is a subjective notion in economics, referring to the amount of satisfaction a person gets from consumption of a certain item.
- The utility that you will drive from the consumption of the product depends on your taste and preference for the commodity.

Answer: C

11. Cardinal utility theory assumes that consumers can
    A. rank baskets of goods as to their preference.

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B. determine the number of utils that can be derived from consuming all goods.
C. determine the marginal rate of substitution between goods.
D. avoid the law of diminishing marginal utility.

***Explanation*:**

- Cardinal utility approach assumes that consumers can measure the satisfaction that they will drive from the consumption of goods and services in cardinal terms using a hypothetical unit of measurement called utils.

Answer: B

12. State the axioms of the cardinal utility approach and hence the equi-marginal principle.

***Explanation*:**

The axioms of cardinal utility theory are:

- Rationality of the consumer
- Disposable income of the consumer is given
- Tastes and preferences are given
- There is perfect competition in the goods market that is prices of goods are given
- Marginal utility is positive
- Utility is measurable/quantifiable

13. The basic hypothesis of utility theory states that as more of a commodity is consumed during a given period of time,
A. total utility and marginal utility will both decrease
B. total utility and marginal utility will both increase
C. total utility will increase but marginal utility will eventually decrease
D. total utility will decrease but marginal utility will increase

***Explanation*:**

Economists assume that the economy is made up of people who would rather have more of something then less. However, beyond a certain point, each additional unit of a good brings less and less satisfaction to a person.

Answer: C

14. Negative marginal utility means that
A. total utility is also negative
B. marginal utility is increasing
C. the price of the product is increasing as additional units are consumed
D. total utility is decreasing as additional units are consumed

***Explanation*:**

- The point where the marginal unit of the product is zero is the point of satisfaction. Any additional consumption after the point of satisfaction will lead to dissatisfaction.

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- Negative marginal utility means that as you consume extra unit of a commodity total utility decreases.

Answer: D

15. For a person consuming a certain good, piece by piece, and who has reached a stage where she/he has no more desire to consume, what is the best conclusion that can be made about marginal and total utility of the person at this stage
    A. Total utility is zero and marginal utility is maximum
    B. Marginal utility is zero and total utility is maximum
    C. Marginal utility has reduced to the average utility
    D. Marginal utility is still greater than average utility

***Explanation*:**

- The person reached at the point of satisfaction.
- The person is fully satisfied
- At the point of full satisfaction total utility is maximum and marginal utility is zero

Answer: B

16. Which of the following statement describes utility best?
    A. The marginal utility derived from the consumption of a good increases as consumption increases
    B. The total utility derived from the consumption of a good does not increase as consumption increase
    C. The average utility derived from the consumption of a good increase as consumption increases linearly
    D. The total utility derived from the consumption of a good increase up to a certain point as consumption increases.

***Explanation*:**

- The total utility refers to the sum total of satisfaction which a consumer receives by consuming the various units of the commodity.
- The more units of a commodity she/he consumes, the greater will be her/his total utility or satisfaction from it, up to a certain point.

Answer: D

17. The total utility for the first five breads consumed is 4, 10, 15, 18, and 17, respectively. Which of the following is true?
    A. The marginal utility of the third bread is 15
    B. The marginal utility of the first bread is 0
    C. The marginal utility of the fifth bread is -1
    D. The marginal utility of the third bread is -3

***Explanation*:**

$MU = TU_n - TU_{n-1}$

$MU \text{ of the third bread} = MU_{3units} - MU_{2units}$

$MU_3 = 15 - 10 = 5$

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$$MU \text{ of the fifth bread} = MU_{5 \text{ units}} - MU_{4 \text{ units}}$$

$$MU_{5 \text{ units}} = 17 - 18 = -1$$

The marginal utility of the fifth bread is -1 **Answer: C**

3.2 The cardinal utility theory:

This theory assumes that consumers can measure the utility they will drive from the consumption of goods and services in cardinal terms like their weight and height.

The Cardinalist school asserts that utility can be measured and quantified. It means, it is possible to express utility that an individual derives from consuming a commodity in quantitative terms. Thus, a person may express the utility he derives from consuming an apple as 10 utils or 20 utils. Moreover, it allows consumers to compare and define the difference in utilities perceived in two commodities. Thus, it allows an individual to state that commodity A (accruing an utility of 20 utils) gives double the utility of commodity B (which accrues an utility of 10 utils).

3.2.1 Assumptions of cardinal utility theory:

Rational consumers: it means that the consumers consume first a commodity which yields the highest utility and the last which gives the least. The consumer is rational in the sense that given his income constraints, he would always attempt to maximize his utility.

Cardinal measurement of utility: Utility is a cardinal concept and it can be measured and expressed in quantitative terms. For convenience, it is expressed in terms of the monetary units that a consumer is willing to pay for the marginal unit of the commodity.

Marginal utility of Money is constant. That is, as one acquires more and more money, the marginal utility of money will remain unchanged. This assumption is critical because money is used as a standard unit of measurement of utility, and, hence, cannot be elastic.

Diminishing marginal utility: the utility gained from successive units of a commodity consume decreases as a consumer consumes a larger quantity of the commodity.

Law of diminishing marginal utility: Each additional unit of a good eventually gives less and less extra utility. This implies that as a consumer increases his consumption of a commodity, the utility accruing from successive units of the commodity decreases. In other words, the marginal utility of a commodity will keep falling as a consumer goes on increasing its consumption.

The total utility of a 'bundle' of goods depends on the quantities of the individual commodities. Thus: $U = f(x1, x2, \ldots\ldots\ldots\ldots, xn)$.

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where U means total utility; x1, x2 .....xn are the quantities of n number of commodities

Illustrative Example

18. The law of diminishing marginal utility states that as an individual increases consumption of a given product within a set period of time, the utility gained from consumption eventually declines.

A. True B. False

***Explanation*:**

- In general, the law of diminishing marginal utility states that as an individual increases consumption of a given product within a set period of time, the marginal utility gained from consumption eventually declines. **Answer: False**

19. Marginal utility measures the added satisfaction derived from a one-unit increase in consumption, holding consumption of other goods and services constant.

A. True B. False

***Explanation*:**

- Total utility measures the consumer's overall level of satisfaction derived
- from consumption activities
- Marginal utility measures the added satisfaction derived from a one-unit increase in consumption of a particular good or service, holding consumption of other goods and services constant. **Answer: True**

20. When goods are relatively scarce, the law of diminishing marginal utility means that the added value of another unit of goods will be small in relation to the added value of another unit of services.

A. True B. False

***Explanation*:**

- When goods are relatively scarce, the law of diminishing marginal utility means that the added value of another unit of goods will be large in relation to the added value of another unit of services

Answer: False

21. Is an increase in total utility or satisfaction following an increase in income inconsistent with the law of diminishing marginal utility?

***Explanation*:**

No, the law of diminishing marginal utility states that the marginal utility derived from consumption will eventually diminish as consumption increases during a given time period.
This means that the addition to total utility per dollar of income will tend to fall as total income rises.

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22. The law of diminishing marginal utility states that
   A. when the extra satisfaction from consuming a good becomes negative, total utility starts falling, other things constant.
   B. eventually total utility falls as more of a good is consumed, other things constant.
   C. the extra satisfaction from consuming a good decreases as more of a good is consumed, other things constant.
   D. the extra satisfaction from consuming a good increases slowly as more of a good is consumed, other things constant.

Explanation: In a short period of time, the marginal utility derived from successive units of a given product will decline. This is known as diminishing marginal utility.

Answer: C

23. Diminishing marginal utility means:
   A. As you consume more, your total utility diminishes
   B. As you decrease consumption, marginal utility diminishes
   C. As you consume more, an additi0nal item is worthless
   D. As you consume less, an additional item is worth less

Explanation: As a consumer consumes successive units of a product, eventually the marginal utility from additional units consumed will start to become smaller than the marginal utility of previous units consumed.

Answer: C

24. Why the marginal utility does diminishes?

Explanation:

- Marginal utility diminishes because of the successively less important uses of additional quantities of a commodity.

3.2.2Consumers' equilibrium:

Initially we derive the equilibrium of the consumer when he spends his money income M on a single commodity x. Here, the consumer will be at equilibrium when the marginal utility of x is equal to its market price.

Symbolically: $MUx = Px$

If:

- $MUx > Px$, then the consumer can increase his welfare by consuming more of x. He will continue to do that until his marginal utility for x falls sufficiently, to be equal with its price.
- $MUx < Px$, then the consumer can enhance his welfare by cutting down on his consumption of x. He will be persisting on doing this, until his MUx increases to equal the price Px. If more commodities are introduced into the model, then the consumer will attain equilibrium when the ratios of the marginal utilities of the individual commodities to their respective prices are equal for all commodities. That is,

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$$\frac{MU_x}{P_x P_y} = MU_y$$

This state is defined by the "Law of equi-marginal utility", which states that a consumer will distribute his money income among different commodities in such a way that the utility derived from the last Birr spent on each commodity is equal.

Now if:

MU x /Px)> (MUy/Py then the consumer will start substituting commodity y with commodity x, causing MUx to fall and MUy to rise. This he will continue until MUx / Px equals MUy / Py

Conversely, if **MUx / Px) < (MUy/Py**, then the consumer will substitute commodity x with commodity y until the equilibrium is restored.

Note:

PX QX+PY QY= I

Where, Q is quantity, P is price and I is income

Example: The consumer has Birr 30 to spend on Apples and Bread. The price of Apple is Birr 6 and the price Bread is Birr 3. The table shows his total utility from different quantities of the two items.

What is the consumers' optimal consumption bundle?

Quantity of Apples	Total Utility	Quantity of Bread	Total utility Utility
1	60	1	40
2	10	2	71
3	13	3	90
4	14	4	10
5	14	5	11
6	13	6	11
7	12	7	11

***Solution*:**

Quantity of Apples	Total utility	MUA	MUA/PA	Quantity of bread	Total utility	MUB	MUB/PB
1	60	-	-	1	40	-	-
2	102	42	42/6=7	2	70	30	30/3=10
3	132	30	30/6=5	3	91	21	21/3=7
4	144	12	12/6=2	4	106	15	15/3=5
5	144	0	0/6=0	5	112	6	6/3=2
6	138	+6	-6/6=-1	6	115	3	3/3=1
7	128	-10	-10/6=-1.7	7	115	0	0/3=0

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To be obtain maximum utility it must be satisfy two conditions

A. MUA/PA= MUB/PB

B. PA x QA+ PB x QB = Income

To obtain maximum utility, the consumer purchase 3 units of Apples and 4 units of Bread

At this combination MUA= 5 and MUB = 5

MUA/PA = MUB/PB

PA x QA + PB x QB= I

6x3 + 3x4= 30

Illustrative Example

25. Based on the table below answer question 20

Quantity of Milk	Total utility	Quantity of Sandwiches	Total utility
1	40	1	45
2	60	2	75
3	72	3	102
4	82	4	120
5	88	5	135
6	90	6	145

The above table shows Kebron's utility from soup and sandwiches. The price of milk is Birr 2 per cup and the price of a sandwich is Birr 3. Kebron has Birr 18 to spend on these two goods.

Find out her maximum utility.

***Solution*:**

Quantity of milk	Total utility	MUM	MUM/PM	Quantity of sandwich	Total utility	MUS	MUS/PS
1	40	-	-	1	45	-	-
2	60	20	20/2=10	2	75	30	30/3=10
3	72	12	12/2=6	3	102	27	27/3=9
4	82	10	10/2=5	4	120	18	18/3=6
5	88	6	6/2=3	5	135	15	15/3=5
6	90	2	2/2=1	6	145	10	10/3=3.3

To maximize utility the consumer will equate MUM/PM = MUS/PS

MUM/PM= 6

MUS/PS= 6

The maximum utility is obtained at 3 units of milk and 4 units of sandwich.

PA x QA +PS x QS = Income

2x 3 + 3x 4= 18

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26. Refer to the above table, Suppose Kebron's income increases from Birr18 to Birr 23 but prices have not changed. What is her utility maximizing bundle now?–

***Solution*:**

At 4 units of milk and 5 units of sandwich,

MUM/PM= MUS/PS = 5

At this bundle:

PM x QM + PS x QS=Income

2x 4+ 3x 5= 23

The maximum utility bundle is 4 units of milk and 5 units of sandwich.

27. Ted has Birr 600 a week to spend on food and clothing. The price of food is Birr 5 and the price of clothing is Birr 30. Which of the following pairs of food and clothing are in the Ted's choice set?

A. 20 units of clothing and 50 units of food
B. 0 unit of clothing and 200 units of food
C. 10 units of clothing and 125 units of food
D. 10 units of clothing and 60 units of food

***Solution*:**

PF x QF + PCXQC= Income

5x60 + 30x10= 600

The possible combination of Ted's is 10 units of cloth and 60 units of food.

Answer: D

28. Richard is consuming X and Y so that MUx/Px = 6 and MUy/Py = 10. To maximize utility, Richard should

A. consume less of both X and Y.
B. consume more of X and less of Y.
C. continue to consume the same amount of X and Y, as the consumer is already maximizing utility.
D. consume less of X and more of Y.

***Explanation*:**

- If MUX/PX > MUY/PY, MUX must be reduced, then the consumer consumes more of good X
- If MUX/PX< MUY/PY, MUY must be reduced and the MUY can be reduced only by consumer's consume more of good Y.

So the consumer should consume less of X and more of Y

Answer: D

29. Suppose you have a fixed budget for two goods, *X* and *Y*, *Px* = 10 and *Py* = 5, *MUx*= 60 utils and *MUy* = 15 utils. Should the consumption of *X* and/or *Y* be higher, lower, or remain the same?

A. The current combination of goods maximizes total utility. Consumption should remain the same.

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B. Consumption of good *X* should decrease, and consumption of good *Y* should increase.
C. Both the consumption of goods *X* and *Y* should decrease.
D. Consumption of good *X* should increase, and consumption of good *Y* should decrease

***Explanation*:**

MUX/PX= 60/ 10= 6
MUY/PY= 15/5 =3
MUX/PX> MUY/PY, So the consumer could increase utility by consuming more of good X and less of good Y. This would reduce the marginal utility of good X and raise the marginal utility of good Y until the two were equal.

Answer: D

30. To maximize total utility, consumption should be arranged such that the
A. the total utility associated with each good consumed is equal for all goods consumed
B. the ratio of the total utility associated with each good consumed to the price of the good is equal for all goods consumed
C. marginal utility associated with the last unit of each good consumed is equal for all goods consumed
D. ratio of the marginal utility associated with the last unit of each good consumed to the price of the good is equal for all goods consumed

***Explanation*:**

A rule for utility maximization is the ratio of the marginal utility to the price for any two goods must be equal.

Answer: D

31. The price of good X is Birr1.50 and that of good Y is Birr1. If a particular consumer's marginal utility for Y is 30, and he is currently maximizing his total utility. Calculate marginal utility of X :

***Solution*:**

MUX/PX= MUY/PY
MUX/1.50= 30/1
MUX= 1.50x 30
MUX= 45

32. Suppose a consumer has an income of Birr 16, the price of A is Birr 2, and the price of B is Birr 1. Which combination is on the consumer's budget line?

A. 4A and 9B B. 5A and 6B C. 6A and 6B D. 8A and 8B

***Solution*:**

PA x QA + PB x QB =Income

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$PA \times QA + PB \times QB = 16$

$2x5 + 1X6 = 16$

The consumer's budget line is 5 units of good A and 6 units of good B

Answer: B

33. If a consumer has an income of Birr 200, the price of X is Birr 5, and the price of Y is Birr 10, the maximum quantity of X the consumer is able to purchase is:

***Solution*:**

$PX \times QX + PY \times QY = \text{Income}$

Maximum quantity of X means the consumer spent all income by consuming only good X. consumption of good Y is zero

$PX \times QX + 0 = \text{Income}$

$QX = I/PX$

$QX = 200/5$

$QX = 40$

34. Ahadu consumes only two goods, X and Y. He spends his entire weekly income on goods X and Y. at his current consumption level the two goods the marginal utility of the last Birr he spent on goods X is 10 and the marginal utility of the last Birr he spent on goods Y is 20.

To maximize his utility, Ahadu would:

A. Maintain his current consumption on the two goods
B. Buy fewer units of X and more units of Y
C. Buy fewer units of Y and more units of X
D. Buy fewer units of both X and Y

***Explanation*:**

To maximize utility,

- Ahadu should equalize the marginal utilities per Birr spent on the last unit of each goods. Since the marginal utility of the last dollar spent on good X, 10, is less than the marginal utility of the last Birr spent on goods Y, 20.
- Ahadu is not maximizing his utility at current consumption level. To maximize his utility, he should reallocate his income so that he buys fewer units of X and more units of Y.
- As he buys fewer units of X and more units of Y, the marginal utility of goods X rises and the marginal utility of goods Y fall. The process will continue until the marginal utilities per Birr spent are equal on each goods.

Answer: B

3.3The ordinal utility theory:

The ordinalist school asserts that utility cannot be measured in quantitative terms. Rather, the consumer can compare the utility accruing from different commodities (as a combination of them) and rank them in accordance with

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the satisfaction each commodity (or combination of commodities) gives him.

Thus, the cardinal approach to the measurement of utility believes that utility derived from the consumption of a commodity can be expressed in quantitative terms. The ordinalist approach rejects this and states that the consumer at best can rank the various commodities (or combination of them) in accordance with the satisfaction that he expects from their consumption.

Utility can be ordinally measured: The consumer can rank various commodities or combination of commodities in accordance with the satisfaction that he derives from them.

The consumer is rational: Given the market prices and his income, a consumer will attempt to maximize his satisfaction when he undertakes consumption.

Additive Utilities: The quantity of the commodities that is consumed determines the total utility of the consumer.

Consistency of choices: The choice of the consumer is consistent in the sense that if he chooses combination A over B in one period, he will not choose B over A in another period.

Symbolically : if $A > B$, then $B < A$.

Transitivity of consumer choice: If a consumer prefers combination A to B, and prefers B to C, then, it can be concluded that he prefers A to C.

Symbolically : If $A > B$, and $B > C$, then $A > C$.

Illustrative Example

35. Hamelimal tells you that he prefers Pepsi to Coke, Coke to 7-UP, and 7-UP to Pepsi. This violates what assumption made when analyzing consumer preferences?

    A. That consumers are rational.

    B. That there is a diminishing marginal rate of substitution.

    C. That more is better.

    D. That less is sometimes better.

***Explanation*:**

- Consumer's choice is characterized by transitivity; if Pepsi is preferred to coke then Pepsi is preferred to 7- up.
- The consumer cannot prefer 7- up to Pepsi. This violates the assumption of the more is better.

Answer: C

36. Consumer must understand how much one product is preferred over another in order to rank-order consumption alternatives.

    A. True B. False

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Explanation:

- The consumers understanding of ordinal utility makes possible a rank ordering of preferred goods and services.
- Notice that knowledge concerning complete consumer preferences does not necessarily include understanding about how much one product is preferred over another.

Answer: False

37. The non-satiation principle abstracts from time and place considerations.

Explanation:

- At any specific place and time, consumers do become sated. Therefore, the nonsatiation principle involves a certain amount of abstraction from time and place considerations
- It is best considered within the context of money income where more money brings additional satisfaction or well-being.

Answer: True

38. Which of the following is not an assumption of ordinal utility analysis?
    A. Consumers are consistent in their preference.
    B. Consumers can measure the total utility received from any given basket of good.
    C. Consumers are non-satiated with respect to the goods they confront.
    D. All are necessary.

Explanation:

- In ordinal utility approach consumers are capable of ranking goods and services based on their order of preferences at first, second, third, etc.
- It is not realistic to assume consumers who are capable of measuring their satisfaction in quantitative terms.

Answer: B

39. If a man prefers spagetti to shiro and shiro to dorowet, and if he is indifferent between spagetti and kitifo, he must
    A. prefer kitifo to dorowet.
    B. prefer shiro to kitifo.
    C. be indifferent between shiro and kitifo.
    D. be indifferent between spagetti and dorowet

Explanation:

- Spagetti and kitifo drive the same satisfaction for the man
- If the man prefers spaghetti to shiro and shiro to dorowet, he must prefers spagettti to dorowet. He must prefer shiro to kitifo.

Answer: A

40. What is Consumer is rational implies?

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***Explanation*:**

'The consumer is rational means what action to take through a process in which he/she:

- Asks himself/herself "What is desirable"?
- Asks himself/herself "What is feasible"?
- Chooses the most desirable from among the feasible alternatives.'

41. Mention the factors that a rational consumer must consider in deciding how much of a good or goods to buy.

***Explanation*:**

The rational consumer buys goods and services by considering:

- The income of the consumer
- The price of the commodity
- The value of the commodity of the consumer

42. Define an inferior good and a normal good

***Explanation*:**

- An inferior good is defined as one of which an individual purchases less when his or her income rises and more when his or her income falls.
- A normal good is defined as one of which an individual purchases more when his or her income rises and less when his or her income falls.

43. Consider the following three market baskets:

Combination	Food	Clothing
A	6	3
B	8	5
C	5	8

If preferences satisfy all three of the usual assumptions:

A. A is on the same indifference curve as B.
B. B is on the same indifference curve as C.
C. A is preferred to C.
D. B is preferred to A

***Explanation*:**

Market basket B contains more food and more clothing compared to market basket A.

If preferences satisfy the "more is better" assumption, B must be preferred to A.

Answer: D

3.3.1 Indifference curve:

An indifference curve is the locus of the various combinations of two commodities that yield the same satisfaction to the consumer, so that he is indifferent to any one particular combination. In other words, all

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combinations of the two commodities in the indifference curve are equally desired by the consumer.

3.3.2 Indifference Map:

An indifference map, on the other hand, shows all the indifference curves which rank the preference of the consumer. While combinations of commodities on the same indifference curve yield the same satisfaction, combinations on a higher indifference curve yield greater satisfaction and combinations on a lower curve yield less satisfaction. In the following figure 3.2, an indifference map has been shown

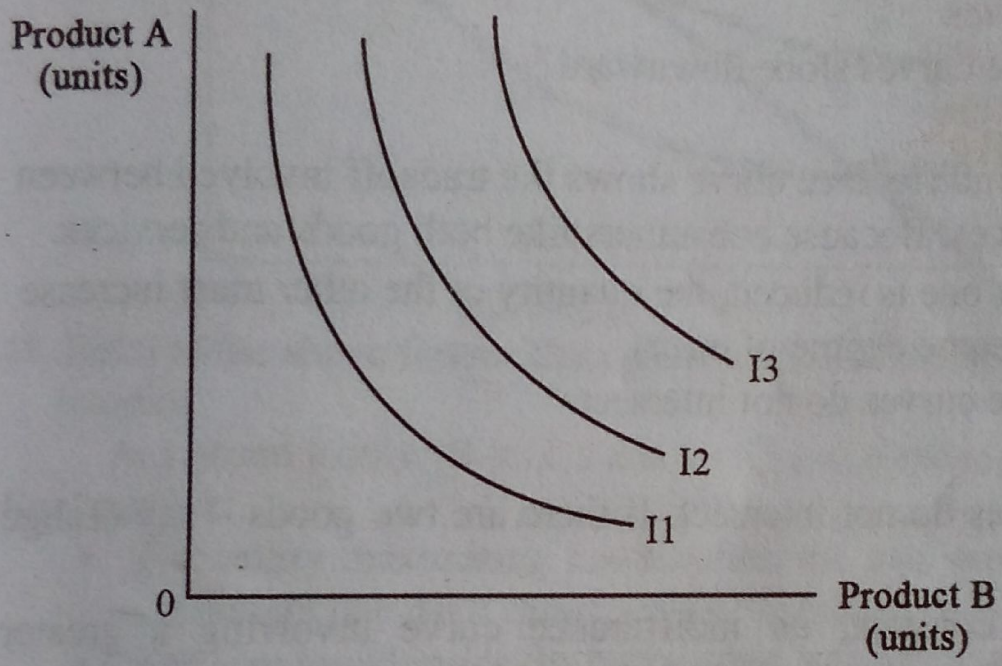

Figure 3.2 indifference maps.

In the figure 3.2, I3 has a higher level of utility than I1. Assuming the aim of consumers is to maximize utility this means they will want to consume on the highest possible indifference curve

3.3.3 Characteristics of indifference curves:

Indifference curves are negatively sloped: because if one basket of goods *X* and *Y* contains more of *X*, it will have to contain less of *Y* than another basket in order for the two baskets to give the same level of satisfaction

Indifference curves cannot intersect: if the y did, the point of their intersection would imply two different level of satisfaction which is impossible

Indifference curves are convex to the origin: Indifference curves will be "bowed in" with respect to the origin. The reason for this is that the more or one good that an individual consumes along a particular indifference curve, the less the marginal utility of that good will become relative to other goods.

Convexity results from or is a reflection of a decreasing marginal rate of substitution.

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Illustrative Example

44. Which indifference curve gives higher satisfaction for consumers

***Explanation*:**

- Consumers prefer more to less, so they prefer higher indifference curves that represent greater combinations of goods and services to lower indifference curves that represent smaller combinations of goods and services.
- Higher indifference curve gives higher satisfaction than the lower indifference curves

45. Why Indifference curves slope downward?

Explanation:

- The slope of an indifference curve shows the tradeoff involved between goods and services. Because consumers like both goods and services.
- If the quantity of one is reduced, the quantity of the other must increase to maintain the same degree of utility.

46. Why indifference curves do not intersect

***Explanation*:**

- Indifference curves do not intersect. If there are two goods – say orange ands of drink,
- Holding orange constant, an indifference curve involving a greater amount of soft drink must give greater satisfaction.
- Similarly, holding soft drink constant, an indifference curve involving a greater amount of orange must give greater satisfaction.
- This stems from the fact that goods and services both provide consumer benefits, and reflects the more is better principle

47. If Chaltu prefers Avocado Juice to Mango Juice and Mango Juice to Orange juice , then if preferences are transitive,

A. at times she could be indifferent among the three items.
B. she could prefer orange to Avocado on some occasions but not mango to orange.
C. she could prefer mango to avocado on some occasions but not orange to avocado.
D. she must prefer Avocado to Orange Juice.

***Explanation*:**

If Avocado prefers to Mango and Mango preferred to Orange, then Avocado Juice is preferred to Orange Juice.

Answer: D

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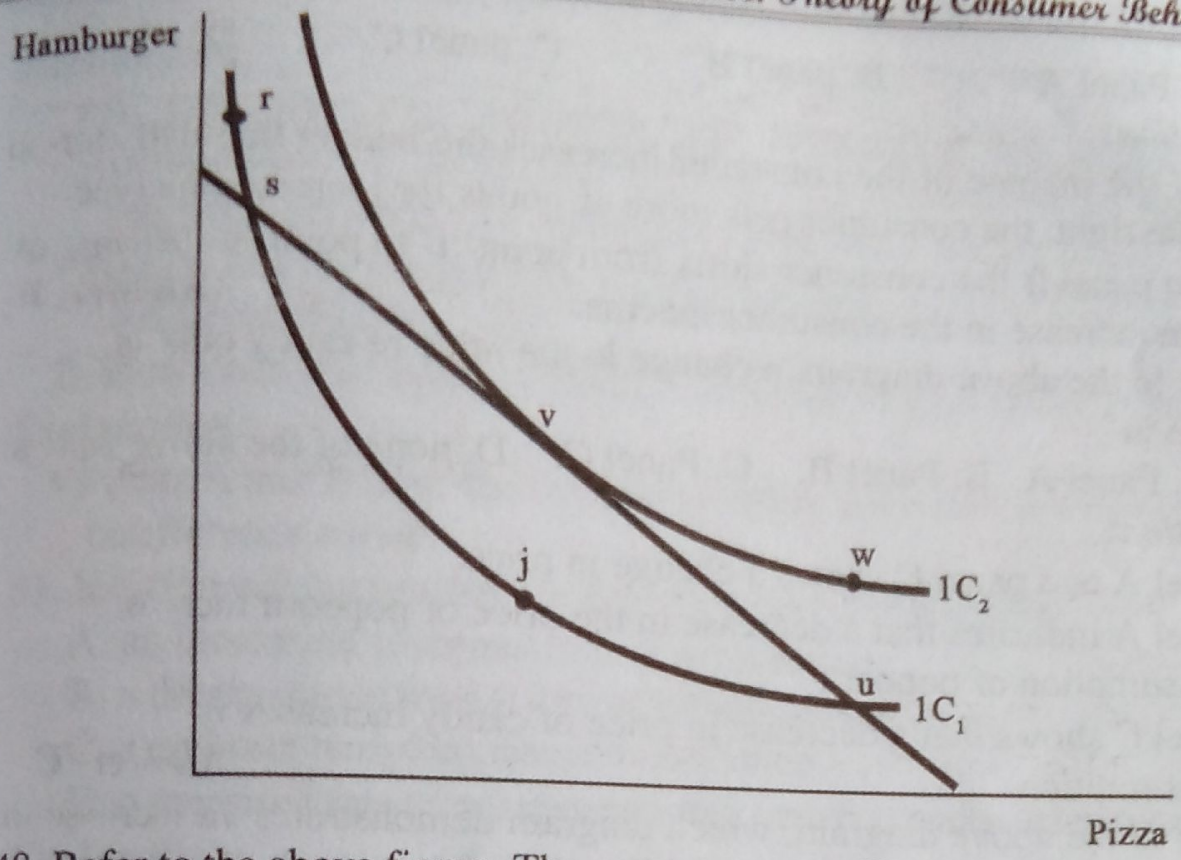


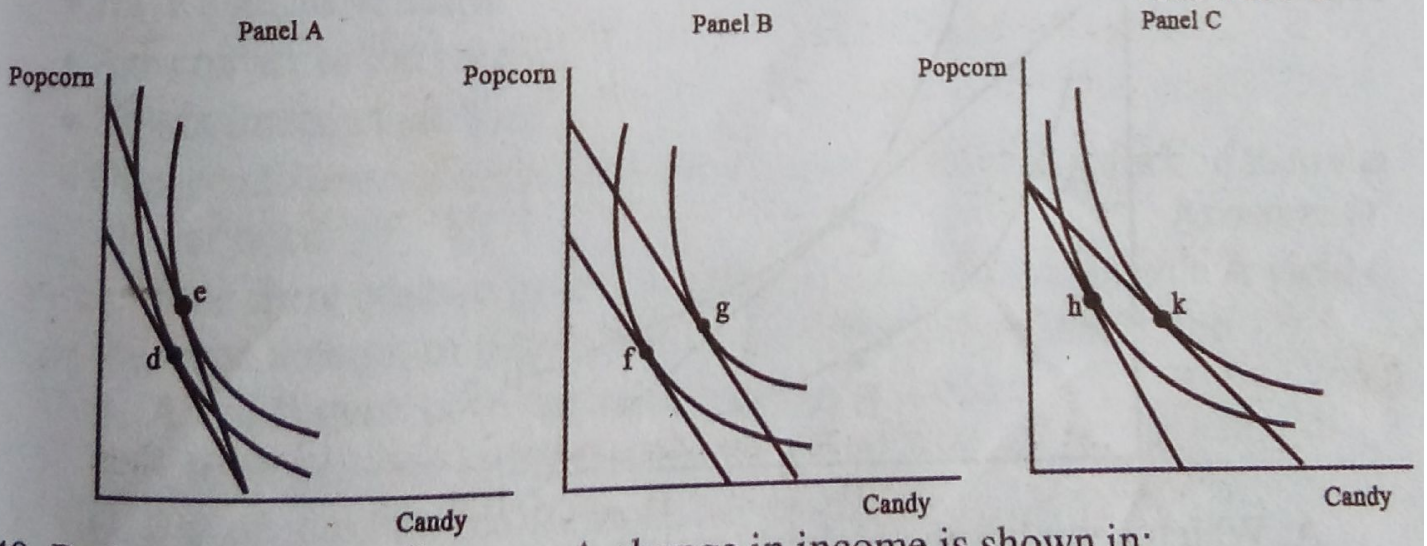
48. Refer to the above figure. The consumer can afford consumption bundles

A. *s, v* and *u* only. B. *r, s, v* and *u*. C. *s, v, t* and *u*. D) *r, s, t* and *u*.

***Explanation*:**

- The utility maximizing combination the two products must lie on the budget line itself rather within the budget line.
- The area found inside the budget line is feasible area (such as point t).
- Any combination of the two goods lie away from the budget line is unattainable.(such as point r and w).
- In combination s, t, u and v the consumer can afford to consume the commodities.
- In combination r and w, the consumer cannot afford to consume a combination of the two goods because of budget constraint.

Answer: C



49. Refer to the above diagram. A change in income is shown in:

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A. Panel A B. panel B C. panel C D. all

***Explanation*:**

- If the income of the consumer increases the budget line will shift to the right, the consumer gets more of goods for increased income
- In panel B the consumer shifts from point' f' to point 'g 'because of an increase in the consumer income.

Answer: B

50. Refer to the above diagram, a change in the price of candy only is shown in

A. Panel A. B. Panel B. C. Panel C. D. none of the above panels

***Explanation*:**

- Panel A and panel C shows a change in price.
- Panel A indicates that a decrease in the price of popcorn increase consumption of popcorn
- Panel C shows that a decrease in price of candy increases it consumption.

Answer: C

51. Refer to the above diagram, which diagram demonstrates an increase in total utility following a decrease in the price of popcorn?

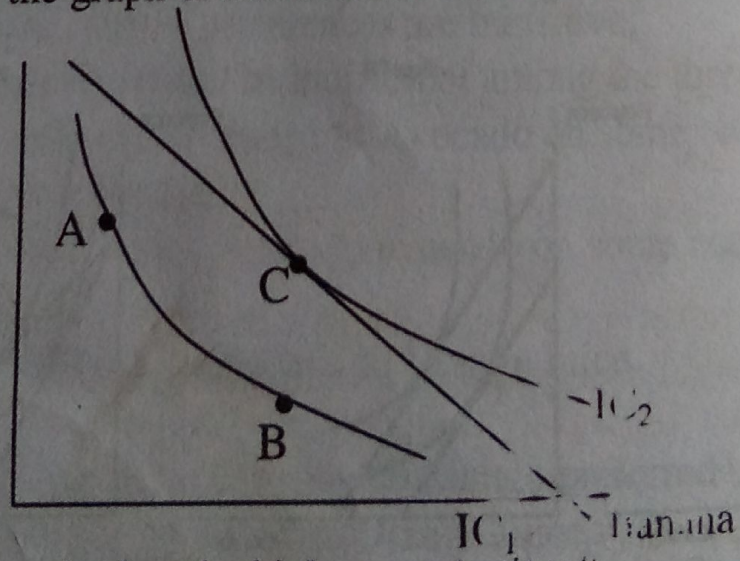
A. the movement from *d* to *e* in Panel A
B. the movement from *f* to *g* in Panel B
C. the movement from *h* to *k* in Panel C
D. none of the above

***Explanation*:**

- When the price of an item declines, the real income (purchasing power) of the consumer increases and the consumer consumes more of the item.
- In panel A, as a result of a decrease in the price of popcorn, the consumer can consume more of popcorn and move from point' d' to point 'e '.

Answer: A

52. Consider the graph of indifference curves below



A. Which point has the highest total utility?

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***Explanation*:**

- A consumer can maximize his satisfaction by consuming the combination of goods and services that correspond to the point of tangency between an indifference curve and a budget line.
- Point C has the highest total utility because it is on the highest indifference curve

B. How does total utility at point A compare to total utility at point B?

***Explanation*:**

- Points A and B have the same total utility since they are on the same indifference curve

53. Indifference curves that are convex to the origin reflect:

A. an increasing marginal rate of substitution
B. a decreasing marginal rate of substitution
C. a constant marginal rate of substitution
D. a marginal rate of substitution that first decreases, then increases

***Explanation*:**

The convexity of the indifference curve implies that the two commodities are imperfect substitution for each other and that the marginal rate of substitution between the two goods decreases as a consumer moves along an indifference curve.

Answer: B

54. The properties of indifference curves include the following:

A. it is downward sloping
B. it exhibits diminishing marginal rate of substitution
C. it never crosses each other
D. all of the above

***Explanation*:**

Properties of indifference curve are:

- Have a negative slope
- Are convex to the origin
- Do not intersect each other
- Upper indifference curve indicate a higher level of satisfaction than the lower ones

Answer: D

55. Suppose there are two goods, x and y. A consumption bundle A yields the same amount of total utility as consumption bundle B, the

A. A and B must have the same amount of x and y
B. A is on a higher indifference curve than B
C. B is on a higher indifference curve than A
D. A and Bare on the same indifference curve

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***Explanation*:**

- Combination of goods on the same indifference curve yield the same level of satisfaction.
- Combination of goods lying on higher indifference curve yield higher level of satisfaction and are preferred
- Combination of goods on a lower indifference curve yield a lower utility

Answer: D

56. What is the meaning of Indifference curve are convex to the origin?

- The indifference curve being convex to origin means that Marginal Rate of Substitution (MRS) between the two goods continuously falls.
- Let the two goods be X and Y shown on the X-axis and the Y-axis respectively.
- It means that the consumer is willing to sacrifice less and less of Y each time he obtains one more unit of X.
- Sacrifice of Y is the price the consumer is willing to pay for obtaining X.
- As he obtains more and more units of X marginal utility of X declines and therefore he is willing to sacrifice only less of Y.

3.4 Marginal rate of substitution:

The marginal rate of substitution (*MRS*) refers to the amount of one good that an individual is willing to give up for an additional unit of another good while maintaining the same level of satisfaction or remaining on the same indifference curve

Illustrative Example

57. The slope of an indifference curve

A. is calculated by dividing the price of good on the vertical axis by price of the good on the horizontal axis.

B. is calculated by dividing the quantity of the good on the vertical axis by the quantity of the good on the horizontal axis.

C. measures the marginal rate of substitution between the two goods in question.

D. measures total utility.

***Explanation*:**

The negative of the slope of the indifference curve at any one point is the marginal rate of substitution between two commodities X and Y

MRS x,y= change in Y/change in X

Answer: C

58. What is the marginal rate of substitution?

***Explanation*:**

- Marginal rate of substitution is the amount of one product that a consumer is willing to sacrifice in order to consume one more unit of another product without changing the overall satisfaction.

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